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CAPSTONE PROJECT REPORT

Report 3 – Software Requirement Specification

– Ho Chi Minh City, September 2026 –

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I. Record of Changes

Date	A* M, D	In charge	Change Description
26/09/2026	A	Nguyen Binh Khiem	Created the SRS overview (v0.9) for Review 1: product overview and context diagram, actors, use case diagrams and descriptions, screen flow, screen descriptions, screen authorization, non-screen functions, conceptual ERD (v3), non-functional requirements and requirement appendix.
26/09/2026	M	Nguyen Binh Khiem	Redrew the use case diagrams by functional area with the system boundary and include/extend relationships, added the overall use case model, renumbered the use case descriptions (UC01-UC76) and re-exported the ERD from draw.io at 900 dpi.
17/10/2026	M	Nguyen Binh Khiem	Renamed the use case Confirm Booking to Review Booking, added the Create Blog Post use case (UC44) for hotel staff to the AI marketing diagram and the overall use case model, and renumbered the use case descriptions (UC01-UC77).
17/10/2026	M	Nguyen Binh Khiem	Updated after the front-end team review: added the partner sign-up and the verification status of each hotel, support for partners with several hotels (My Hotels, Add Hotel), the partner views of bookings and of the hotel wallet, and renamed the staff booking screen to Booking Review. Rewrote BR-05, BR-06 and BR-09 for prepaid final prices, limited booking changes to details that do not affect the price, added BR-19 to BR-21 with the refund rejection reasons, aligned the security requirements and BR-17 with the implementation, and renumbered the use cases (UC01-UC81) and screens.
17/10/2026	M	Nguyen Binh Khiem	Added the withdrawal flow: the partner requests a withdrawal from the hotel wallet to the personal bank account and the platform manager processes it (Request Withdrawal, Process Withdrawal Request, BR-22, two screens). Added the Email Verification screen and BR-23 for the 6-digit email code, included Verify Email in the partner registration, and renumbered the use cases (UC01-UC83) and screens.
17/10/2026	A	Nguyen Binh Khiem	Released SRS v1.0: added the detailed functional specifications (sections 3.2 to 3.16) for every screen and non-screen function, with trigger, description and function details. Screen layouts are marked for the front-end team to add from the UI prototype.
17/10/2026	M	Nguyen Binh Khiem	Corrected the description of the physical account model in 3.1.5 to match the implementation: a shared

			USER_ACCOUNT table plus one table per role (class table inheritance).
18/10/2026	M	Nguyen Binh Khiem	Aligned the SRS with the code: Next.js and NestJS; SePay with VietQR and webhook, refunds and payouts as recorded bank transfers (UC35 renamed Record Refund Transfer, use case diagram updated); Nodemailer over SMTP; Cloudinary storage; the common API response envelope; the four-step partner wizard with legal entity and beneficiary details; required verification documents; per-hotel review and re-verification; room configuration and room statuses; five staff permission groups; staff invitations, login lockout, sessions and rate limits; money rounding and the commission range (BR-13, BR-18, BR-19, BR-22, BR-24 to BR-29).
26/10/2026	M	Nguyen Binh Khiem	Added the screen layouts to every screen function in sections 3.2 to 3.16: web screenshots for all screens and mobile screenshots for the customer screens that exist in the app.
29/10/2026	M	Nguyen Binh Khiem	Changed the approver of AI marketing content to the hotel partner, as in the Review 1 slide: BR-14, UC44, UC47 (now an actor of the hotel partner), the actor description of the hotel partner, Blog List, Content Editor, the content generation service, the MARKETING_CONTENT entity, the screen authorization and the AI marketing use case diagram.
29/10/2026	M	Nguyen Binh Khiem	Aligned the refund decision with the code: staff check a refund request and forward it, the hotel partner approves or rejects it, and a request still undecided at the check-in time is approved automatically (UC34, BR-20, Refund Requests, Refund Review, actor descriptions, screen authorization and the cancellation and refund use case diagram).
29/10/2026	M	Nguyen Binh Khiem	Aligned refunds with the code: an approved refund is credited to the customer wallet instead of a bank transfer (UC35 renamed Credit Refund to Wallet, UC34, Refund Review, the Payment Gateway actor and the use case diagram).
02/11/2026	M	Nguyen Binh Khiem	Aligned the SRS with the decisions of the Report 5 review: the password reset link expires after 15 minutes (3.2.4, BR-17); the partner dashboard defaults to the month to date (3.11.2); commission is also charged on no-show bookings (BR-13); staff no longer confirm or decline bookings, since a booking is confirmed by its payment (3.7.3, UC27, Booking Review screen, screen authorization, usability goal); the mobile application also serves hotel staff with the same functions as on the web (screen overview, user

			interfaces); the hotel partner approves or rejects AI content in the Content Approval view of Content Editor, and rejected content has its own tab in Content Library (3.9.1, 3.9.2).

*A - Added M - Modified D - Deleted

II. Software Requirement Specification

1. Product Overview

StayReco is a multi-tenant web and mobile platform that connects small and mid-sized hotels in Vietnam with travellers. Travellers search and book rooms across all partner hotels, pay through a payment gateway, chat with an AI assistant and receive personalized recommendations. Each hotel partner works inside an isolated tenant workspace to manage its hotel, rooms, pricing, staff and analytics, while hotel staff handle bookings, front desk, customer care and AI-assisted marketing. The platform manager reviews partners and follows commission and revenue, and the admin keeps accounts, payment configuration, logs and AI settings in order.

The context diagram below shows StayReco as a single process and the information exchanged with its six user roles and with the external payment gateway. Other external services used by the system (OpenAI API, Cloudinary, Firebase Cloud Messaging and the email provider) are described in section 4.1.

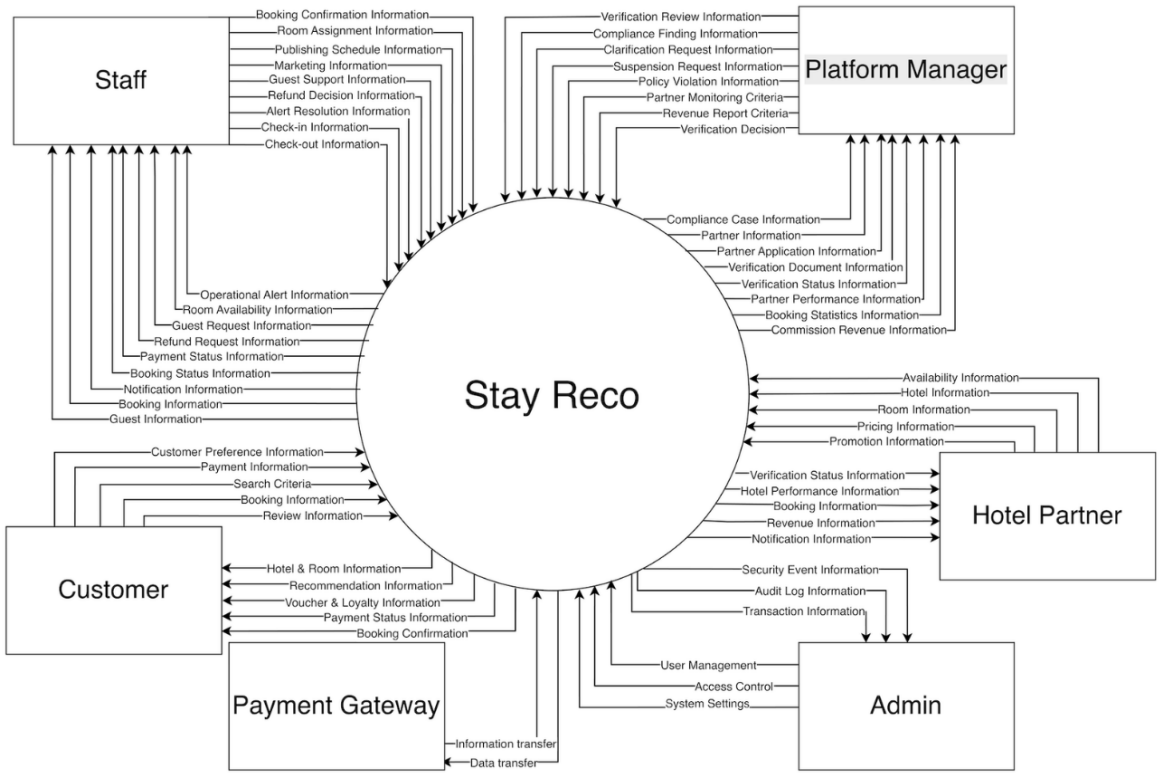


Figure 1. Context diagram of StayReco

2. User Requirements

2.1 Actors

StayReco has six user roles registered in form FA26SE056 and one external system actor.

#	Actor	Description
1	Guest	An anonymous visitor of the web or mobile application. A guest can search and browse hotels and rooms, read the hotel blog, chat with the AI assistant for suggestions based on the current session, and register or log in.
2	Customer	A registered traveller. Besides everything a guest can do, a customer books and pays for rooms, manages the bookings, receives personalized recommendations, writes reviews, and uses vouchers and loyalty points.
3	Hotel Staff	An employee of one partner hotel, working only with the data of that hotel. Hotel staff review bookings, handle check-in and check-out, answer guests, take over AI conversations, check and forward refund requests, handle alerts and prepare marketing content according to the permissions granted by the partner.
4	Hotel Partner	The owner or operator of one or more hotels. A partner signs up, gets each hotel verified, then manages the information, rooms, availability, pricing, promotions, loyalty program, staff and AI content settings of each hotel inside an isolated tenant workspace, approves the AI marketing content prepared by the staff, decides the refund requests, and follows the bookings, analytics and wallet of its hotels.
5	Platform Manager	The operator of the StayReco marketplace. The platform manager approves partner verification requests, monitors partner activity, opens policy-violation cases and requests suspension, and follows commission, platform revenue and growth reports.
6	Admin	The system administrator. The admin manages user accounts and access decisions, the payment gateway configuration, transaction logs, audit and security logs, and the platform-level AI settings.
7	Payment Gateway	The external payment service SePay. The customer pays by bank transfer with a VietQR code into the bank account of the platform; SePay watches that account and sends a webhook for each incoming transfer so the system can confirm the booking. Refunds are credited to the customer wallet in StayReco; partner payouts are real bank transfers made by hand and recorded in the system. StayReco never stores card data.

2.2 Use Cases

Figure 2 gives the overall use case model of the system. The use cases are then detailed in eight diagrams by functional area. Each diagram shows the StayReco system boundary, the actors outside it and the include and extend relationships between use cases. "Authorized User" stands for any logged-in role (Customer, Hotel Staff, Hotel Partner, Platform Manager and Admin), and the Customer inherits

every use case of the Guest. Use cases of the Guest and the Customer are available on both the web and the mobile application.

2.2.1 Diagram(s)



Figure 2. Overall use case model of StayReco

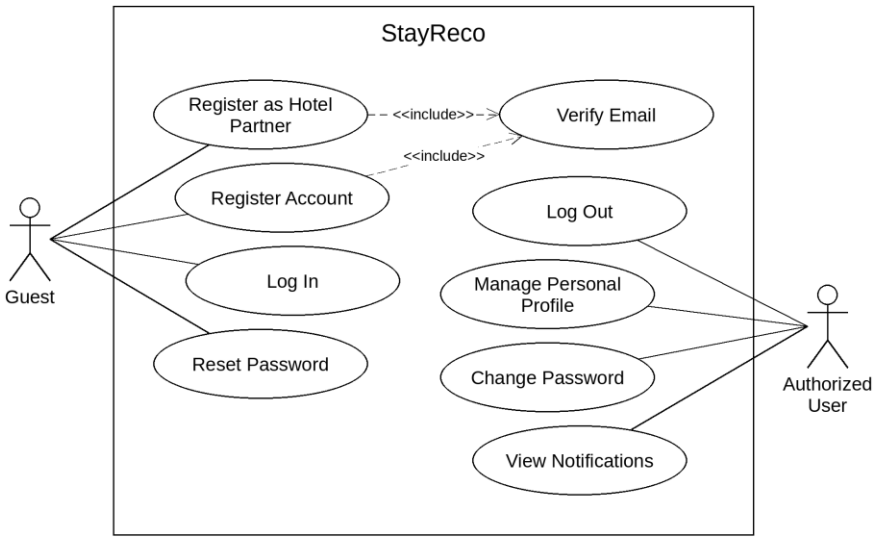


Figure 3. Use case diagram - Account and access

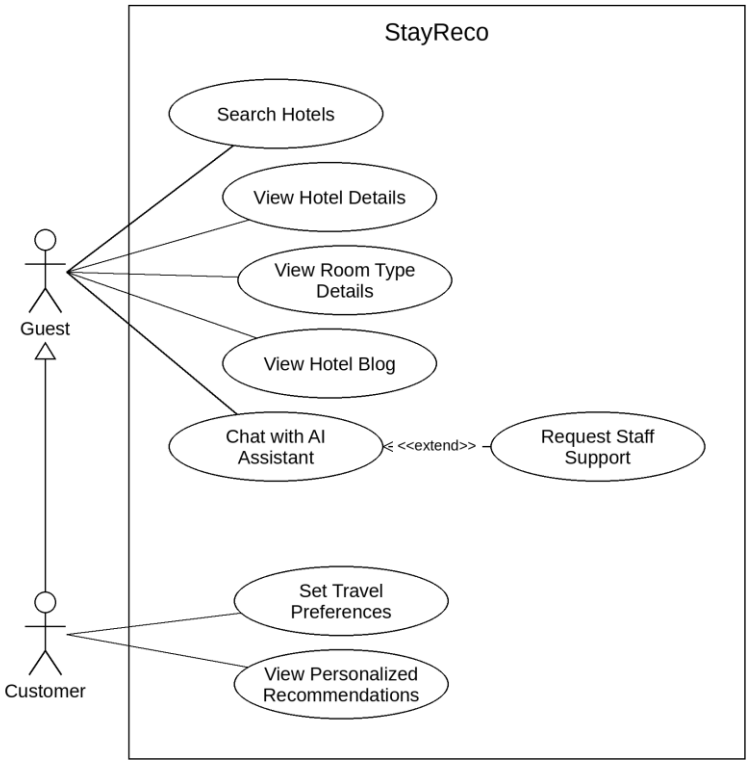


Figure 4. Use case diagram - Hotel discovery, AI assistant and recommendations

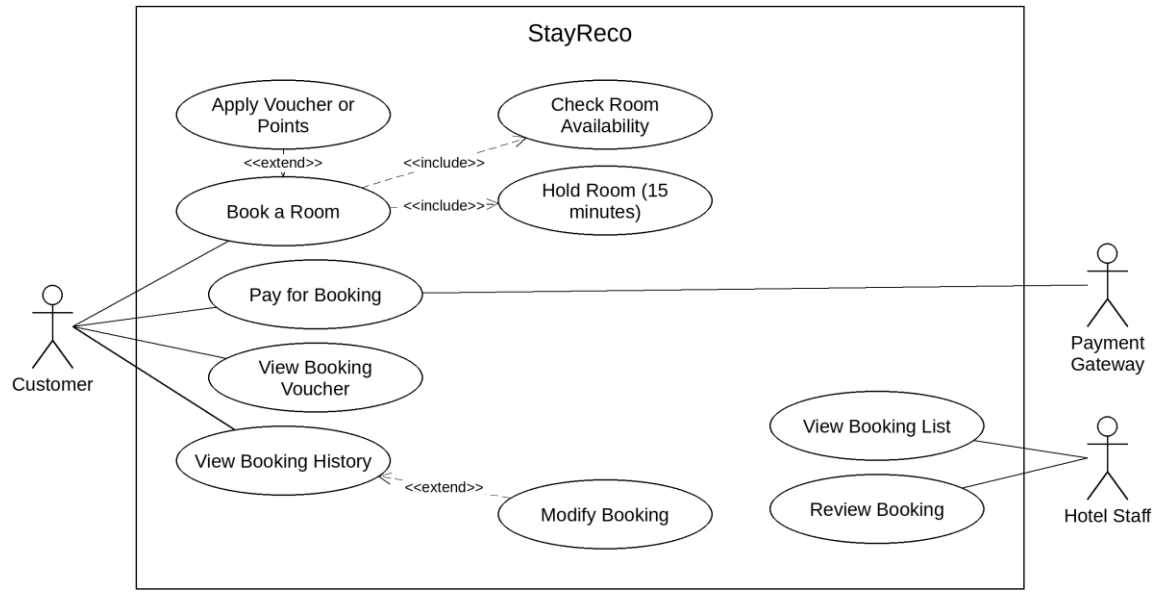


Figure 5. Use case diagram - Booking and payment

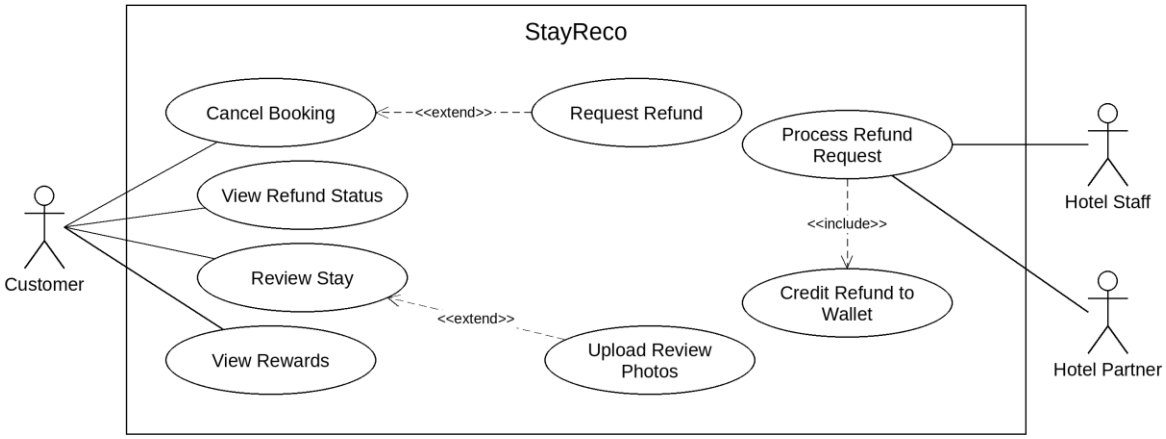


Figure 6. Use case diagram - Cancellation, refund and review

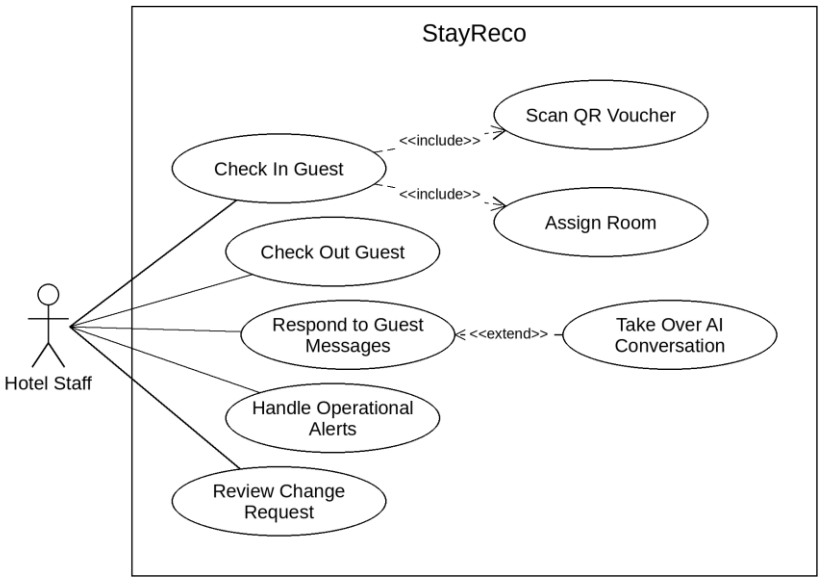


Figure 7. Use case diagram - Front desk and customer care

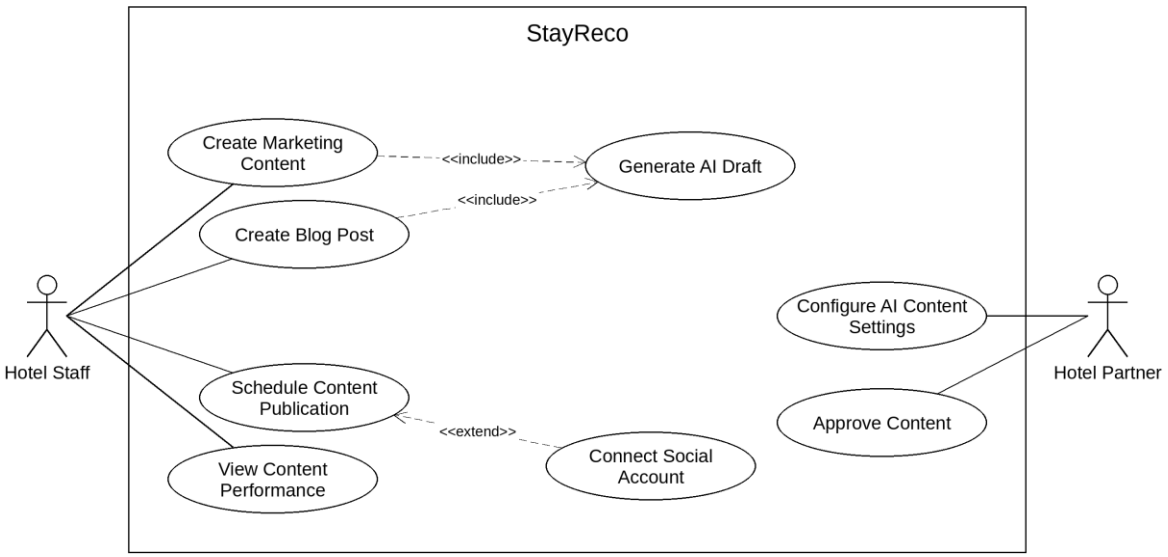


Figure 8. Use case diagram - AI marketing support

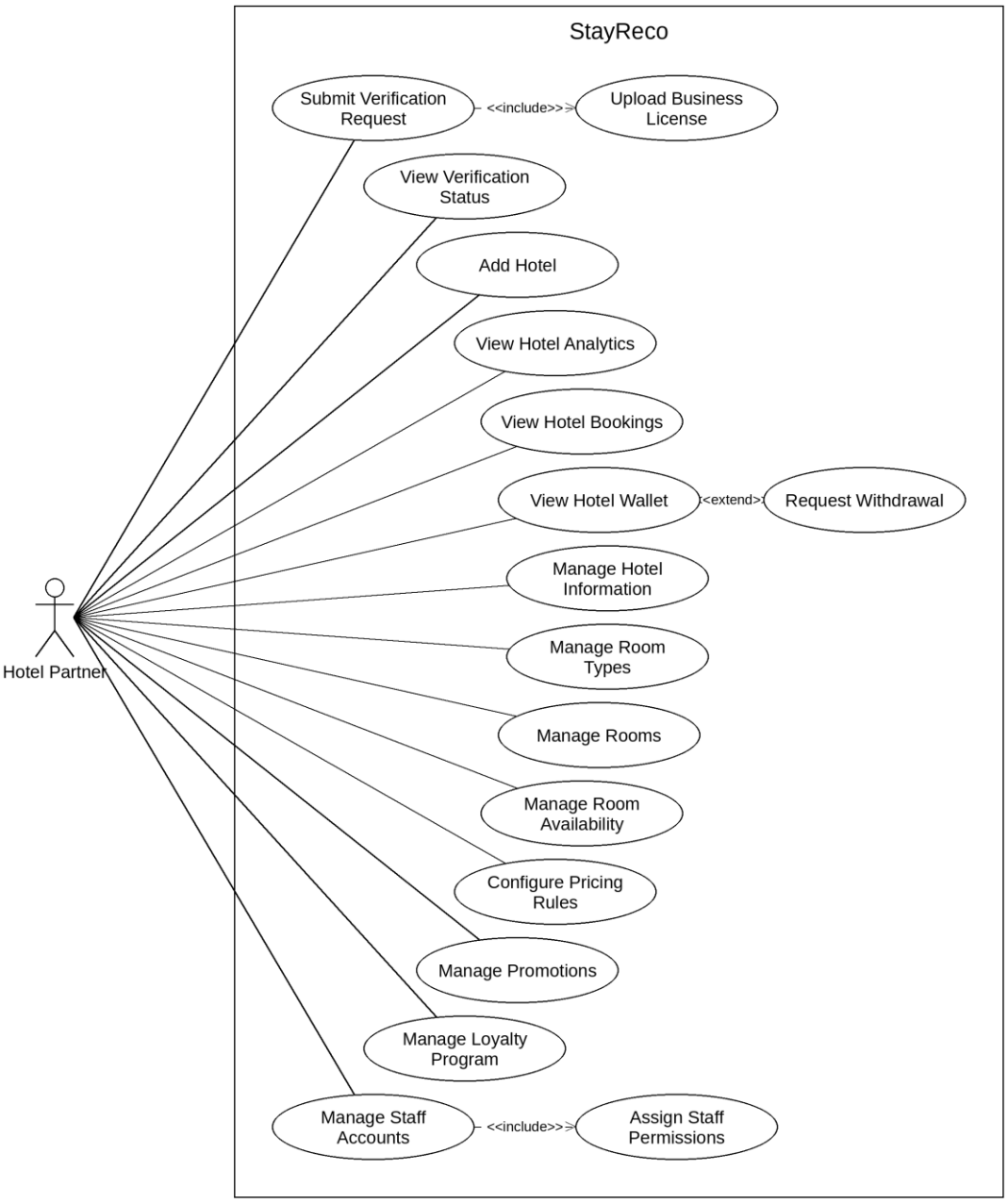


Figure 9. Use case diagram - Hotel partner workspace

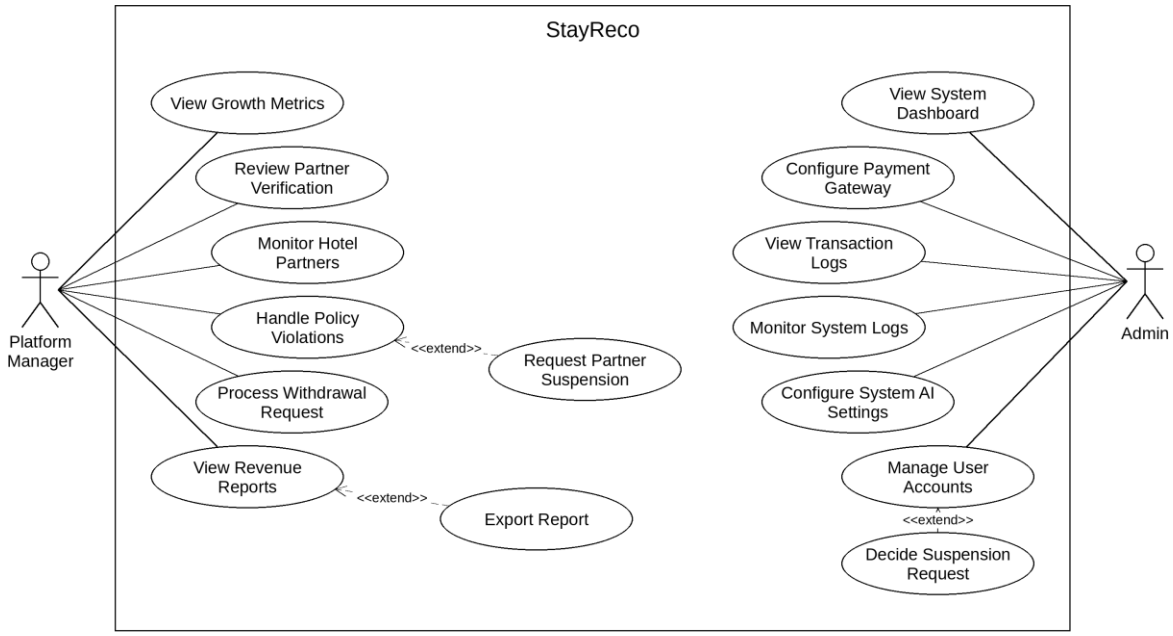


Figure 10. Use case diagram - Platform operations and administration

2.2.2 Descriptions

The table below describes each use case shown in the diagrams.

ID	Use Case	Actors	Use Case Description
UC01	Register Account	Guest	The guest enters name, email and password to create a customer account and confirms the email address. A hotel owner uses the partner sign-up instead.
UC02	Register as Hotel Partner	Guest	A hotel owner enters name, email, phone number, password and, optionally, the company name to create a partner account, then activates it with the code sent by email. After that the partner can sign in, but only the onboarding and verification screens are open until one of its hotels is approved.
UC03	Verify Email	Guest	The system sends a 6-digit code to the registered email. The account stays inactive and cannot sign in until the code is entered; a correct code activates the account and signs the user in. The code is valid for 10 minutes, a new code can be requested, and the code is cancelled after 5 wrong entries.
UC04	Log In	Guest	The user logs in with email and password. The system checks the account status and opens the workspace of the role of that account; a pending booking selection is kept after login.
UC05	Reset Password	Guest	The user asks for a reset link by email and sets a new password with a valid link. The system never reveals whether an email is registered.

UC06	Log Out	Authorized User	The signed-in user ends the session; the access token is revoked on the device.
UC07	Manage Personal Profile	Authorized User	The user views and updates name, avatar, phone and contact details. The role is shown read-only.
UC08	Change Password	Authorized User	The user enters the current password and a new password that meets the password rules.
UC09	View Notifications	Authorized User	The user views system notifications, marks them as read and opens the related record allowed for the role.
UC10	Search Hotels	Guest, Customer	The user searches hotels of all partners by destination, dates and number of guests, then filters and sorts the results by price, rating, amenities and cancellation terms.
UC11	View Hotel Details	Guest, Customer	The user views photos, location, amenities, reviews, policies and the room offers available for the selected dates of one hotel.
UC12	View Room Type Details	Guest, Customer	The user views the details of a room type, compares rate plans and selects the room to book. A guest is asked to log in before continuing.
UC13	View Hotel Blog	Guest, Customer	The user reads the list of travel and hotel articles, filters them by category and opens an article with its related hotel.
UC14	Chat with AI Assistant	Guest, Customer	The user asks the AI assistant about hotels, rooms and policies in Vietnamese or English. The assistant answers from the hotel data and suggests rooms; a guest gets suggestions from the current session only, a customer also from preferences and history. The user can ask to talk to hotel staff.
UC15	Request Staff Support	Guest, Customer	From the AI chat the user asks to talk to a person; the conversation is passed to the staff of the related hotel.
UC16	Set Travel Preferences	Customer	The customer optionally selects trip purpose, budget, favourite destinations, amenities and accommodation style. The preferences are used by the recommender and can be edited at any time.
UC17	View Personalized Recommendations	Customer	The customer sees hotels and rooms recommended from preferences, search behaviour and booking history, with the reason for each suggestion.
UC18	Book a Room	Customer	The customer enters guest details and special requests, applies an optional voucher or loyalty points and confirms. The system rechecks availability and holds the room while the payment is completed.

UC19	Check Room Availability	Customer	The system rechecks that the selected room type still has a free room for every night of the stay.
UC20	Hold Room (15 minutes)	Customer	The system holds the room for the booking while the payment is completed and releases it when the time runs out.
UC21	Apply Voucher or Points	Customer	The customer applies an eligible voucher, promotion code or loyalty points of the hotel to the booking.
UC22	Pay for Booking	Customer, Payment Gateway	The customer pays by bank transfer to the VietQR code shown for the booking. The booking becomes confirmed only after SePay reports the matching transfer; if no payment arrives before the hold expires, the held room is released.
UC23	View Booking Voucher	Customer	The customer views and downloads the QR-code e-voucher of a confirmed booking to show at check-in.
UC24	View Booking History	Customer	The customer views upcoming, in-house, completed and cancelled bookings and opens the details and timeline of each booking.
UC25	Modify Booking	Customer	The customer asks to change details that do not affect the price, such as the guest name, contact details, arrival time or special requests; the change takes effect after the hotel accepts it. New dates, another room type or another number of guests need a cancellation and a new booking.
UC26	View Booking List	Hotel Staff	The staff member searches and filters the bookings of the hotel by code, guest, date and status.
UC27	Review Booking	Hotel Staff	The staff member reviews a new booking of the hotel, checks the guest details, payment status and special requests, and handles its change requests.
UC28	Cancel Booking	Customer	The customer cancels a booking with a reason. The system applies the cancellation policy of the booking, shows the fee and the refundable amount, and creates a refund request when money is due.
UC29	Request Refund	Customer	When the cancelled booking has a refundable amount, the system creates a refund request for the hotel.
UC30	View Refund Status	Customer	The customer follows the refund of a cancelled booking until it is completed.
UC31	Review Stay	Customer	After a completed stay the customer rates the hotel, writes a review and uploads photos.
UC32	Upload Review Photos	Customer	The customer attaches photos of the stay to the review.

UC33	View Rewards	Customer	The customer views the loyalty points and vouchers of each hotel with history and conditions.
UC34	Process Refund Request	Hotel Staff, Hotel Partner	The staff member checks a refund request whose amount is calculated by the system and forwards it to the hotel partner with a note. The hotel partner approves it, or rejects it for one of the reasons in BR-20, and the customer is informed; a request still undecided at the check-in time is approved automatically. An approved refund is credited to the customer wallet and its status is shown to the customer.
UC35	Credit Refund to Wallet	Hotel Partner	The approved refund is moved from the pending balance of the hotel wallet to the available balance of the customer wallet, and the refund status is updated.
UC36	Check In Guest	Hotel Staff	The staff member scans the QR voucher or enters the booking code, assigns a ready physical room of the booked room type and confirms the check-in.
UC37	Scan QR Voucher	Hotel Staff	The staff member scans the QR voucher or types the booking code to open the booking.
UC38	Assign Room	Hotel Staff	The staff member assigns a ready physical room of the booked room type to the stay.
UC39	Check Out Guest	Hotel Staff	The staff member reviews the charges and payments of the stay and completes the check-out; the stay becomes completed and eligible for review and loyalty points.
UC40	Respond to Guest Messages	Hotel Staff	The staff member answers guest inquiries and complaints in the unified inbox, with the related booking shown next to the conversation.
UC41	Take Over AI Conversation	Hotel Staff	The staff member monitors live AI conversations and takes over one that needs a person; the AI stops replying in that conversation until it is handed back.
UC42	Handle Operational Alerts	Hotel Staff	The staff member reviews alerts raised for unusual booking patterns or negative reviews, takes them over and closes them with a resolution note.
UC43	Review Change Request	Hotel Staff	The staff member accepts or declines a change request of a customer (guest details, arrival time or special requests) with a reason.
UC44	Create Marketing Content	Hotel Staff	The staff member asks the AI to draft a caption or promotion, edits the draft, checks the facts and submits it to the hotel partner for approval.
UC45	Create Blog Post	Hotel Staff	The staff member asks the AI to draft a blog article from a short brief, edits it, adds

			photos and the related hotel, and submits it for approval. Once approved, the article appears in the hotel blog for guests and customers.
UC46	Generate AI Draft	Hotel Staff	The AI drafts a caption, blog post or promotion from the brief and the brand settings of the hotel.
UC47	Approve Content	Hotel Partner	The hotel partner approves the content for publication or rejects it with feedback; later edits need a new approval.
UC48	Schedule Content Publication	Hotel Staff	The staff member connects the supported social media accounts of the hotel and schedules approved content on the publishing calendar.
UC49	Connect Social Account	Hotel Staff	The staff member connects or reconnects a supported social media account of the hotel.
UC50	View Content Performance	Hotel Staff	The staff member views reach, engagement and attributed bookings of published content.
UC51	Configure AI Content Settings	Hotel Partner	The partner sets the brand voice, tone and content templates that the AI uses when drafting content for the hotel.
UC52	Submit Verification Request	Hotel Partner	For each hotel, the partner enters business and hotel information, uploads the business license and submits the verification request; a draft can be saved.
UC53	Upload Business License	Hotel Partner	The partner uploads the required documents (business license, fire safety certificate and security and order certificate) and optional ones such as the star rating certificate (BR-26).
UC54	View Verification Status	Hotel Partner	The partner follows the review of each hotel, reads the reasons when changes are requested and resubmits.
UC55	Add Hotel	Hotel Partner	The partner adds another hotel to the account. The new hotel goes through its own verification request and is listed only after it is approved.
UC56	View Hotel Analytics	Hotel Partner	The partner views revenue, occupancy, booking sources and satisfaction trends of the selected hotel, or of all its hotels together, for a chosen period.
UC57	View Hotel Bookings	Hotel Partner	The partner views and searches the bookings of the selected hotel with their booking, payment and refund status in read-only mode; handling the bookings stays with the hotel staff.
UC58	View Hotel Wallet	Hotel Partner	The partner views the pending and available balance of the wallet of the selected hotel and its transactions: booking payments, commission, refunds and withdrawals.

UC59	Request Withdrawal	Hotel Partner	The partner asks to withdraw an amount from the available balance of a hotel wallet to the beneficiary bank account declared by the partner. The amount is held until the platform manager processes the request.
UC60	Manage Hotel Information	Hotel Partner	The partner updates hotel description, address, amenities, images and policies shown to guests.
UC61	Manage Room Types	Hotel Partner	The partner creates and edits room types with base price, capacity, area, bed types and quantities, amenities and photos. A room type that still has physical rooms cannot be deleted.
UC62	Manage Rooms	Hotel Partner	The partner creates and edits the physical rooms of each room type within the declared total, and locks a room for maintenance or unlocks it.
UC63	Manage Room Availability	Hotel Partner	The partner views and updates the number of rooms open for sale per room type and date, and closes dates when needed.
UC64	Configure Pricing Rules	Hotel Partner	The partner defines rule-based, seasonal and early-booking rates (by how many days before arrival the booking is made) for chosen room types and date ranges and previews the resulting price.
UC65	Manage Promotions	Hotel Partner	The partner creates campaigns and discount codes with validity, eligible room types, limits and discount value, applied on top of the base price.
UC66	Manage Loyalty Program	Hotel Partner	The partner configures point-earning rules, redemption conditions and hotel vouchers.
UC67	Manage Staff Accounts	Hotel Partner	The partner invites staff by email, choosing the hotel and the position, deactivates staff accounts and assigns permissions; each staff account belongs to exactly one hotel of the partner.
UC68	Assign Staff Permissions	Hotel Partner	The partner grants permission groups (booking, front desk, customer care, refund, marketing) to a staff account; without any group the staff member can only view.
UC69	View Growth Metrics	Platform Manager	The manager views the number of partners, bookings, customers and their growth on the platform dashboard.
UC70	Review Partner Verification	Platform Manager	The manager opens the verification request of one hotel, checks the documents and approves it, or requests changes or rejects it with a reason. The manager can also ask an active hotel to verify again.
UC71	Monitor Hotel Partners	Platform Manager	The manager views the booking activity, cancellation rate and satisfaction of each partner hotel.

UC72	Handle Policy Violations	Platform Manager	The manager records a policy-violation case with evidence and sends a suspension request to the admin, who makes the account decision.
UC73	Request Partner Suspension	Platform Manager	The manager sends a suspension request with reason and evidence to the admin.
UC74	Process Withdrawal Request	Platform Manager	The manager checks a withdrawal request against the wallet balance and the bank details, transfers the money to the bank account of the partner and records the transfer reference, or rejects the request with a reason. The partner is notified either way.
UC75	View Revenue Reports	Platform Manager	The manager views commission, platform revenue, revenue by hotel and money flow reports for a chosen period and exports them.
UC76	Export Report	Platform Manager	The manager exports the filtered report with its date range and scope.
UC77	View System Dashboard	Admin	The admin views account statistics, pending account decisions, payment issues and recent security events.
UC78	Configure Payment Gateway	Admin	The admin maintains the gateway configuration with masked credentials and tests the connection.
UC79	View Transaction Logs	Admin	The admin views payment and refund transactions, checks their status with the gateway and investigates mismatches.
UC80	Monitor System Logs	Admin	The admin reads the audit trail and security events with filters by time, actor, action and severity.
UC81	Configure System AI Settings	Admin	The admin selects the AI model, maintains the API key, sets usage limits and safety filters for the platform.
UC82	Manage User Accounts	Admin	The admin searches accounts, invites platform managers, and activates or deactivates accounts with a reason, including the decision on a suspension request from the platform manager. Accounts have no approval step; approval applies only to hotel verification.
UC83	Decide Suspension Request	Admin	The admin reviews the evidence of a suspension request and approves or rejects the account decision.

3. Functional Requirements

3.1 System Functional Overview

This section gives the functional overview of StayReco: the screen flow, the screen descriptions, the authorization of each role on the screens, the functions that run without a screen, and the entity relationship diagram.

3.1.1 Screens Flow

The screen flow is drawn per role. Blue text marks screens reachable without logging in, a beige background marks screens shared by several roles, a rounded box is a pop-up or drawer, a box with inner items is a screen with tabs, and a dashed box points to screens drawn in another figure. After login, PostLogin sends each user to the start screen of the role.

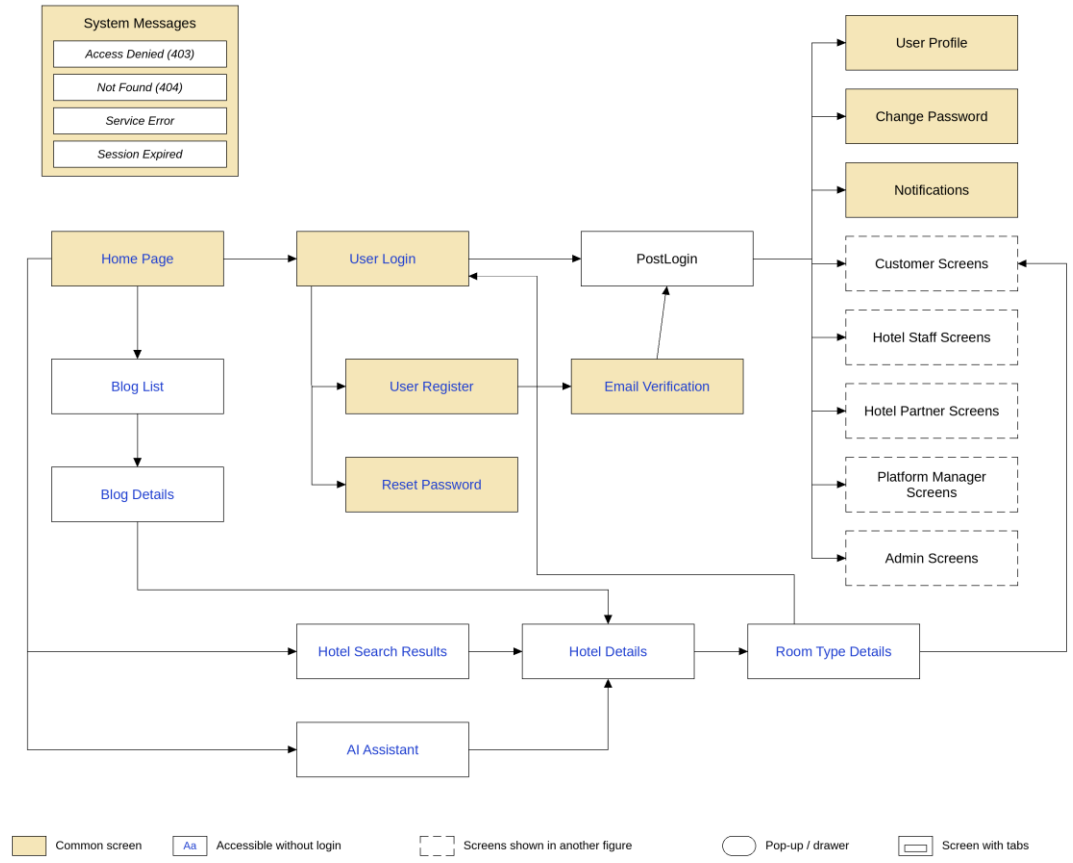


Figure 11. Screen flow of the Guest and common screens

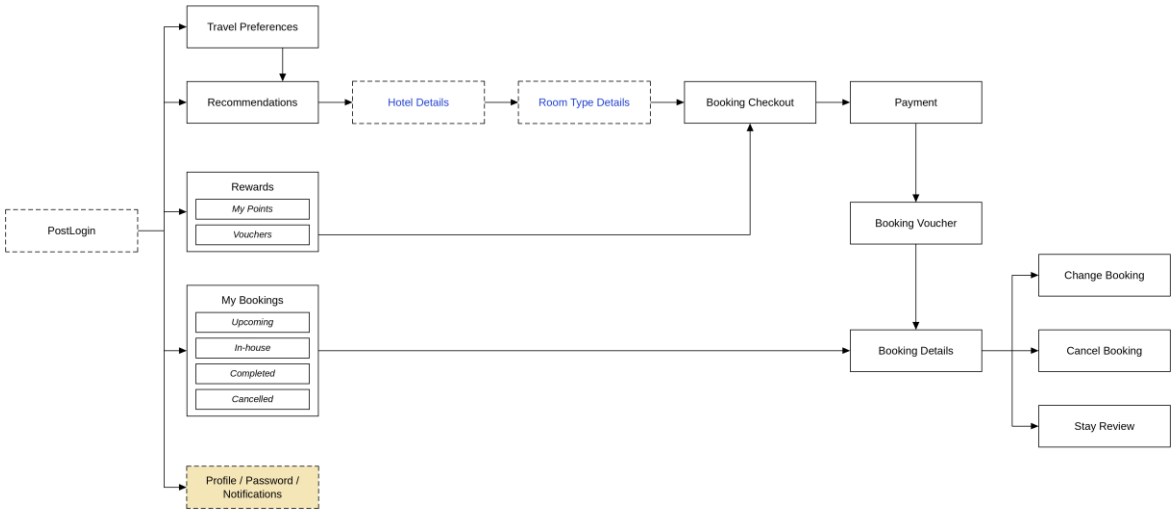


Figure 12. Screen flow of the Customer

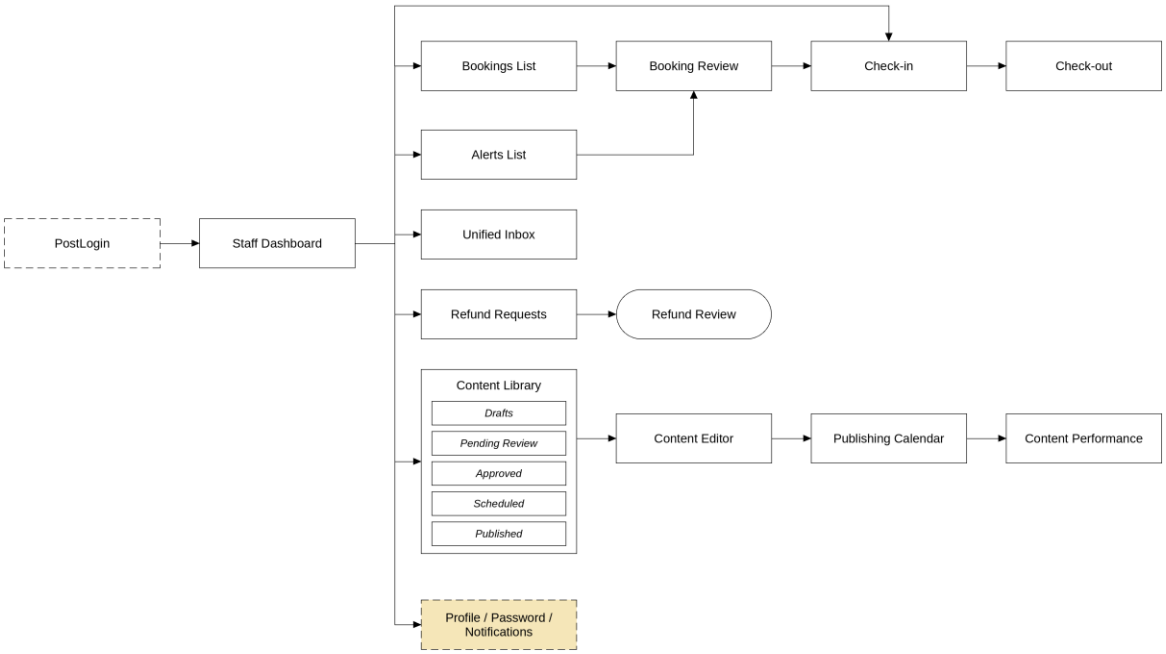


Figure 13. Screen flow of the Hotel Staff

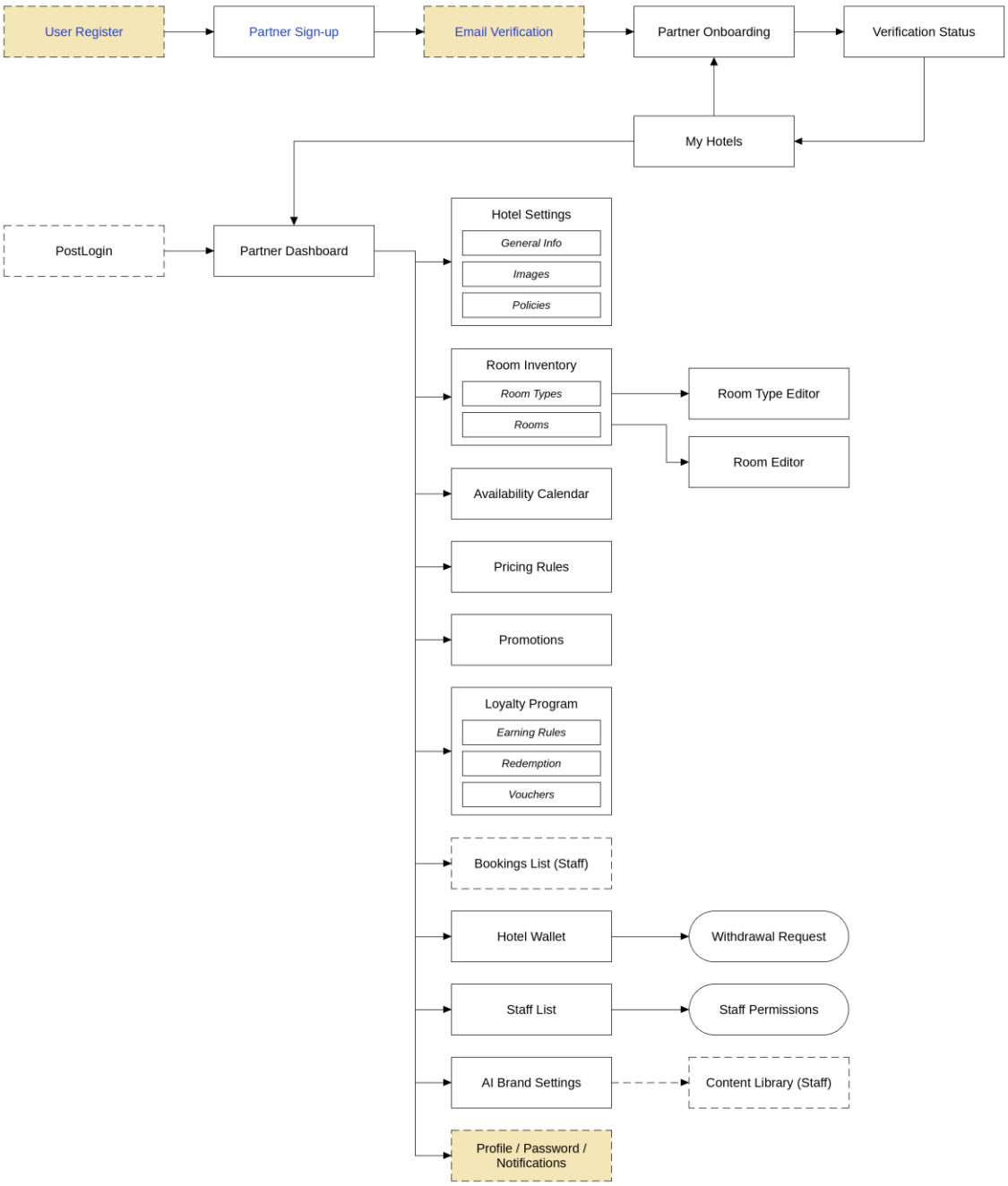


Figure 14. Screen flow of the Hotel Partner

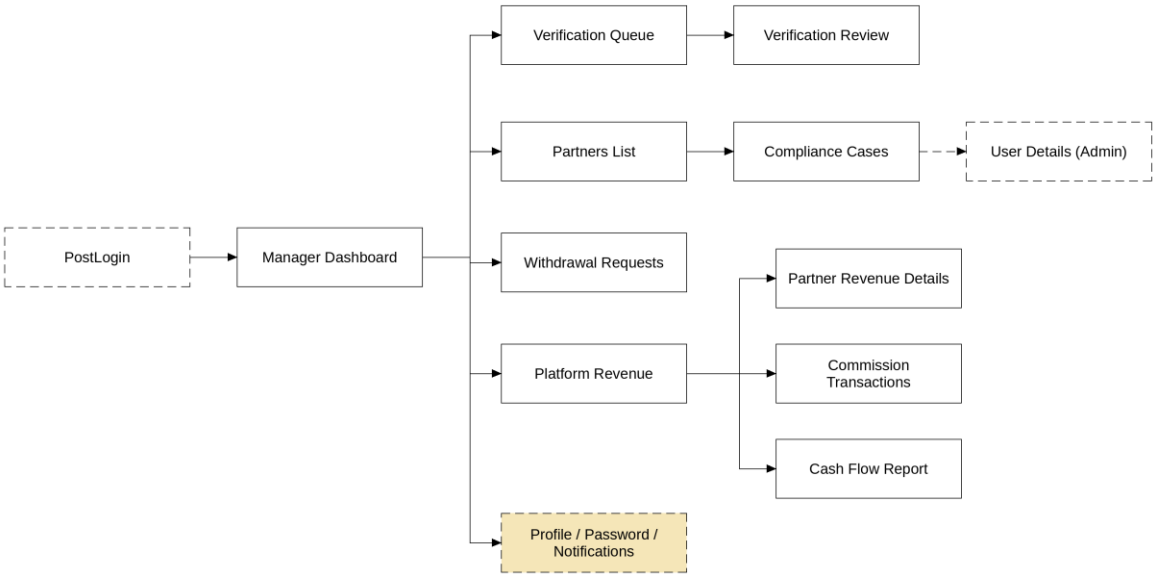


Figure 15. Screen flow of the Platform Manager

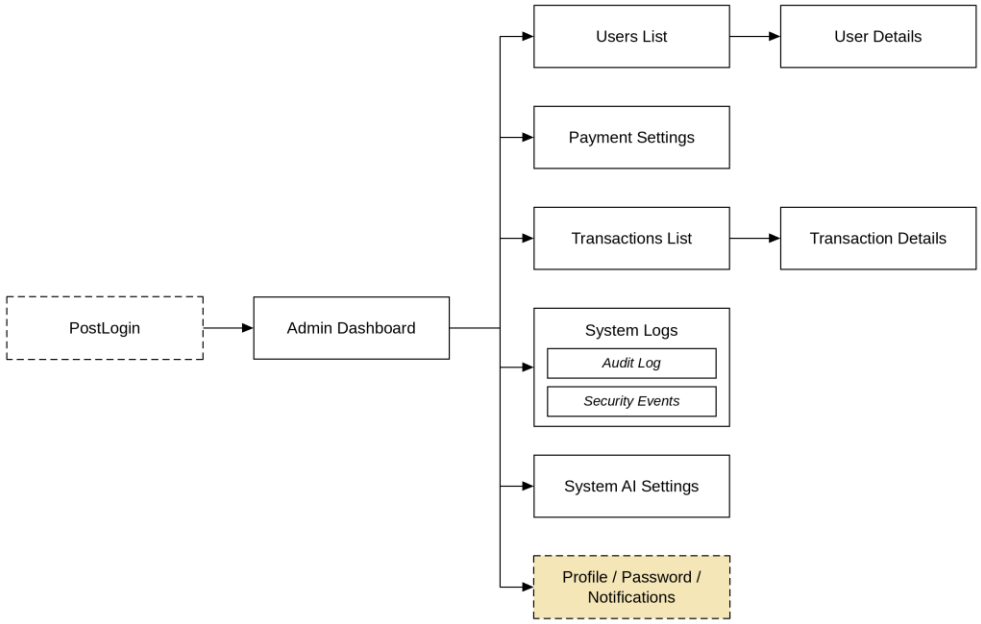


Figure 16. Screen flow of the Admin

3.1.2 Screen Descriptions

The screens below are the web screens of the UI prototype. The mobile application reuses the Guest, Customer and Hotel Staff screens in a mobile layout; hotel staff have the same functions on mobile as on the web.

#	Feature	Screen	Description
1	Account & Access	User Login	Login form with email and password, links to registration and password recovery. After login the user is sent to the workspace of the role, and a booking selection made before login is kept.

2	Account & Access	User Register	Registration form for a traveller account (name, email, password, confirmation) with email verification, and a link to the partner sign-up for hotel owners.
3	Account & Access	Email Verification	Form to enter the 6-digit code sent to the email, with a countdown and the resend action. A correct code activates the account and opens the workspace of the role.
4	Account & Access	Reset Password	Email form that sends a reset link, followed by the new-password form opened from a valid link.
5	Account & Access	User Profile	Personal information of the signed-in user: avatar, full name, phone and contact details; the role is read-only.
6	Account & Access	Change Password	Form with current password, new password and confirmation; the account menu also holds the logout action.
7	Account & Access	Notifications	List of notifications grouped by date with All and Unread tabs, mark-as-read and a link to the related record.
8	Account & Access	System Messages	Shared pages for access denied (403), not found (404), temporary service error and expired session.
9	Hotel Discovery	Home Page	Public home page with the search box (destination, dates, guests), an entry to the AI assistant, featured hotels, destinations and recent blog articles.
10	Hotel Discovery	Hotel Search Results	Result list of hotels with filters (budget, star class, guest rating, amenities, breakfast, cancellation terms), sorting and the total price for the stay.
11	Hotel Discovery	Hotel Details	Hotel page with gallery, location, amenities, reviews, policies and the room offers for the selected dates.
12	Hotel Discovery	Room Type Details	Room type page with photos, size, beds, occupancy, amenities, rate plans and the "Select this room" action.
13	Hotel Discovery	Blog List	List of travel and hotel articles with category filter, search and pagination.
14	Hotel Discovery	Blog Details	Article page with table of contents, related hotel and related articles.
15	AI Assistant & Recommender	AI Assistant	Chat workspace with the AI assistant and matching hotel and room cards, editable trip criteria and the request to talk to hotel staff.
16	AI Assistant & Recommender	Travel Preferences	Optional preference setup: trip purpose, budget, destinations, amenities and accommodation style.
17	AI Assistant & Recommender	Recommendations	Personalized hotel and room suggestions with the reason of each suggestion and a shortcut to edit preferences.

18	Booking & Payment	Booking Checkout	Checkout form with booker and guest details, special requests, voucher and loyalty points, cancellation terms and the order summary.
19	Booking & Payment	Payment	Payment page with the VietQR code, the amount and the transfer content of the booking, then the result: verifying, paid, failed, cancelled or expired.
20	Booking & Payment	Booking Voucher	Confirmation page with booking code, stay details, paid amount and the QR-code e-voucher to download.
21	Trip Management	My Bookings	Booking list with tabs Upcoming, In-house, Completed and Cancelled, search by booking code and the main action for each booking.
22	Trip Management	Booking Details	Details of one booking: booking, payment and refund status, price breakdown, cancellation terms, timeline and the allowed actions.
23	Trip Management	Change Booking	Form to change the guest details, contact details, arrival time or special requests and send the change request. The price does not change; a note explains that new dates or another room need a new booking.
24	Trip Management	Cancel Booking	Cancellation review with policy deadline, cancellation fee, refundable amount and reason.
25	Trip Management	Stay Review	Review form for a completed stay: rating, review text and photo upload.
26	Trip Management	Rewards	Points and vouchers of the customer per hotel (tabs My Points and Vouchers) with history and conditions.
27	Front Desk	Staff Dashboard	Daily overview of the assigned hotel: arrivals, departures, rooms, pending tasks and alerts.
28	Front Desk	Bookings List	Searchable table of the bookings of the hotel with booking and payment status filters. Staff work on the bookings of their hotel; a partner sees the bookings of the selected hotel in read-only mode.
29	Front Desk	Booking Review	Booking record: guest, stay, payment status, requests and change requests, with the accept or decline action on change requests for staff. A partner sees the same record without actions.
30	Front Desk	Check-in	QR scan or booking-code lookup, list of ready physical rooms of the booked type and the check-in confirmation.
31	Front Desk	Check-out	Summary of charges, payments and refunds of the stay with the check-out confirmation.
32	Customer Care	Unified Inbox	Three-column workspace: conversation list, message thread (AI and staff messages) and the related booking; includes the take-over action.
33	Customer Care	Refund Requests	Queue of refund requests with booking code, date, reason, amount and status.
34	Customer Care	Refund Review	Pop-up drawer with the original payment, applicable policy, fee and maximum refundable

			amount, with the forward action for staff and the approve or reject actions for the hotel partner.
35	Customer Care	Alerts List	Queue of alerts for unusual bookings and negative reviews with severity, evidence and resolution note.
36	AI Marketing Support	Content Library	Marketing content of the hotel in tabs Drafts, Pending Review, Approved, Rejected, Scheduled and Published.
37	AI Marketing Support	Content Editor	Brief form, AI draft editor with fact check, channel preview and the submit for approval action.
38	AI Marketing Support	Publishing Calendar	Month, week and list views of scheduled posts, and the drawer to connect supported social media accounts.
39	AI Marketing Support	Content Performance	Reach, engagement and attributed bookings of published content with date and channel filters.
40	Partner Onboarding	Partner Sign-up	Registration form for hotel owners (name, email, phone number, password, optional company name); after sign-up the partner enters the email code and then goes straight to onboarding.
41	Partner Onboarding	Partner Onboarding	Four-step wizard for one hotel: legal entity (with the beneficiary details), accommodation, documents, and confirmation of commitments. The partner uses the same wizard when adding another hotel.
42	Partner Onboarding	Verification Status	Status and timeline of the verification request of each hotel (draft, pending review, changes requested, approved or rejected), the reasons for requested changes and the resubmit action.
43	Hotel & Room Management	My Hotels	Hotels of the partner with verification and listing status, the hotel switcher of the workspace and the add-hotel action.
44	Hotel & Room Management	Partner Dashboard	Analytics of the selected hotel, or of all hotels of the partner: bookings, occupancy, revenue, booking sources and satisfaction for a chosen period.
45	Hotel & Room Management	Hotel Settings	Hotel information in tabs General Info, Images (ordered photos with one cover) and Policies, with the room configuration of the hotel.
46	Hotel & Room Management	Room Inventory	Room types and physical rooms of the hotel in two tabs, with search and status filters.
47	Hotel & Room Management	Room Type Editor	Form for a room type: name, base price, occupancy, area, bed types with quantities, amenities and photos.
48	Hotel & Room Management	Room Editor	Form for a physical room: room number, floor, room type and status (available, maintenance, cleaning, occupied), with lock and unlock.
49	Hotel & Room Management	Availability Calendar	Date grid of available and total rooms per room type with bulk edit of selected dates.

50	Pricing & Promotion	Pricing Rules	Table of base-rate and seasonal rules with an editor drawer and a price preview for a room and date.
51	Pricing & Promotion	Promotions	Campaigns and discount codes with validity, eligible room types, limits and a pricing example.
52	Pricing & Promotion	Loyalty Program	Loyalty settings of the hotel in tabs Earning Rules, Redemption and Vouchers.
53	Partner Administration	Staff List	Directory of the staff accounts of the selected hotel with status and the invite and deactivate actions.
54	Partner Administration	Staff Permissions	Pop-up drawer with five permission groups: Booking, Front desk, Customer care, Refund and Marketing.
55	Partner Administration	AI Brand Settings	Brand description, tone, vocabulary, content templates and a preview of sample output.
56	Partner Administration	Hotel Wallet	Pending and available balance of the selected hotel, its wallet transactions (booking payments, commission, refunds and withdrawals) with date and type filters, and the withdraw action.
57	Partner Administration	Withdrawal Request	Pop-up form with the available balance, the amount to withdraw and the bank account of the partner, followed by the status of earlier requests.
58	Platform Operations	Manager Dashboard	Pending applications, active partners, bookings, commission revenue and customer growth with a short action queue.
59	Platform Operations	Verification Queue	List of partner applications with search, status and date filters.
60	Platform Operations	Verification Review	Document viewer and review panel with checklist and the approve, request changes or reject decisions.
61	Platform Operations	Partners List	Partner hotels with verification and listing state, bookings, cancellation rate and satisfaction.
62	Platform Operations	Compliance Cases	Policy-violation cases with evidence, findings and the suspension request sent to the admin.
63	Platform Revenue	Platform Revenue	Commission earned, refunds adjustments and gross booking value with trend chart, drill-down table and export.
64	Platform Revenue	Partner Revenue Details	Bookings and money flow of one partner hotel in the selected period.
65	Platform Revenue	Commission Transactions	Commission and deduction records per booking with rate, amount and status.
66	Platform Revenue	Cash Flow Report	Breakdown of the money received through the payment gateway into hotel share, platform commission and refunds.
67	Platform Revenue	Withdrawal Requests	Queue of partner withdrawal requests with hotel, amount, bank account and status, and the drawer to confirm the transfer with its reference or reject the request with a reason.
68	System Administration	Admin Dashboard	Account statistics, pending account decisions, payment issues and recent security events.

69	System Administration	Users List	Table of all accounts with role, organization, status and created date, with search and filters.
70	System Administration	User Details	Account summary, audit timeline, a linked suspension request if any, and the activate, deactivate or decide-suspension actions.
71	System Administration	Payment Settings	SePay configuration: the bank account that receives payments, the masked API key, the webhook URL and a connection test.
72	System Administration	Transactions List	Payment and refund transactions with filters by date, status, booking code and transaction ID, and export.
73	System Administration	Transaction Details	One transaction with linked booking, expected and confirmed amounts, event history and status check with the gateway.
74	System Administration	System Logs	Read-only logs in tabs Audit Log and Security Events with filters by time, actor, action and severity.
75	System Administration	System AI Settings	Model selection, masked API credential, usage limits and safety filters for the whole platform.

3.1.3 Screen Authorization

An X means the role can open the screen or perform the activity. Hotel staff only see the data of the hotel they work for and a hotel partner only sees the data of the hotels it owns; the last rows describe the data scope of each role.

Screen	Guest	Customer	Hotel Staff	Hotel Partner	Platform Manager	Admin
User Login	X					
User Register	X					
Email Verification	X					
Reset Password	X					
User Profile		X	X	X	X	X
Change Password		X	X	X	X	X
Notifications		X	X	X	X	X
System Messages	X	X	X	X	X	X
Home Page	X	X				
Hotel Search Results	X	X				
Hotel Details	X	X				
Room Type Details	X	X				
Blog List	X	X				
Blog Details	X	X				
AI Assistant	X	X				
Travel Preferences		X				
Recommendations		X				
Booking Checkout		X				
Apply voucher or loyalty points		X				
Payment		X				

Booking Voucher		X				
My Bookings		X				
Booking Details		X				
Change Booking		X				
Cancel Booking		X				
Stay Review		X				
Rewards		X				
Staff Dashboard			X			
Bookings List			X	X		
Booking Review			X	X		
Review booking / accept or decline a change request			X			
Check-in			X			
Check-out			X			
Unified Inbox			X			
Take over an AI conversation			X			
Refund Requests			X	X		
Refund Review			X	X		
Approve or reject a refund				X		
Alerts List			X			
Content Library			X			
Content Editor			X			
Approve content for publication				X		
Publishing Calendar			X			
Content Performance			X			
Partner Sign-up	X					
Partner Onboarding				X		
Verification Status				X		
My Hotels				X		
Partner Dashboard				X		
Hotel Settings				X		
Room Inventory				X		
Room Type Editor				X		
Room Editor				X		
Availability Calendar				X		
Pricing Rules				X		
Promotions				X		
Loyalty Program				X		
Staff List				X		
Staff Permissions				X		
AI Brand Settings				X		
Hotel Wallet				X		

Withdrawal Request				X		
Manager Dashboard					X	
Verification Queue					X	
Verification Review					X	
Approve, request changes or reject					X	
Partners List					X	
Compliance Cases					X	
Send a suspension request					X	
Platform Revenue					X	
Export report					X	
Partner Revenue Details					X	
Commission Transactions					X	
Cash Flow Report					X	
Withdrawal Requests					X	
Admin Dashboard						X
Users List						X
User Details						X
Activate or deactivate an account						X
Payment Settings						X
Transactions List						X
Transaction Details						X
System Logs						X
System AI Settings						X
Data scope						
Query All Data						X
Query Own Data		X	X	X	X	X
Query Managed Data (assigned hotel / own hotels / all partners)			X	X	X	
Add New Data		X	X	X		X
Update All Data						
Update Own Data		X	X	X	X	X
Update Managed Data			X	X		X
Delete Data (soft delete / deactivate)				X		X

3.1.4 Non-Screen Functions

The functions below run on the server without a screen of their own: services, scheduled jobs and integration handlers.

#	Feature	System Function	Description

1	Account & Access	Authentication & Authorization	Issues and verifies JWT access tokens and checks the role and tenant of every API request, so each role reaches only its own functions and a partner or staff member only the data of the hotels it may manage.
2	Account & Access	Email Service	Sends verification codes, password reset links, invitations, booking confirmations and refund notices with Nodemailer over SMTP.
3	Account & Access	Notification Service	Creates in-app notifications and sends push notifications to the mobile app through Firebase Cloud Messaging.
4	Booking & Payment	Booking Hold Expiry Job	Runs every minute and releases rooms held by bookings whose payment was not completed within the hold time.
5	Booking & Payment	Payment Callback Handler	Receives the SePay webhook for each transfer into the account of the platform, checks the API key, matches the transfer with a booking by content and amount, updates the payment and booking status and ignores duplicate notifications.
6	Booking & Payment	QR Voucher Generator	Generates the signed QR-code e-voucher when a booking is confirmed and invalidates it when the booking is cancelled.
7	Booking & Payment	Commission Recording	Records the platform commission of each completed booking and the adjustment when a refund is made, for the revenue reports.
8	Trip Management	Loyalty Point Accrual	Adds loyalty points to the account of the customer after check-out according to the program of the hotel, and expires points and vouchers when configured.
9	Customer Care	Alert Detection Job	Scans new bookings and reviews and raises alerts for unusual booking patterns and negative reviews.
10	AI Assistant & Recommender	AI Chat Service	Python FastAPI service that answers guest questions with the OpenAI API, retrieves hotel knowledge from pgvector (RAG) and calls system functions such as hotel search and availability check.
11	AI Assistant & Recommender	Embedding Pipeline	Turns hotel, room type and article content into embeddings and refreshes them when the content changes.

12	AI Assistant & Recommender	Recommendation Service	Computes the hybrid recommendation score from content similarity, interactions and ratings, adapts the weights to the data density of the user and the hotel, and serves both the chat and the browse pages.
13	AI Assistant & Recommender	Model Training & Evaluation Job	Retrains the recommendation models on a schedule, stores the predicted ratings and records Precision@K, Recall@K, NDCG, MAE, RMSE and cold-start coverage of each run.
14	AI Marketing Support	Content Generation Service	Generates caption, blog and promotion drafts with the brand settings of the hotel; drafts always wait for the approval of the hotel partner.
15	AI Marketing Support	Scheduled Publisher	Publishes approved content to the connected social media accounts at the scheduled time and collects reach and engagement metrics.
16	System Administration	Audit Logging	Writes an audit record for every create, update, approval and security-relevant action with actor, time and target.

3.1.5 Entity Relationship Diagram

The conceptual ERD below is version 3 of the StayReco data model. At the conceptual level each user role is drawn as its own entity; at the physical level the shared account data (email, password, status and role) is stored in one USER_ACCOUNT table, and each role keeps its own table (CUSTOMER, HOTEL_PARTNER, HOTEL_STAFF, PLATFORM_MANAGER, ADMIN) for its specific attributes, sharing the primary key with USER_ACCOUNT (class table inheritance). The recommender entities follow the supervisor guidance: ratings are stored per criterion together with the predicted ratings of each model run so that MAE and RMSE can be computed.

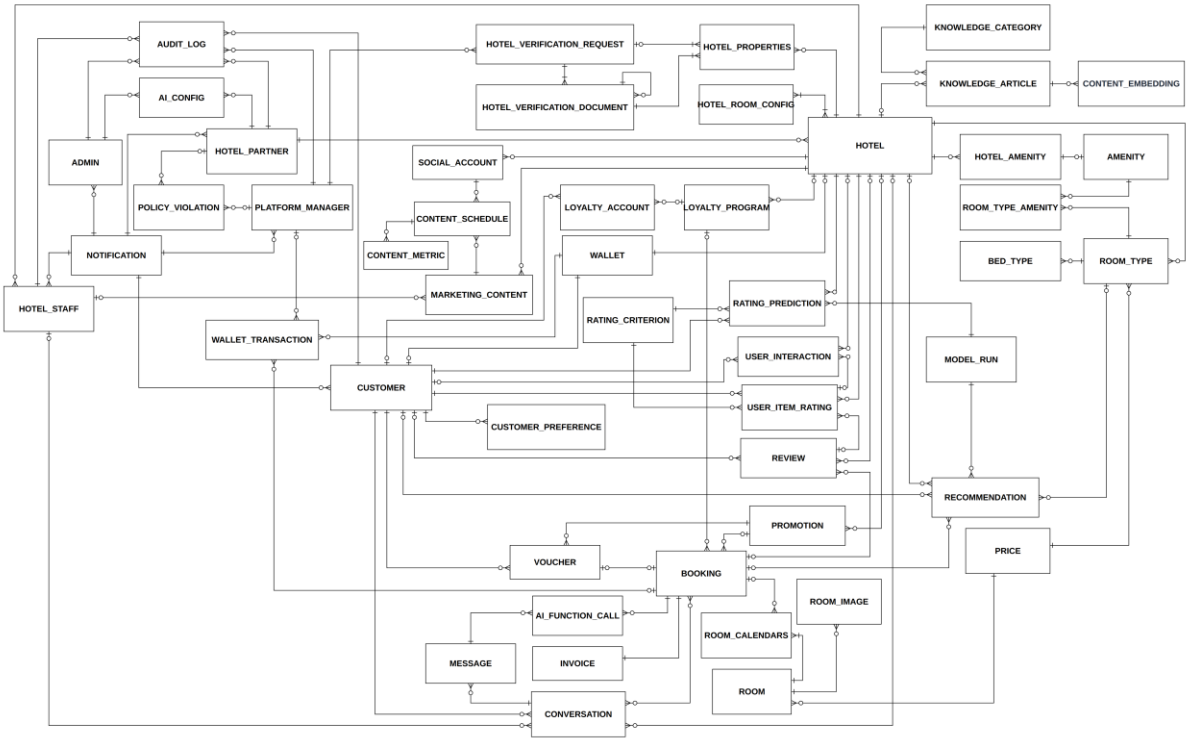


Figure 17. Conceptual entity relationship diagram of StayReco

Entities Description

#	Entity	Description
1	CUSTOMER	A registered traveller who books rooms, writes reviews and receives recommendations.
2	HOTEL_PARTNER	The owner or operator account that manages one or more hotels and keeps the bank account used for withdrawals.
3	HOTEL_STAFF	An employee account attached to one hotel, with the permissions granted by the partner.
4	PLATFORM_MANAGER	The account that reviews partners, handles violations and follows platform revenue.
5	ADMIN	The system administrator account.
6	NOTIFICATION	A message sent by the system to a user, with read status.
7	AUDIT_LOG	A record of an action performed by a user: actor, action, target and time.
8	AI_CONFIG	AI settings: platform-level model and limits set by the admin, and brand voice and templates set by a partner.
9	POLICY_VIOLATION	A compliance case opened against a partner, with evidence, findings and suspension request.
10	HOTEL	A hotel listed on StayReco; the tenant boundary for rooms, bookings, staff and content.
11	HOTEL_PROPERTIES	Descriptive information and policies of a hotel submitted and maintained by the partner.
12	HOTEL_VERIFICATION_REQUEST	A verification request for one hotel, submitted by its partner and decided by the platform manager.

13	HOTEL_VERIFICATION_DOCUMENT	A document attached to a verification request, such as the business license; a newer version can replace an older one.
14	HOTEL_ROOM_CONFIG	Configuration values shared by the rooms of a hotel.
15	AMENITY	A catalogue item of amenities (pool, Wi-Fi, parking, breakfast...).
16	HOTEL_AMENITY	An amenity offered by a hotel.
17	ROOM_TYPE	A sellable room category of a hotel with capacity, area and base price.
18	ROOM_TYPE_AMENITY	An amenity available in a room type.
19	BED_TYPE	The bed configuration of a room type.
20	ROOM	A physical room with its number and floor, assigned to guests at check-in.
21	ROOM_IMAGE	A photo of a room.
22	ROOM_CALENDARS	The per-date state of a physical room (available, held, booked, closed) linked to the booking that uses it.
23	PRICE	A price definition (base or seasonal rate) applied to room types and rooms.
24	BOOKING	A reservation made by a customer: stay dates, guests, amounts, booking and payment status.
25	INVOICE	The invoice issued for a booking.
26	PROMOTION	A campaign or discount code created by a hotel, applied on top of the base price.
27	VOUCHER	A voucher issued to a customer from a promotion and redeemed on a booking.
28	LOYALTY_PROGRAM	The loyalty rules of a hotel: earning basis, redemption value and expiry.
29	LOYALTY_ACCOUNT	The point balance of a customer in the program of a hotel.
30	WALLET	A balance record owned by a customer, a hotel or the platform, used for refunds, hotel share and commission.
31	WALLET_TRANSACTION	A money movement on a wallet: payment, refund, commission, withdrawal (with its status and bank transfer reference) or adjustment.
32	REVIEW	A rating and review with photos written by a customer after a completed stay.
33	CONVERSATION	A chat session between a user and the AI assistant or hotel staff, possibly linked to bookings.
34	MESSAGE	A message in a conversation, sent by the user, the AI or a staff member.
35	AI_FUNCTION_CALL	A system function called by the AI assistant during a conversation, with arguments and result.
36	KNOWLEDGE_CATEGORY	A category of knowledge articles used by the AI assistant.
37	KNOWLEDGE_ARTICLE	An FAQ, policy or blog article, shared or specific to a hotel.
38	CONTENT_EMBEDDING	The vector embedding of a content chunk stored in pgvector for semantic search.

39	MARKETING_CONTENT	A caption, blog post or promotion drafted with AI and approved by the hotel partner.
40	SOCIAL_ACCOUNT	A social media account connected by a hotel for publishing.
41	CONTENT_SCHEDULE	The planned publication of a content item on a social account.
42	CONTENT_METRIC	Reach, engagement and conversion collected for a published item.
43	CUSTOMER_PREFERENCE	A travel preference stated by a customer (budget, destination, amenities...).
44	USER_INTERACTION	A behaviour event such as view, search, click or booking, from a customer or an anonymous session.
45	RATING_CRITERION	One of the rating criteria used to split a review into sub-ratings.
46	USER_ITEM_RATING	The rating of a customer for a hotel on one criterion, taken from reviews or interactions.
47	RATING_PREDICTION	The rating predicted by a model run for a customer, a hotel and a criterion, used for MAE and RMSE.
48	MODEL_RUN	One training or evaluation run of a recommendation model with its version and metrics.
49	RECOMMENDATION	A hotel or room suggested to a customer or session, with score, reason and click or booking feedback.

3.2 Account & Access

3.2.1 User Login

- **Function trigger:** The user clicks "Log in" on the header of any public page, or is sent to this screen when opening a page that needs a signed-in account (for example "Book now").
- **Function description:** Actors: all registered roles. Purpose: sign in with email and password and open the workspace of the role. The screen shows the email and password fields, a "Show password" toggle, the links "Forgot password?" and "Create an account", and the "Log in" button. After a successful login the system creates a session and redirects the user by role (customer to the page they came from or the home page, staff to Staff Dashboard, partner to Partner Dashboard, platform manager to Manager Dashboard, admin to Admin Dashboard).
- **Screen layout: Figure 18 (web) and Figure 19 (mobile app).**

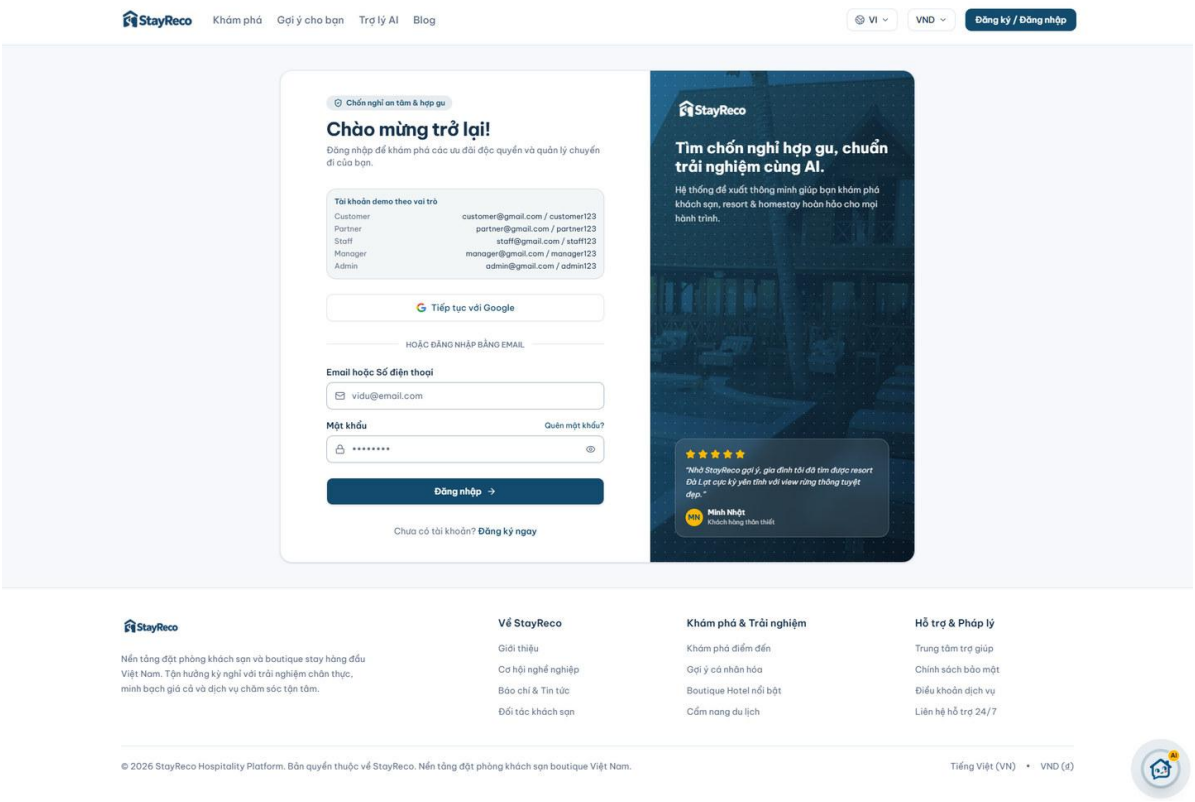


Figure 18. User Login screen (web)

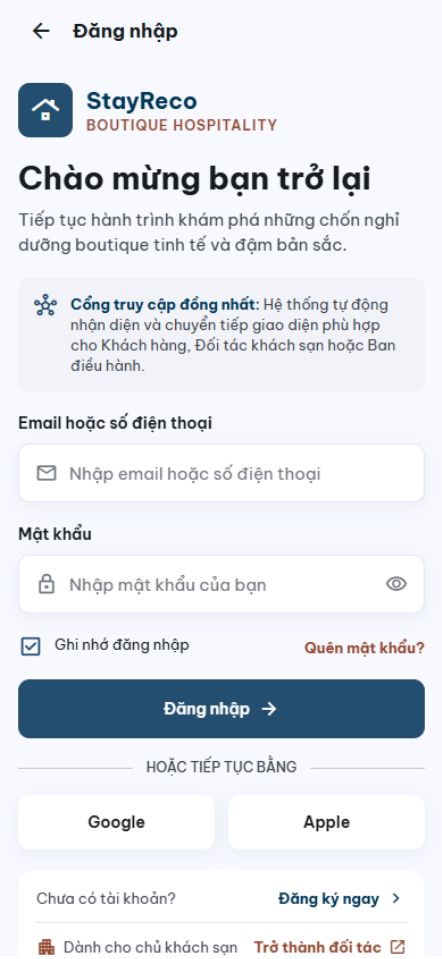


Figure 19. User Login screen (mobile app)

- **Function details:**
  - Email: required, valid email format (MSG02, MSG04). Password: required.
  - Wrong email or wrong password shows the same message MSG05, so the system never reveals whether an email exists.
  - After 5 wrong passwords in a row the account is locked for 15 minutes and MSG26 is shown; login requests are limited to 5 per minute (BR-24).
  - An account that has not entered its email code gets MSG28 and is taken to Email Verification; a deactivated or suspended account gets MSG06.
  - If the user chose a room before logging in, the booking selection is kept and the user returns to Booking Checkout (UC04).

3.2.2 User Register

- **Function trigger:** The guest clicks "Create an account" on User Login or "Sign up" on the header.
- **Function description:** Actor: Guest. Purpose: create a customer account. The form has full name, email, password, confirm password and the "Create account" button, plus a link "Are you a hotel owner? Register as a partner" that opens Partner Sign-up. On success the account is created in the inactive state and the user goes to Email Verification (UC01).
- **Screen layout: Figure 20 (web) and Figure 21 (mobile app).**

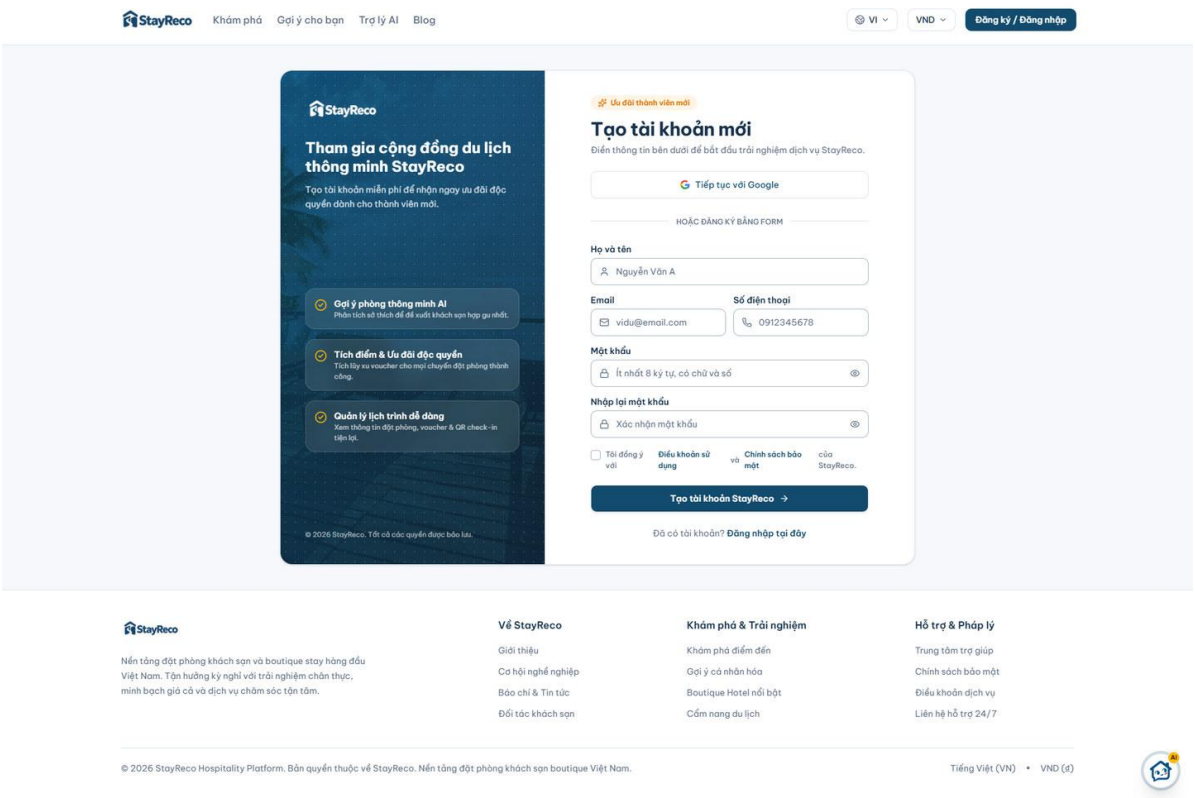


Figure 20. User Register screen (web)

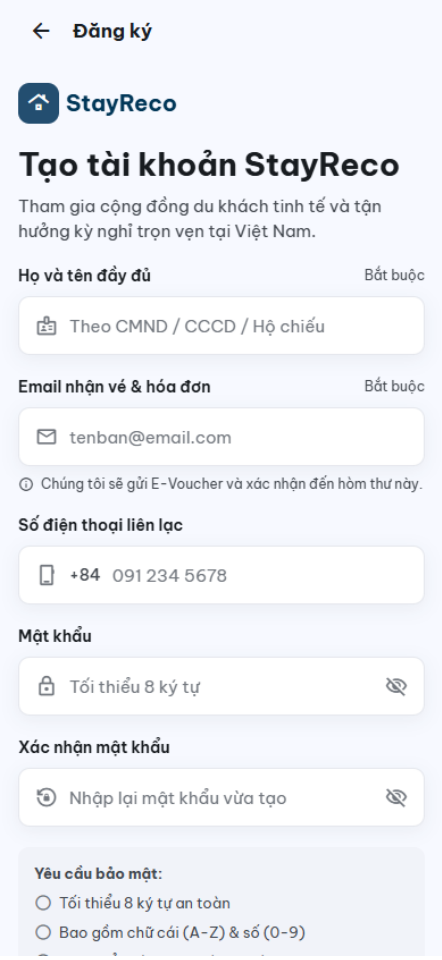


Figure 21. User Register screen (mobile app)

- **Function details:**
  - Full name: required, at most 100 characters. Email: required, valid format, stored in lower case.
  - Password: required, at least 8 characters with a letter and a digit (BR-17, MSG08); confirm password must match.
  - An email that is already registered shows MSG07 under the email field.
  - No session is created at this step; the account stays inactive until the code is entered (BR-23).

3.2.3 Email Verification

- **Function trigger:** Opened automatically after User Register or Partner Sign-up, and after a login attempt by an account that has not been verified.
- **Function description:** Actors: Guest (new customer or new partner). Purpose: activate the account with the 6-digit code sent to the registered email. The screen shows the email the code was sent to, six input boxes, a countdown of the code validity and the "Resend code" button (UC03).
- **Screen layout: Figure 22 (web).**

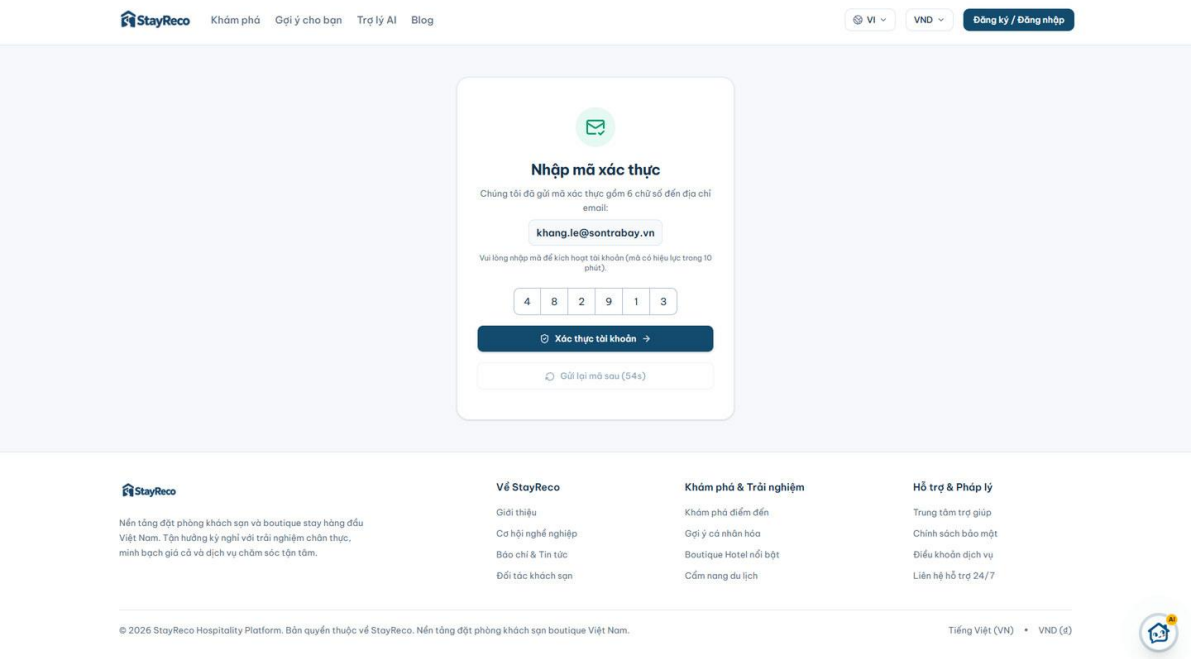


Figure 22. Email Verification screen (web)

- **Function details:**
  - The code has 6 digits and is valid for 10 minutes (BR-23).
  - A wrong or expired code shows MSG29. After 5 wrong entries the code is cancelled and the user must request a new one.
  - "Resend code" sends a new code and cancels the previous one; sending emails is rate limited.
  - A correct code activates the account, signs the user in and opens the workspace of the role (a partner goes to Partner Onboarding).

3.2.4 Reset Password

- **Function trigger:** The user clicks "Forgot password?" on User Login; the second step opens from the link in the reset email.
- **Function description:** Actors: all registered roles. Purpose: set a new password without signing in. Step 1 asks for the email and sends a reset link. Step 2, opened from the link, asks for the new password and its confirmation (UC05).
- **Screen layout: Figure 23 (web), Figure 24 (web) and Figure 25 (mobile app).**

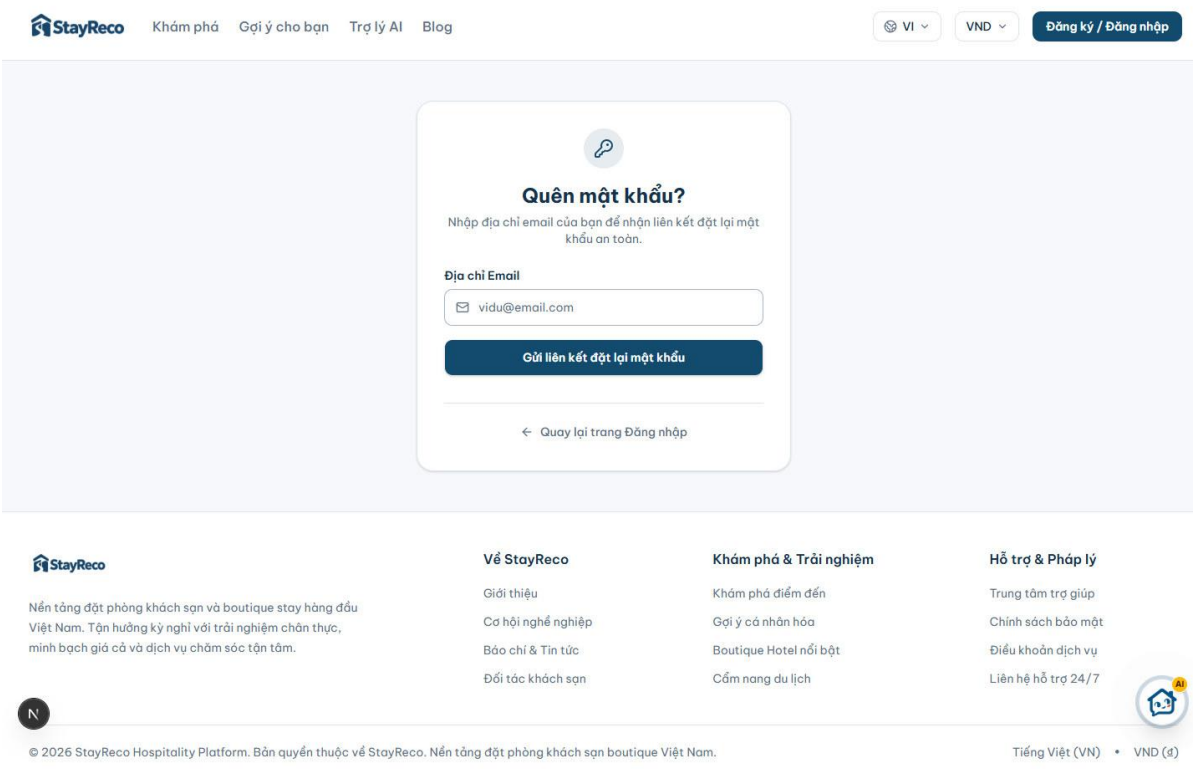


Figure 23. Reset Password screen: request the reset link (web)

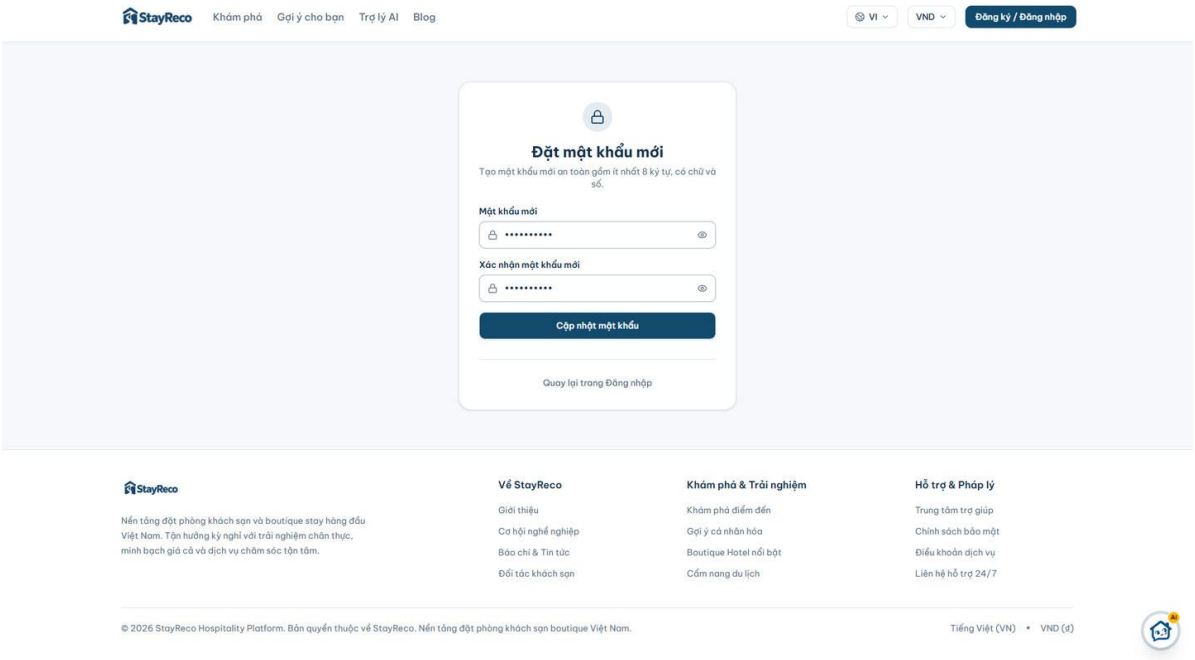


Figure 24. Reset Password screen: set a new password (web)

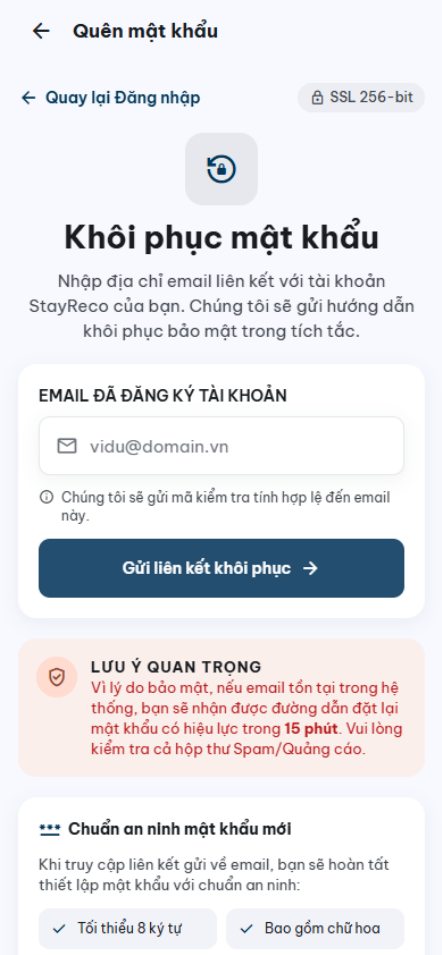


Figure 25. Reset Password screen (mobile app)

- **Function details:**
  - Step 1 always shows MSG09 whether or not the email exists.
  - The link can be used once and expires after 15 minutes (BR-17); an invalid or used link shows an error with a button to request a new link.
  - The new password follows the password rule (MSG08). After the reset every session of the account is signed out and the user logs in again.

3.2.5 User Profile

- **Function trigger:** The signed-in user opens the account menu at the top right and chooses "Profile".
- **Function description:** Actors: all signed-in roles. Purpose: view and update personal information. The screen shows the avatar, full name, email (read-only), phone number, role (read-only) and the "Save" button (UC07).
- **Screen layout: Figure 26 (web), Figure 27 (mobile app) and Figure 28 (mobile app).**

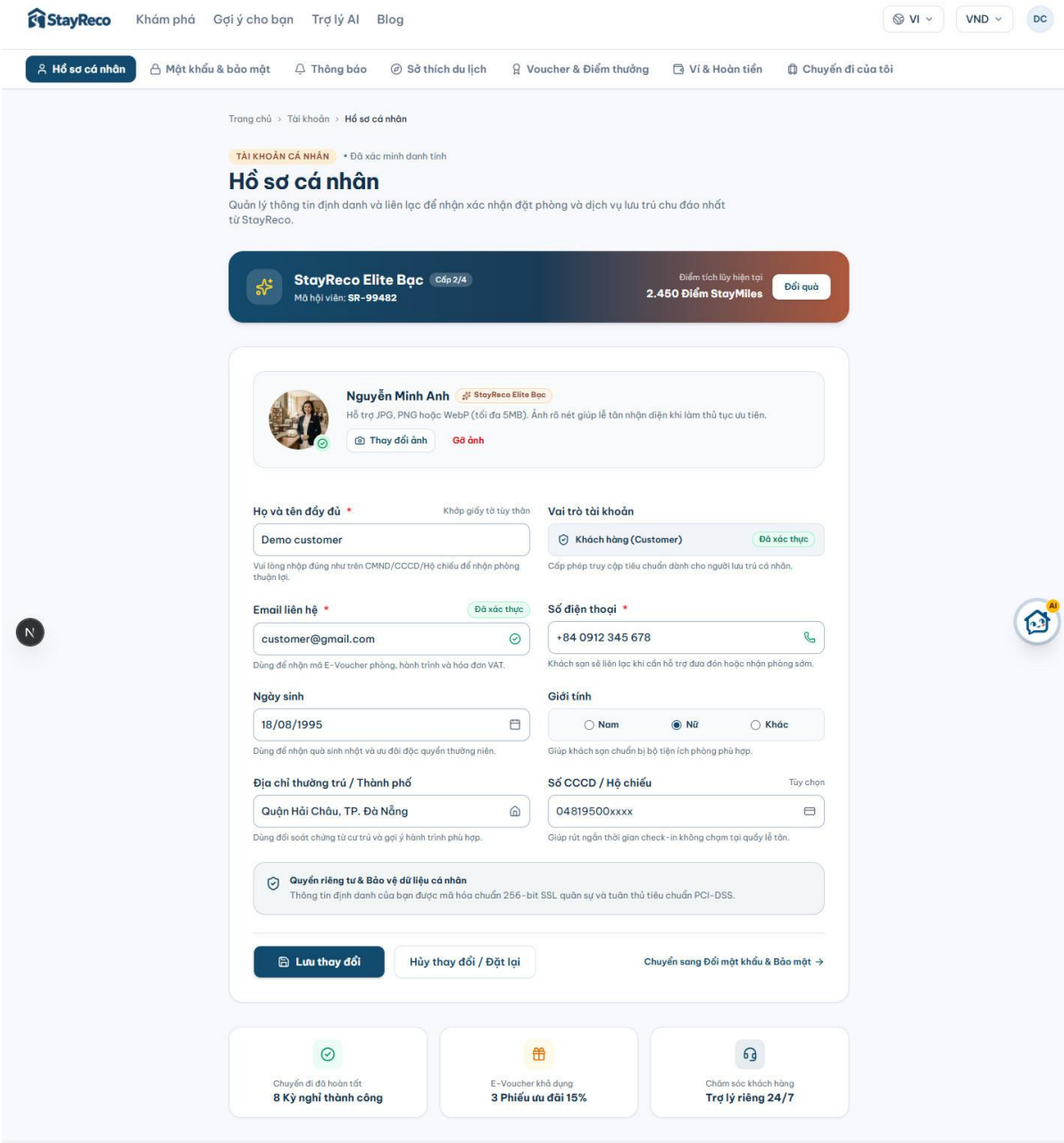


Figure 26. User Profile screen (web)

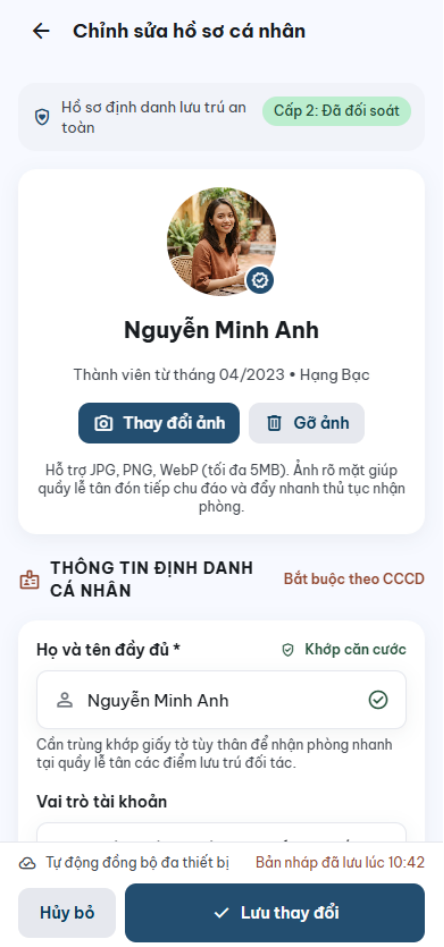


Figure 27. User Profile screen (mobile app)

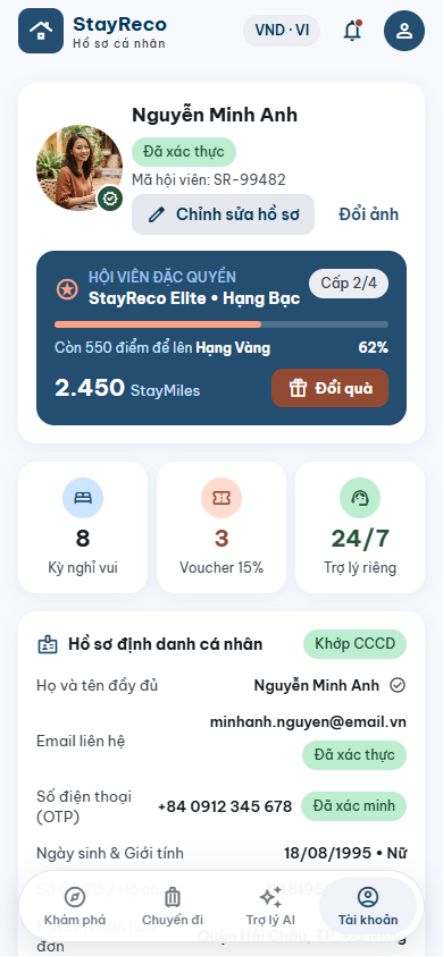


Figure 28. User Profile screen: account menu (mobile app)

- **Function details:**
  - Full name: required, at most 100 characters. Phone: optional, 9 to 15 digits with an optional leading "+".
  - Avatar: image file checked against the allowed types and size limit (MSG23).
  - Saving shows MSG10. The email and role cannot be changed here.

3.2.6 Change Password

- **Function trigger:** Account menu, "Change password".
- **Function description:** Actors: all signed-in roles. Purpose: change the password while signed in and manage the sessions of the account. The form has current password, new password and confirm new password (UC08). The "Signed-in devices" list on the same page shows each session with its device, IP address and last use, and lets the user sign out of one device; the account menu holds "Log out" and "Log out of all devices" (UC06).
- **Screen layout: Figure 29 (web).**

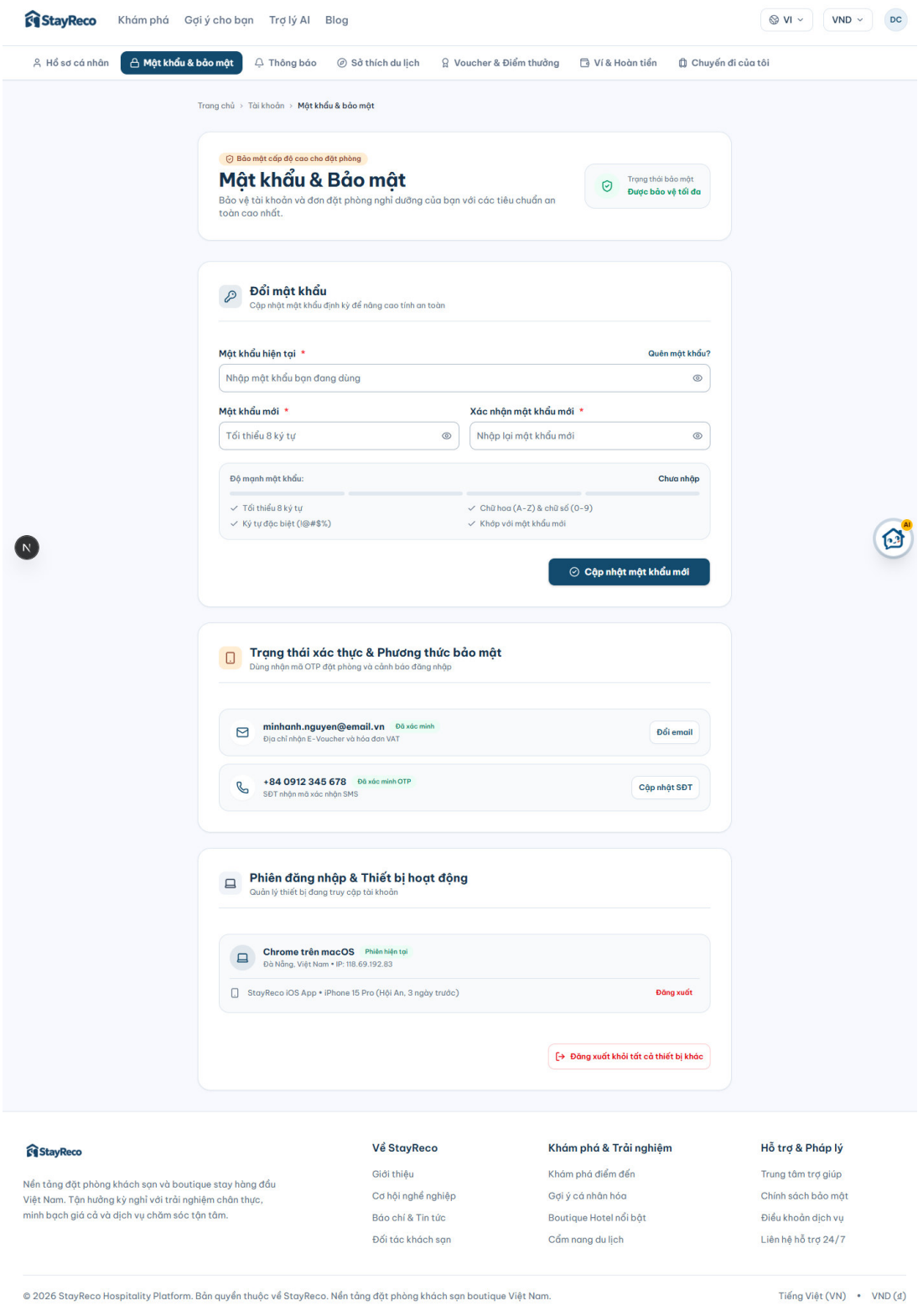


Figure 29. Change Password screen (web)

- **Function details:**
  - A wrong current password shows an error under that field.
  - The new password follows BR-17 (MSG08) and must differ from the current one.
  - After the change the current device stays signed in and every other session is signed out.

3.2.7 Notifications

- **Function trigger:** The bell icon on the header; clicking "View all" opens the full list.
- **Function description:** Actors: all signed-in roles. Purpose: read the notifications of the account. The list is grouped by date with the tabs All and Unread, each item links to the related record (booking, refund, verification request, withdrawal...) (UC09).
- **Screen layout: Figure 30 (web).**

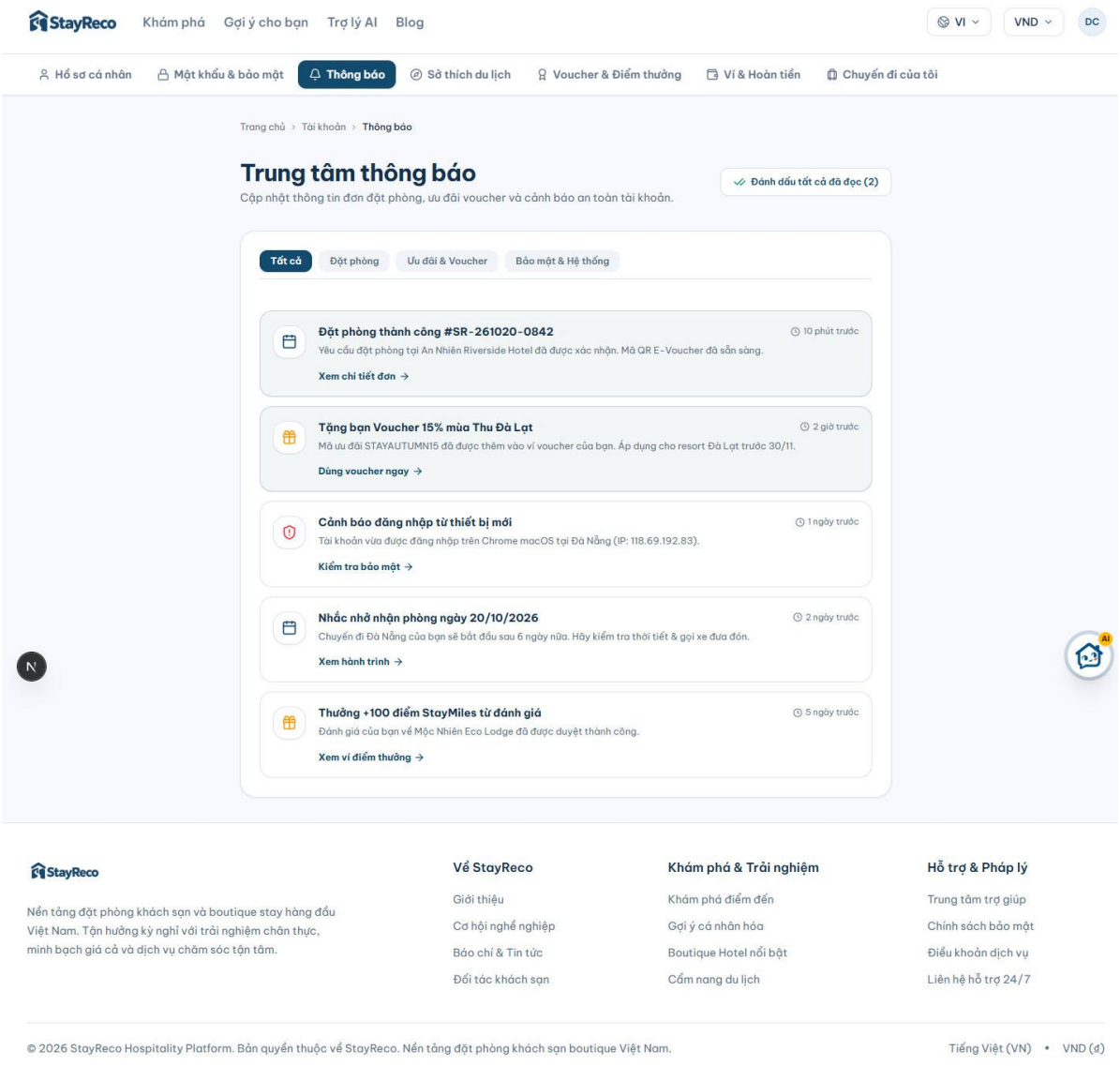


Figure 30. Notifications screen (web)

- **Function details:**
  - The bell shows the number of unread notifications.
  - Opening an item marks it as read; "Mark all as read" clears the counter.
  - A notification whose record the user can no longer access opens MSG25.

3.2.8 System Messages

- **Function trigger:** Shown by the system when a request fails.

- **Function description:** Actors: all users. Purpose: tell the user clearly what went wrong. The pages are Access Denied (403), Not Found (404), Service Error and Session Expired, each with a short message and a button back to a safe page.
- **Screen layout: Figure 31 (web).**

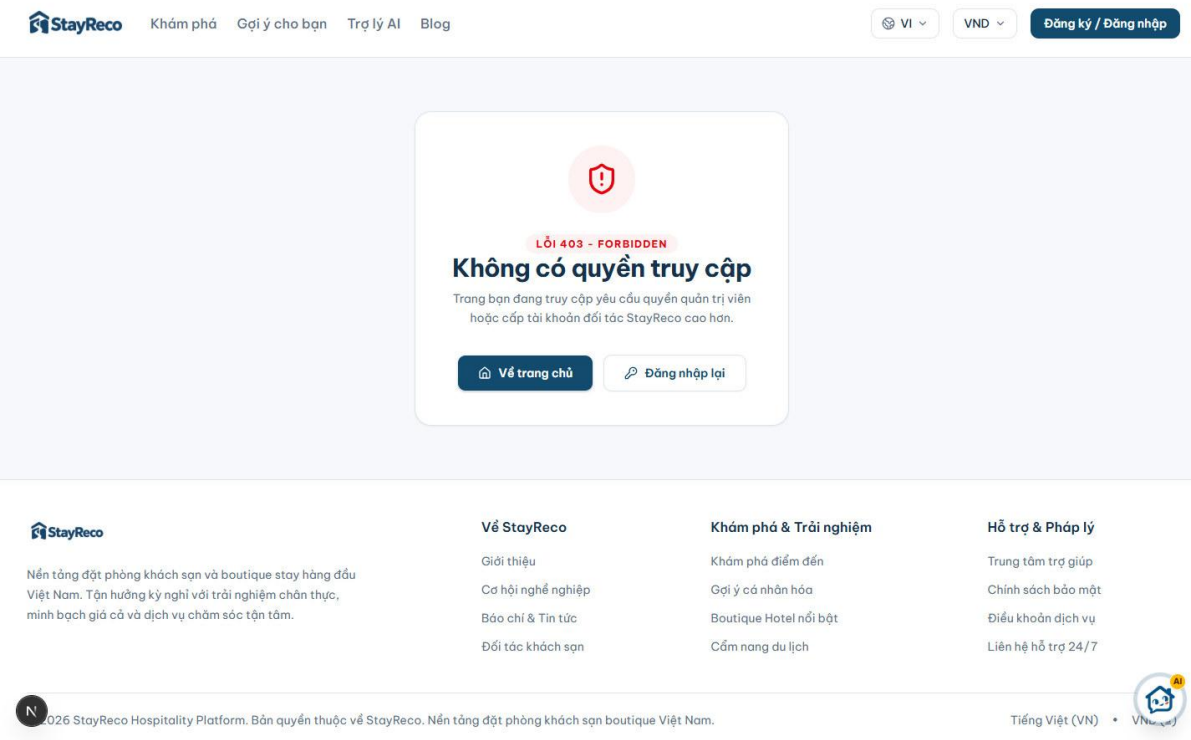


Figure 31. System Messages screen: access denied (403) (web)

- **Function details:**
  - 403 shows MSG25 and a button to the workspace of the role.
  - Session Expired appears when the refresh token is no longer valid; the user is asked to log in again and then returns to the page they were on.
  - Service Error never shows technical details to the user.

3.2.9 Authentication & Authorization

- **Function trigger:** Runs on every API request.
- **Function description:** Non-screen function. Purpose: make sure each request comes from a valid session and that the caller may use the function and the data. It issues short-lived access tokens and rotating refresh tokens, and checks the role and the hotel (tenant) of every request.
- **Screen layout:** *not applicable, this function runs without a screen.*
- **Function details:**
  - Access token valid for 15 minutes; refresh token valid for 30 days, replaced on every use and revoked at logout. Web clients keep the refresh token in an httpOnly cookie.
  - A refresh token used twice revokes the whole session (reuse detection).
  - Every hotel-scoped query is filtered by the hotel id of the caller; staff see one hotel, a partner sees only the hotels it owns (BR-21).
  - A call outside the role returns 403 (MSG25).

3.2.10 Email Service

- **Function trigger:** Called by other functions when an email must be sent.
- **Function description:** Non-screen function. Purpose: send verification codes, password reset links, invitations, booking confirmations and refund notices with Nodemailer over SMTP.
- **Screen layout:** *not applicable, this function runs without a screen.*
- **Function details:**
    - Each email is sent from a template in the language of the user.
    - Sending is retried when the provider fails; a failure never blocks the business action that triggered it.
    - Email-sending requests are limited to 3 per 15 minutes for the same user to prevent abuse (BR-24).

3.2.11 Notification Service

- **Function trigger:** Called when a booking, refund, verification request, content item or withdrawal changes status.
- **Function description:** Non-screen function. Purpose: create in-app notifications and send push notifications to the mobile app through Firebase Cloud Messaging.
- **Screen layout:** *not applicable, this function runs without a screen.*
- **Function details:**
    - A notification stores the receiver, title, short text, link to the record and read status.
    - Push messages are sent only to devices the user has signed in on and only for the notification types the user has turned on.

3.3 Hotel Discovery

3.3.1 Home Page

- **Function trigger:** Opened from the root address of the web site or the logo.
- **Function description:** Actors: Guest, Customer. Purpose: start a search. The page shows the search box (destination, check-in and check-out dates, number of guests and rooms), the entry to the AI assistant, featured hotels, popular destinations and recent blog articles.
- **Screen layout: Figure 32 (web) and Figure 33 (mobile app).**

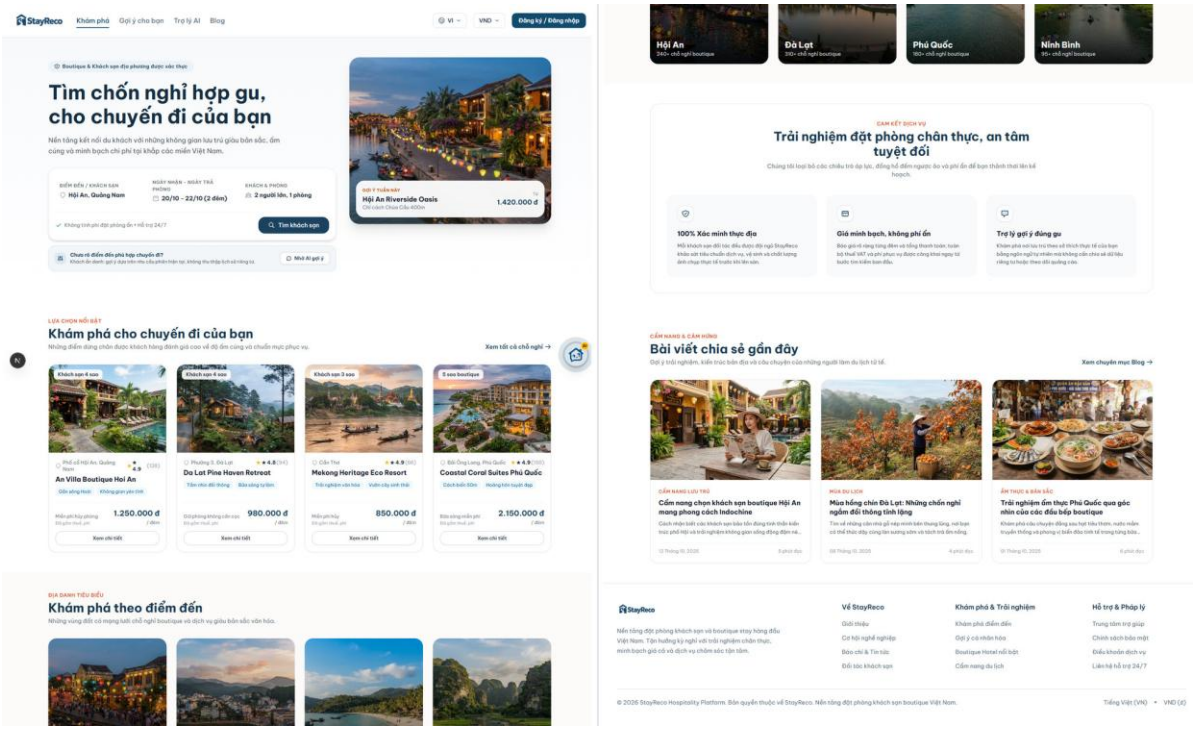


Figure 32. Home Page screen (web)

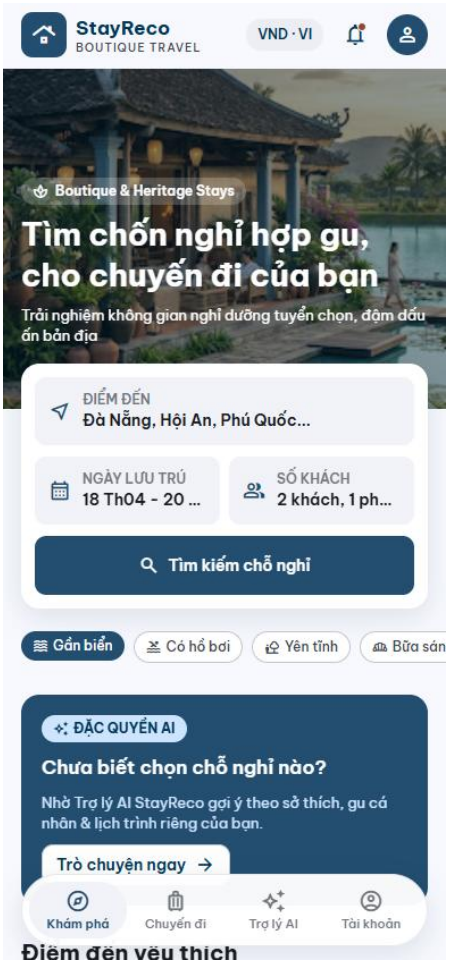


Figure 33. Home Page screen (mobile app)

- **Function details:**

- Destination is required; the check-out date must be after the check-in date and the check-in date cannot be in the past.
- Only approved and active hotels appear in the featured list (BR-01).
- A signed-in customer also sees a short block of personal recommendations.

3.3.2 Hotel Search Results

- **Function trigger:** The user submits the search box on Home Page, or a search suggested by the AI assistant.
- **Function description:** Actors: Guest, Customer. Purpose: compare hotels that have rooms for the chosen dates (UC10). Each card shows photo, name, star class, guest rating, location and the total price for the stay. Filters: budget, star class, guest rating, amenities, breakfast and cancellation terms; sorting by recommendation, lowest price or best rating.
- **Screen layout: Figure 34 (web) and Figure 35 (mobile app).**

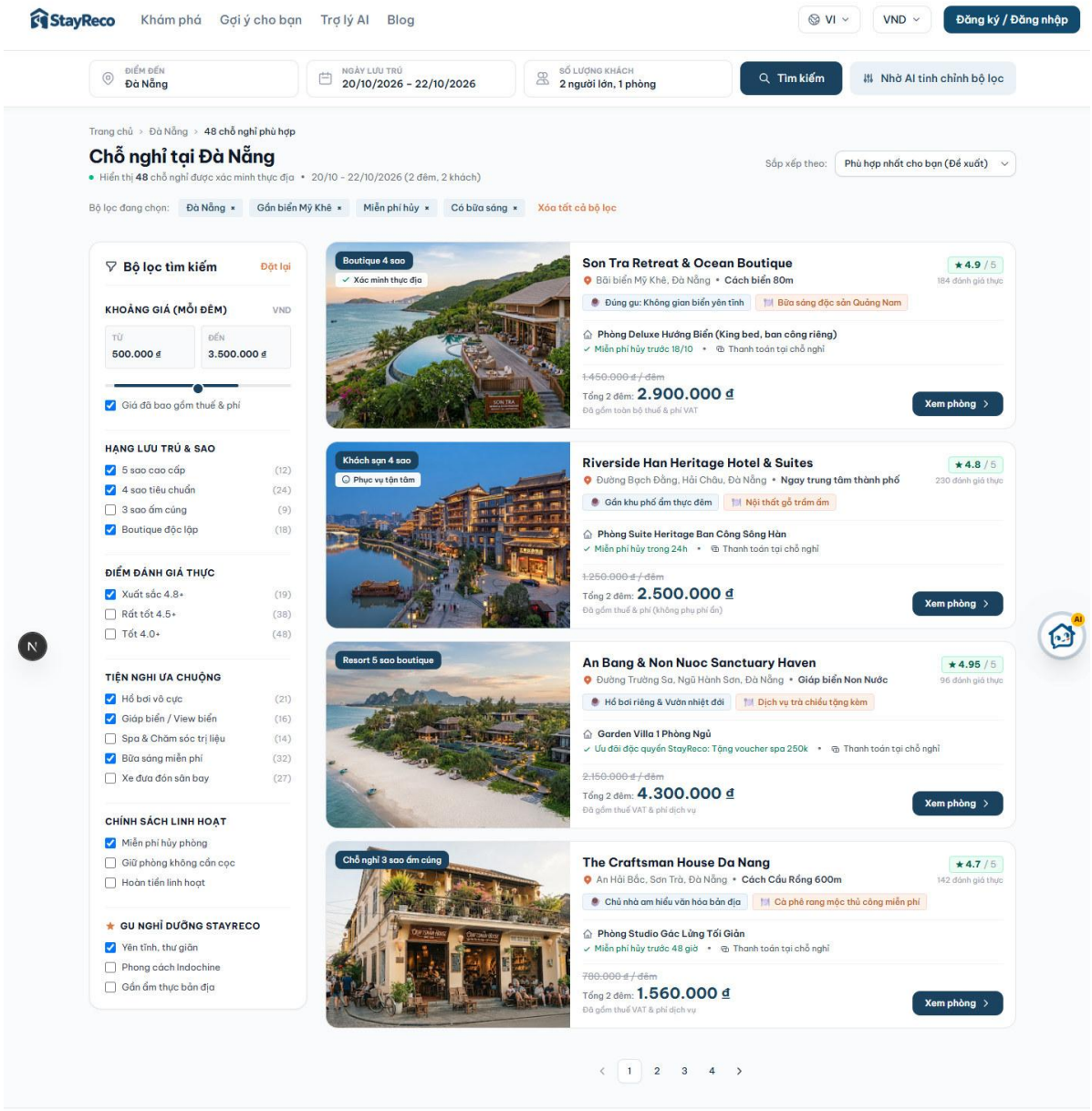


Figure 34. Hotel Search Results screen (web)

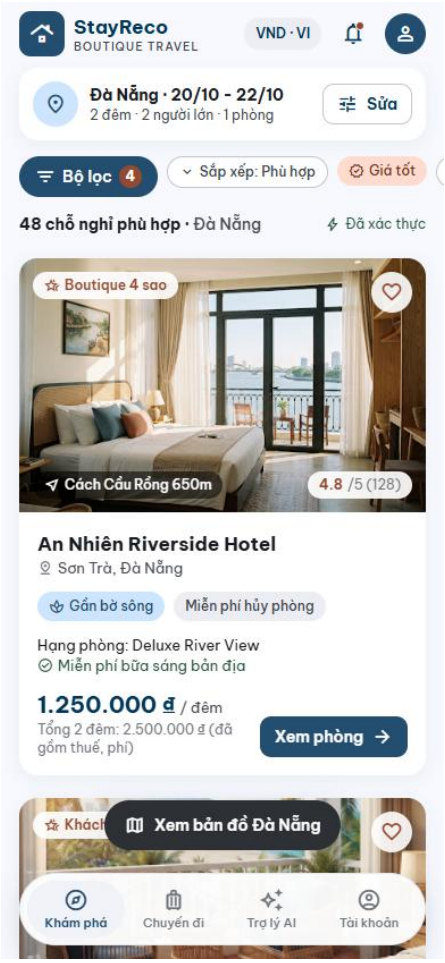


Figure 35. Hotel Search Results screen (mobile app)

- **Function details:**
  - Only hotels with at least one available room for every night of the stay are listed (BR-02).
  - The price shown is the final price for the stay, already including VAT and service charge (BR-05).
  - No result shows MSG01. Results are paginated.

3.3.3 Hotel Details

- **Function trigger:** Click on a hotel card in Hotel Search Results, Recommendations, the AI assistant or a blog article.
- **Function description:** Actors: Guest, Customer. Purpose: show everything needed to choose a room (UC11). The page has the photo gallery, address and map, amenities, policies (check-in and check-out times, cancellation), reviews with the score per criterion and the list of room offers for the selected dates.
- **Screen layout: Figure 36 (web) and Figure 37 (mobile app).**

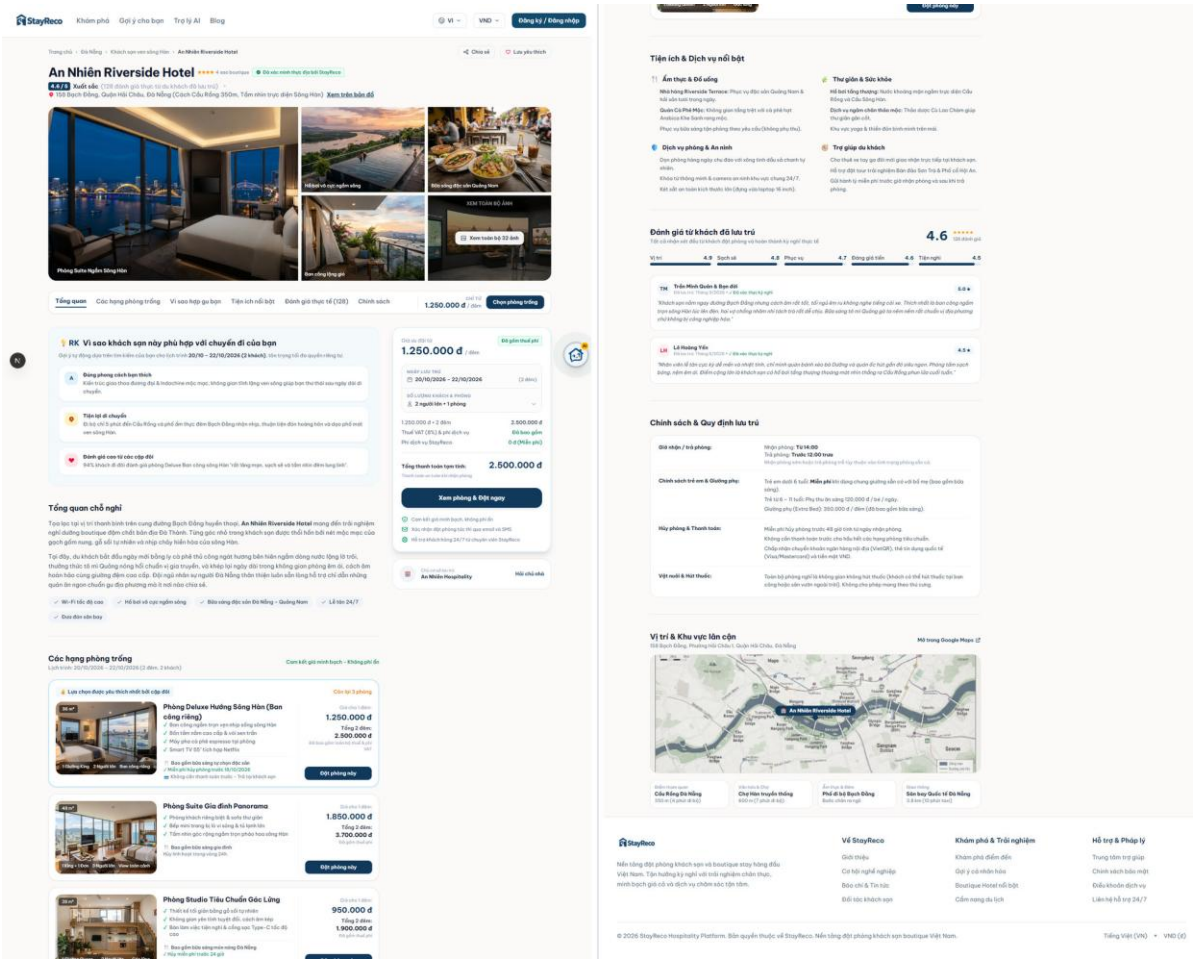


Figure 36. Hotel Details screen (web)

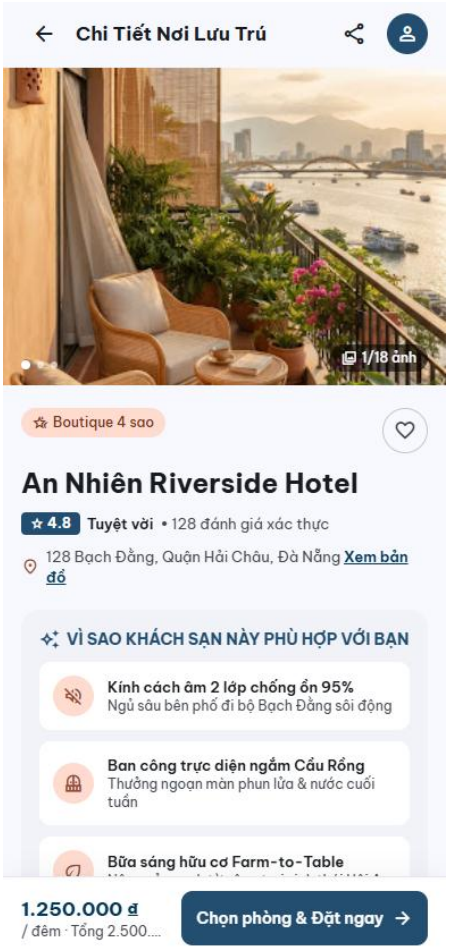


Figure 37. Hotel Details screen (mobile app)

- **Function details:**
  - Changing the dates refreshes the room offers and prices.
  - A room type with no availability for the dates is shown as sold out.
  - A hotel that is not approved or is suspended returns Not Found (404).

3.3.4 Room Type Details

- **Function trigger:** Click on a room offer on Hotel Details.
- **Function description:** Actors: Guest, Customer. Purpose: show one room type in detail (UC12). The page shows photos, area, bed type, maximum occupancy, amenities, rate plans with their cancellation terms and the "Select this room" button.
- **Screen layout: Figure 38 (web).**

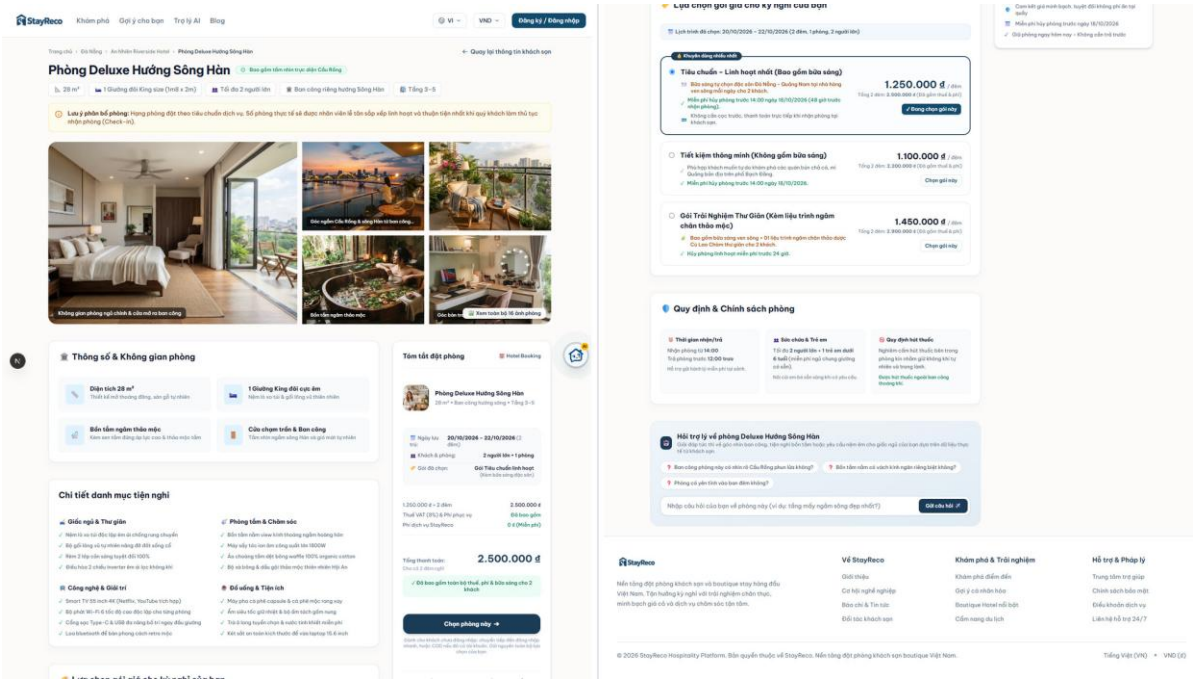


Figure 38. Room Type Details screen (web)

- **Function details:**
  - "Select this room" checks availability again (UC19) and opens Booking Checkout; a guest is first asked to log in, and the selection is kept.
  - The number of guests cannot exceed the occupancy of the room type.

3.3.5 Blog List

- **Function trigger:** The "Blog" link on the header or the article block on Home Page.
- **Function description:** Actors: Guest, Customer. Purpose: browse travel and hotel articles (UC13). The list has a category filter, keyword search and pagination.
- **Screen layout: Figure 39 (web).**

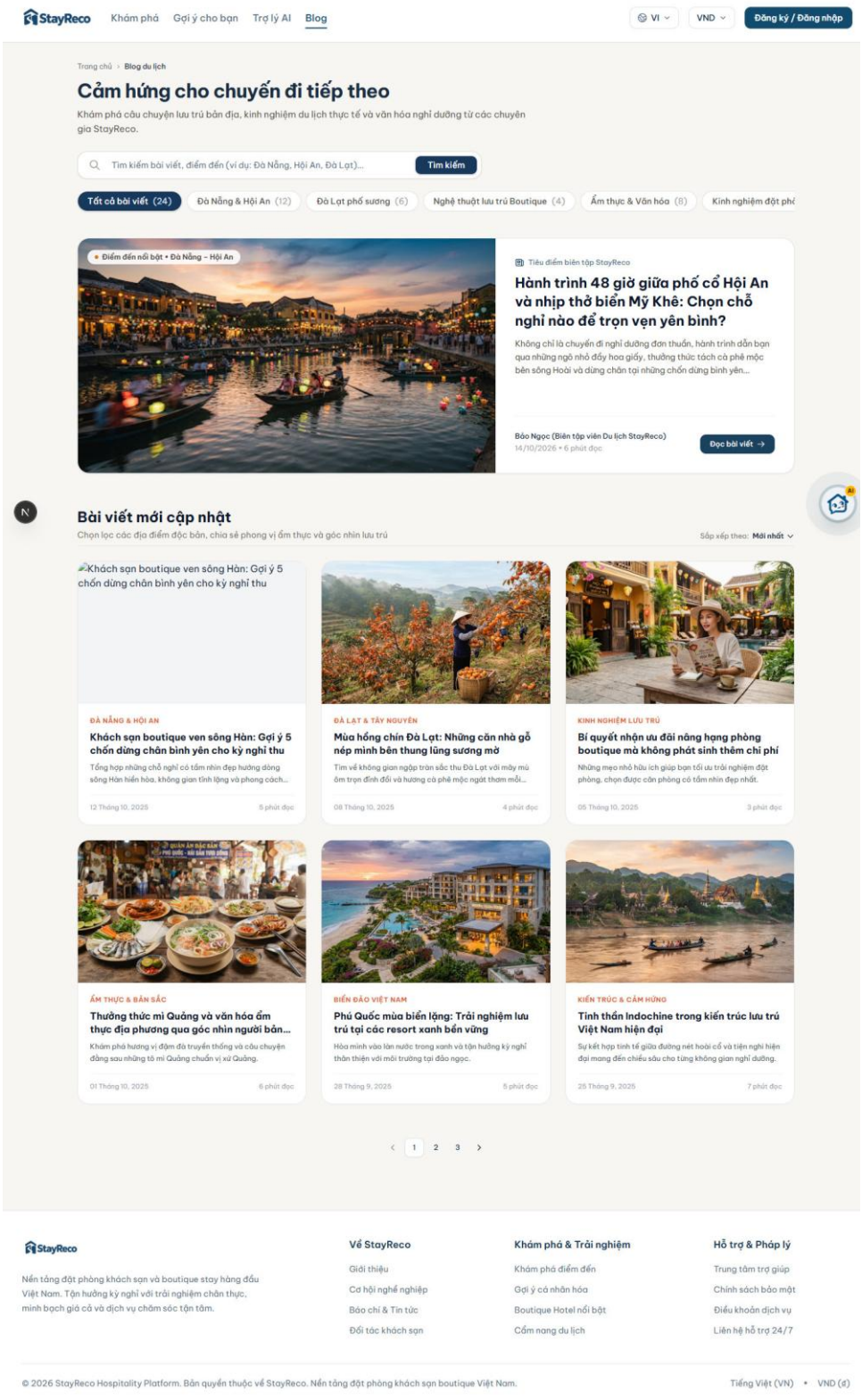


Figure 39. Blog List screen (web)

- **Function details:**
  - Only articles approved by the hotel partner are listed (BR-14).
  - No result shows MSG01.

3.3.6 Blog Details

- **Function trigger:** Click on an article in Blog List or on Home Page.
- **Function description:** Actors: Guest, Customer. Purpose: read one article. The page shows the table of contents, the content, the related hotel with a link to Hotel Details and related articles.
- **Screen layout: Figure 40 (web).**

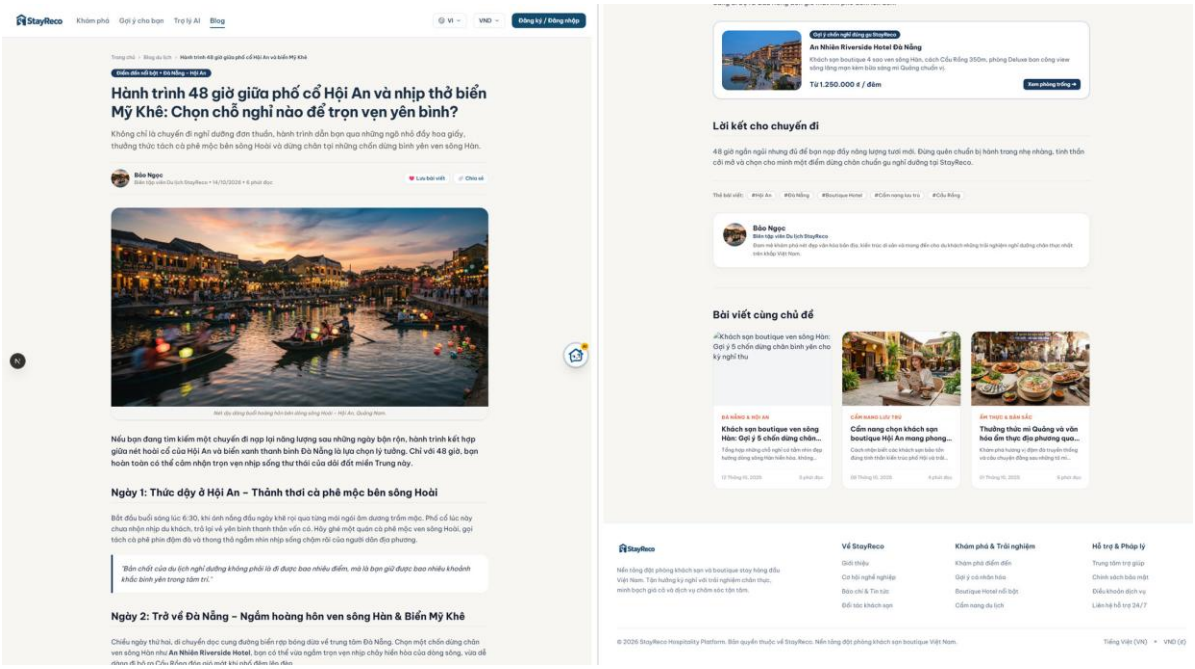


Figure 40. Blog Details screen (web)

- **Function details:**
  - An article that is unpublished or deleted returns Not Found (404).

3.4 AI Assistant & Recommender

3.4.1 AI Assistant

- **Function trigger:** The AI entry on Home Page, the chat button on every public page, or the chat tab of the mobile app.
- **Function description:** Actors: Guest, Customer. Purpose: let the user ask questions and find rooms in natural language (UC14, UC15). The workspace shows the conversation, hotel and room cards suggested by the assistant, the trip criteria it understood (destination, dates, guests, budget) that the user can edit, and the "Talk to hotel staff" action.
- **Screen layout: Figure 41 (web) and Figure 42 (mobile app).**

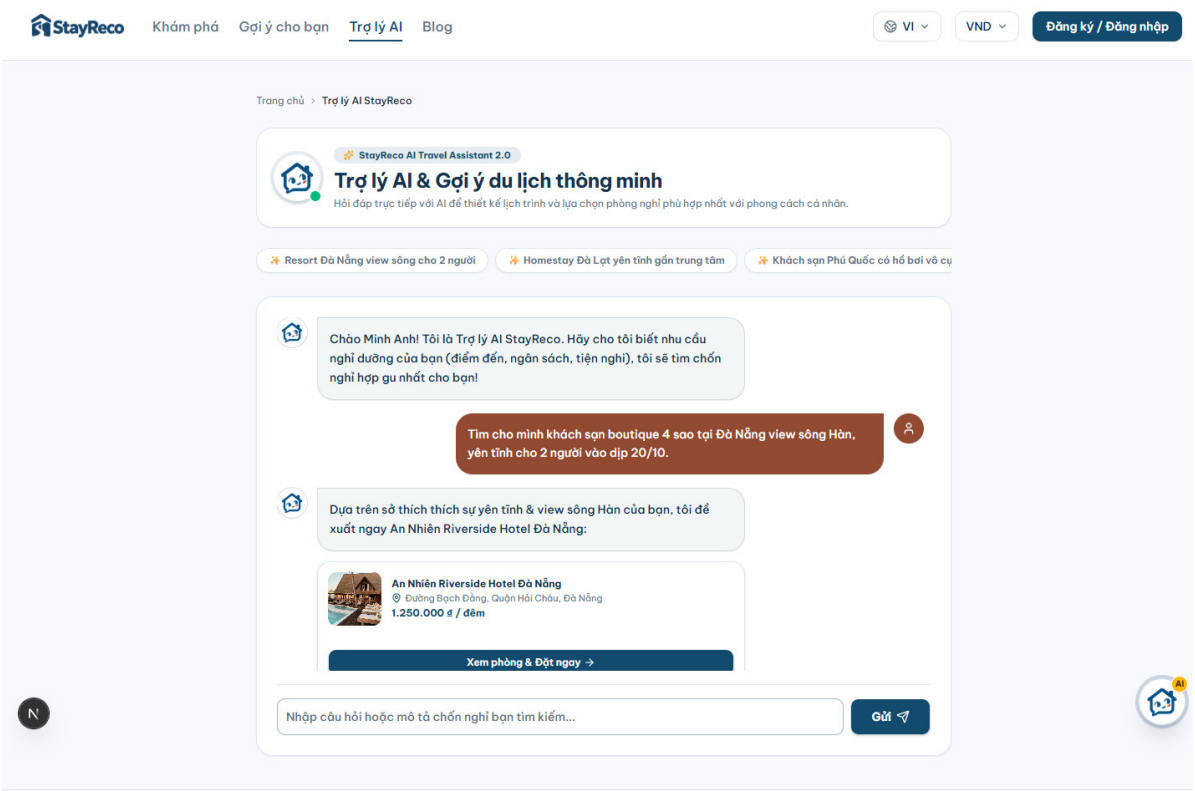


Figure 41. AI Assistant screen (web)



Figure 42. AI Assistant screen (mobile app)

- **Function details:**
  - Answers about policies and FAQs come from hotel knowledge retrieved with pgvector; availability and prices always come from the booking system, never from the model.
  - A guest gets suggestions based on the current session only; a customer also gets personal suggestions (BR-18).
  - "Talk to hotel staff" hands the conversation to the Unified Inbox of the hotel; the assistant stops answering in that conversation.
  - When the AI service is unavailable MSG24 is shown and normal search still works.

3.4.2 Travel Preferences

- **Function trigger:** Account menu, "Travel preferences", or the shortcut on Recommendations.
- **Function description:** Actor: Customer. Purpose: tell the system what kind of stay the customer likes (UC16). The form has trip purpose, budget range, preferred destinations, important amenities and accommodation style. All fields are optional.
- **Screen layout: Figure 43 (web).**

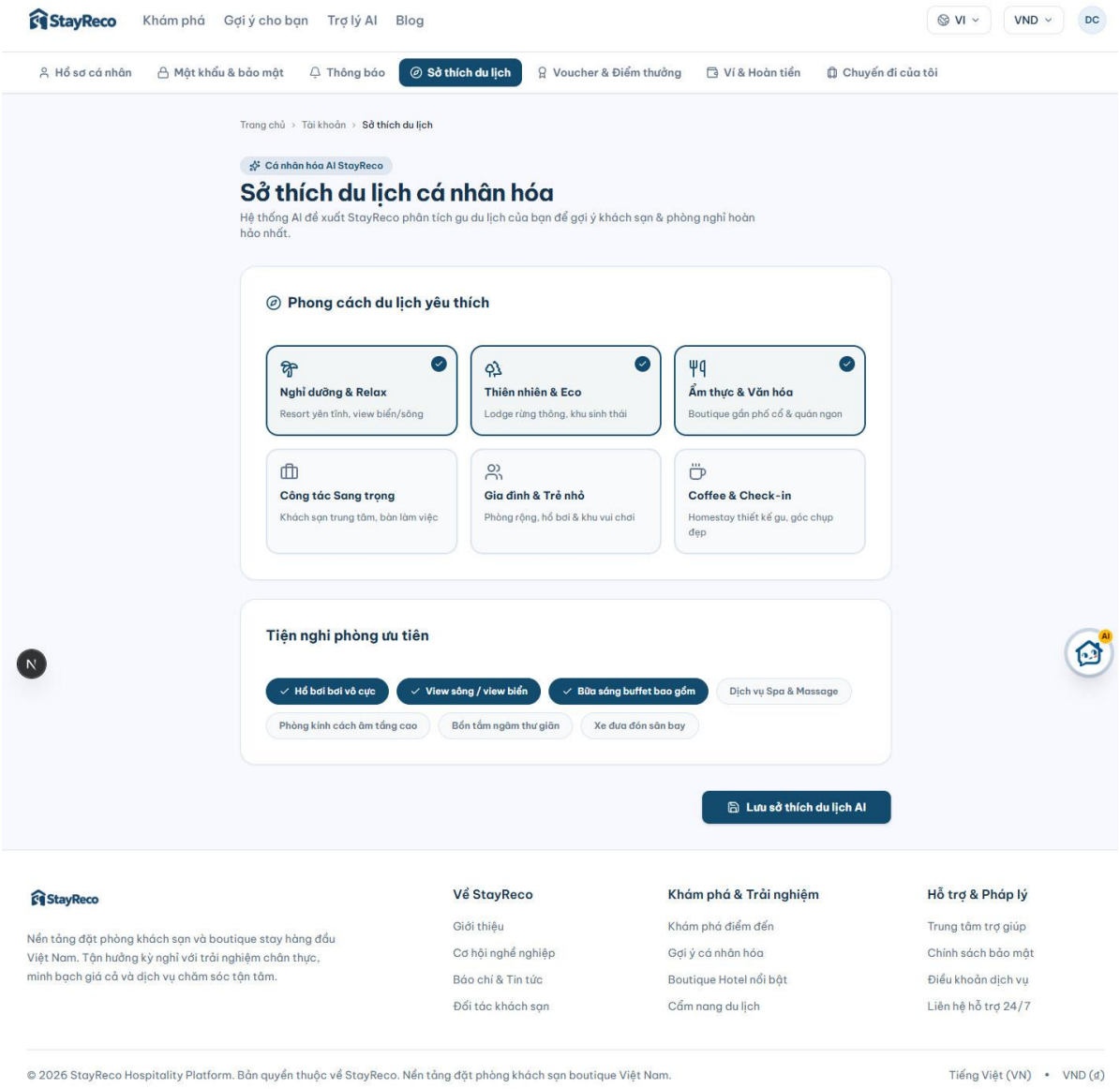


Figure 43. Travel Preferences screen (web)

- **Function details:**
  - The budget minimum cannot be greater than the maximum.
  - Saving shows MSG10; the new preferences are used from the next recommendation request.

3.4.3 Recommendations

- **Function trigger:** The "For you" link in the account menu, or the recommendation block on Home Page.
- **Function description:** Actor: Customer. Purpose: show personal hotel and room suggestions (UC17). Each card shows the reason of the suggestion (for example "similar to hotels you booked" or "matches your budget") and a link to edit preferences.
- **Screen layout: Figure 44 (web).**

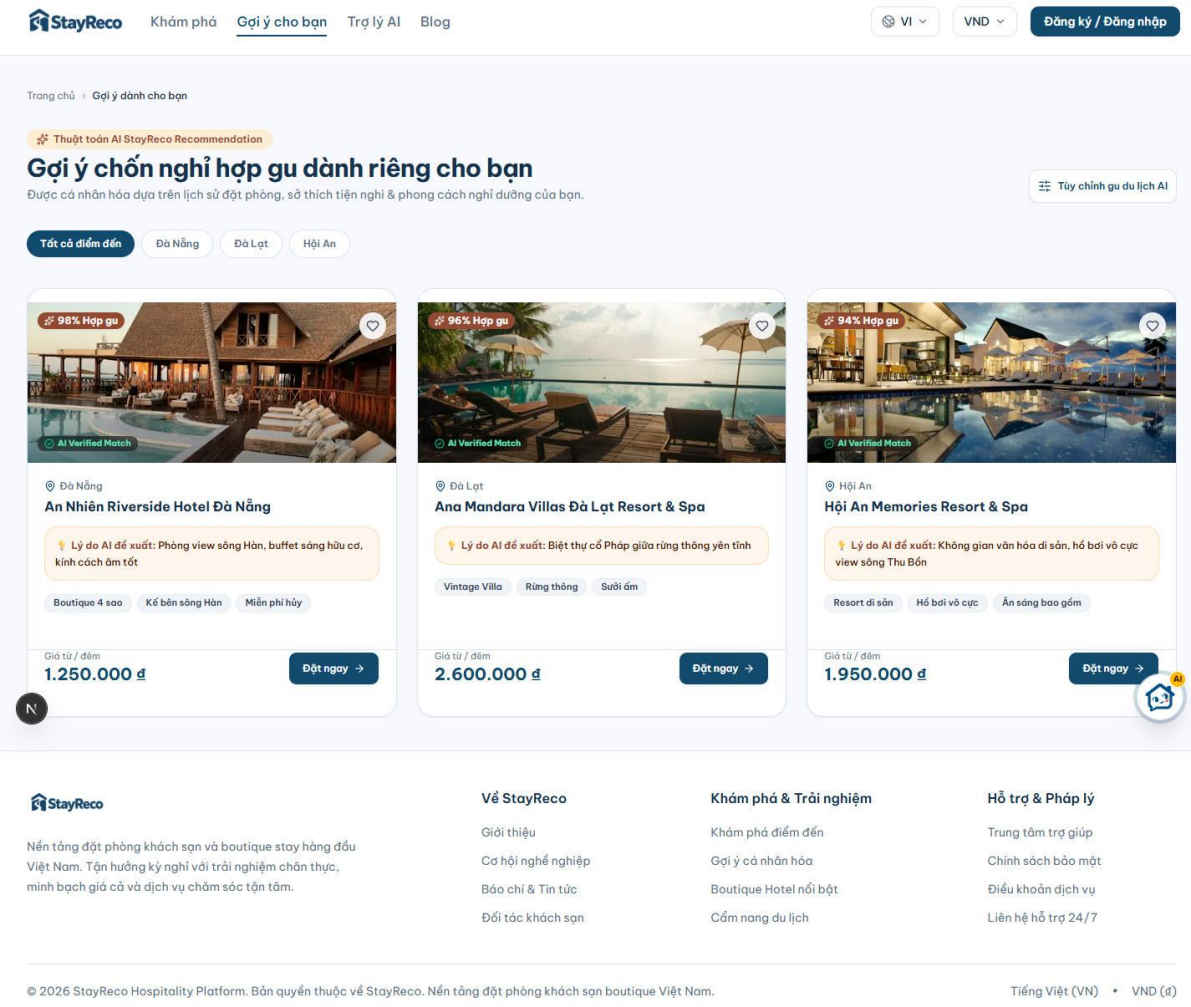


Figure 44. Recommendations screen (web)

- **Function details:**
  - A new customer without history gets suggestions based on preferences and content similarity (cold start).
  - Clicks and bookings made from this page are recorded as feedback for the recommender.

3.4.4 AI Chat Service

- **Function trigger:** Called by AI Assistant for every message.
- **Function description:** Non-screen function. Purpose: answer guest questions and call system functions. It retrieves hotel knowledge from pgvector (RAG) and calls functions such as hotel search and availability check, then writes the answer.
- **Screen layout:** *not applicable, this function runs without a screen.*
- **Function details:**
  - Every function call and its result are stored with the conversation for later review.
  - The service never confirms a booking or a price by itself; it only returns cards that link to the booking flow.
  - On timeout or error the client shows MSG24.

3.4.5 Embedding Pipeline

- **Function trigger:** Runs when hotel, room type or article content is created or changed.
- **Function description:** Non-screen function. Purpose: keep the vector index used by the assistant and the recommender up to date.
- **Screen layout:** *not applicable, this function runs without a screen.*
- **Function details:**
    - Content is split into chunks, turned into embeddings and stored in pgvector with a link to its source record.
    - Old embeddings of a changed record are replaced; embeddings of a deleted record are removed.

3.4.6 Recommendation Service

- **Function trigger:** Called by Recommendations, Home Page and AI Assistant.
- **Function description:** Non-screen function. Purpose: compute the hybrid recommendation score from content similarity, user interactions and ratings, and adapt the weights to how much data exists for the user and the hotel.
- **Screen layout:** *not applicable, this function runs without a screen.*
- **Function details:**
    - Sparse data (new user or new hotel) gives more weight to content similarity; dense data gives more weight to behaviour.
    - Each recommendation is stored with its score and reason, and clicks and bookings are recorded for evaluation.

3.4.7 Model Training & Evaluation Job

- **Function trigger:** Runs on a schedule (nightly) and on demand by the team.
- **Function description:** Non-screen function. Purpose: retrain the recommendation models and measure their quality.
- **Screen layout:** *not applicable, this function runs without a screen.*
- **Function details:**
    - Each run stores the model version, the predicted ratings and the metrics Precision@K, Recall@K, NDCG, MAE, RMSE and cold-start coverage.
    - A new model is used only when its metrics are not worse than the current one.

3.5 Booking & Payment

3.5.1 Booking Checkout

- **Function trigger:** The customer clicks "Select this room" on Room Type Details or on a room card of the AI assistant.
- **Function description:** Actor: Customer. Purpose: enter the booking details and confirm the price (UC18, UC19, UC20, UC21). The form has booker details (prefilled from the profile), guest name, special requests, estimated arrival time, a discount code box, the loyalty points to use, the cancellation terms and the order summary with the final price.
- **Screen layout: Figure 45 (web) and Figure 46 (mobile app).**

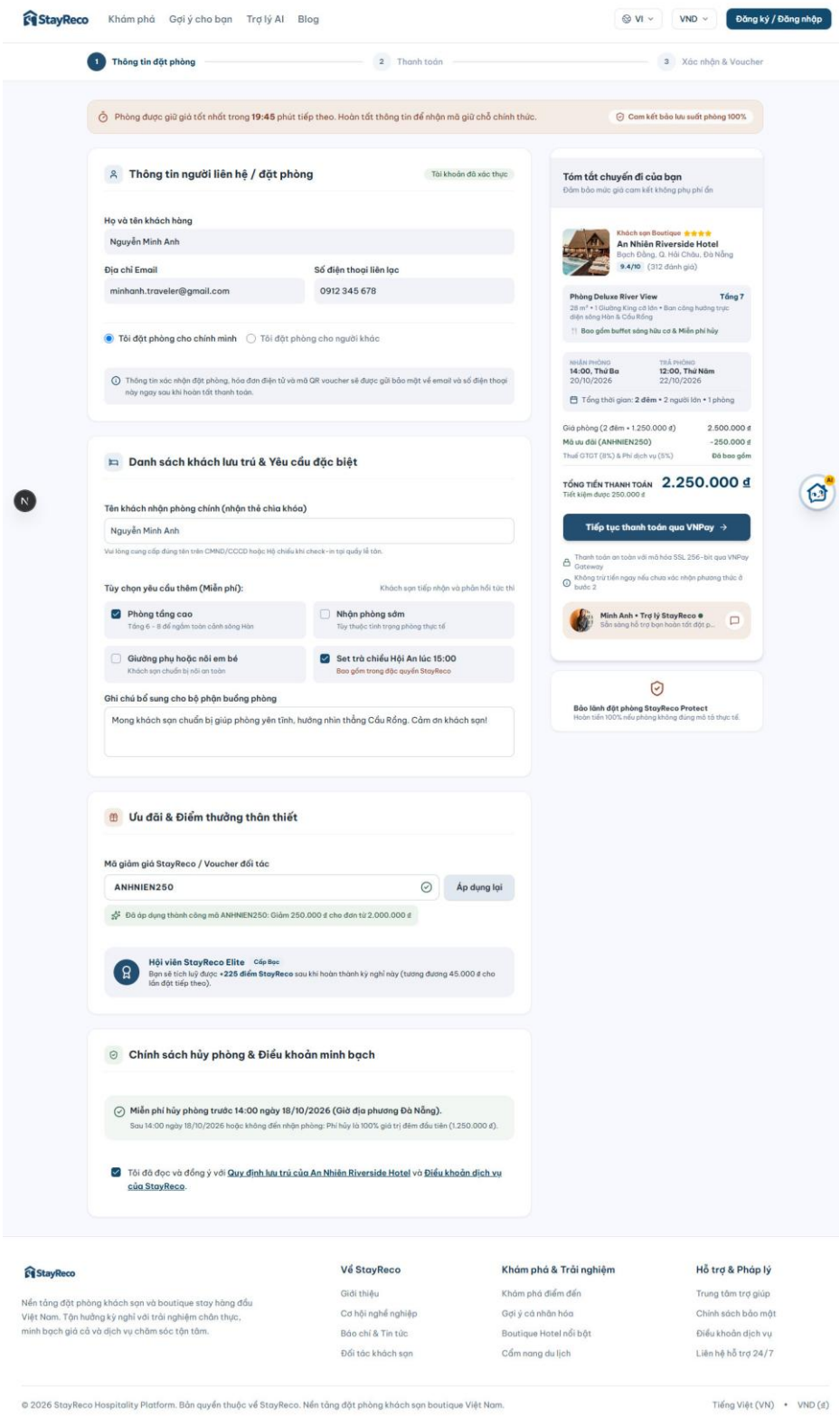


Figure 45. Booking Checkout screen (web)

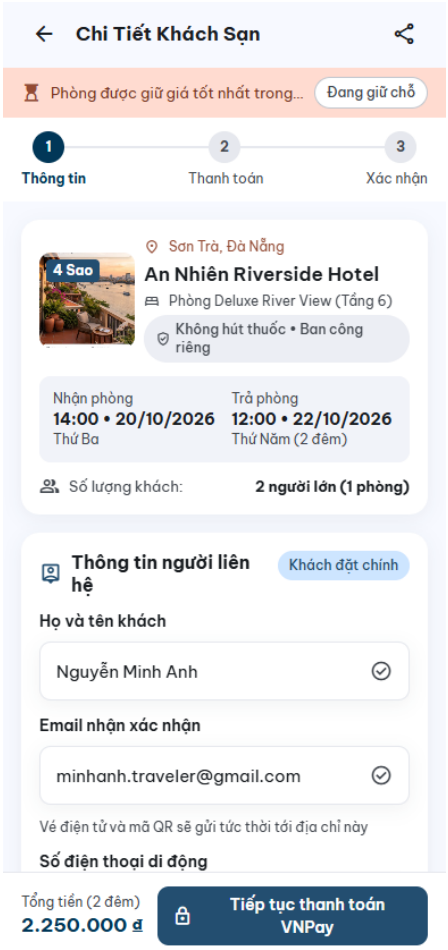


Figure 46. Booking Checkout screen (mobile app)

- **Function details:**
  - Guest name and contact phone are required; special requests are at most 500 characters.
  - Pressing "Continue to payment" checks availability again and holds the room for 15 minutes (BR-03). If the room is gone, MSG12 is shown; if the price changed, MSG13 is shown and the customer confirms the new total.
  - Only one discount code (promotion code or voucher) can be used together with loyalty points, applied in the order of BR-06; an invalid code shows MSG14.
  - The final price already includes VAT and service charge and is never below zero (BR-05).

3.5.2 Payment

- **Function trigger:** Opened right after Booking Checkout, or from "Pay now" on Booking Details while the hold is still valid.
- **Function description:** Actors: Customer, Payment Gateway. Purpose: pay the booking through SePay (UC22). The page shows a VietQR code with the amount and the transfer content of the booking; the customer pays by bank transfer into the account of the platform, and the page then shows the result: verifying, paid, failed, cancelled or expired.
- **Screen layout: Figure 47 (web) and Figure 48 (mobile app).**

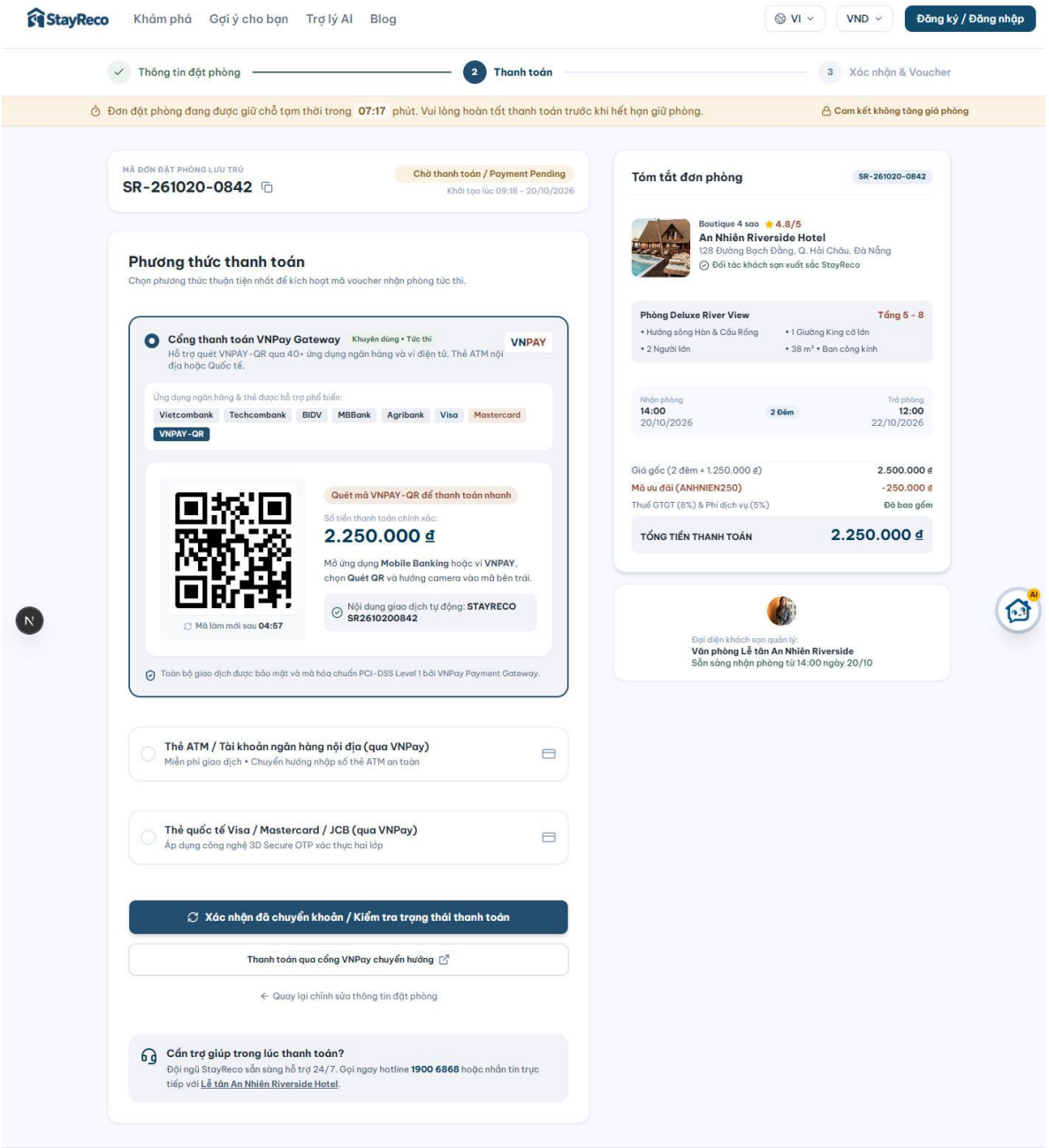


Figure 47. Payment screen (web)

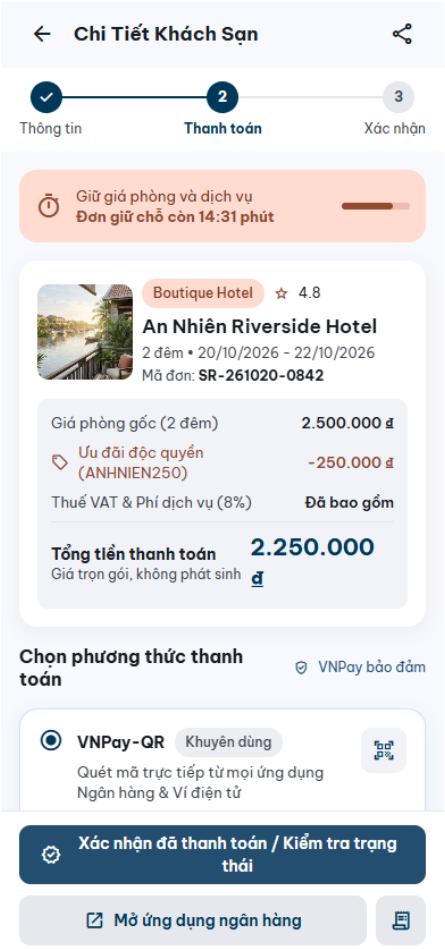


Figure 48. Payment screen (mobile app)

- **Function details:**
  - The booking becomes confirmed only after SePay reports the matching transfer (BR-04); while waiting, MSG15 is shown.
  - A failed or cancelled payment shows MSG16 with the deadline of the hold; after the hold expires the room is released and the booking is marked expired.
  - On success MSG17 is shown and the customer goes to Booking Voucher.

3.5.3 Booking Voucher

- **Function trigger:** Opened after a successful payment, or from Booking Details of a confirmed booking.
- **Function description:** Actor: Customer. Purpose: show the confirmation and the QR-code e-voucher (UC23). The page shows the booking code, hotel, room type, dates, guests, paid amount and the QR code with a "Download" button.
- **Screen layout: Figure 49 (web) and Figure 50 (mobile app).**

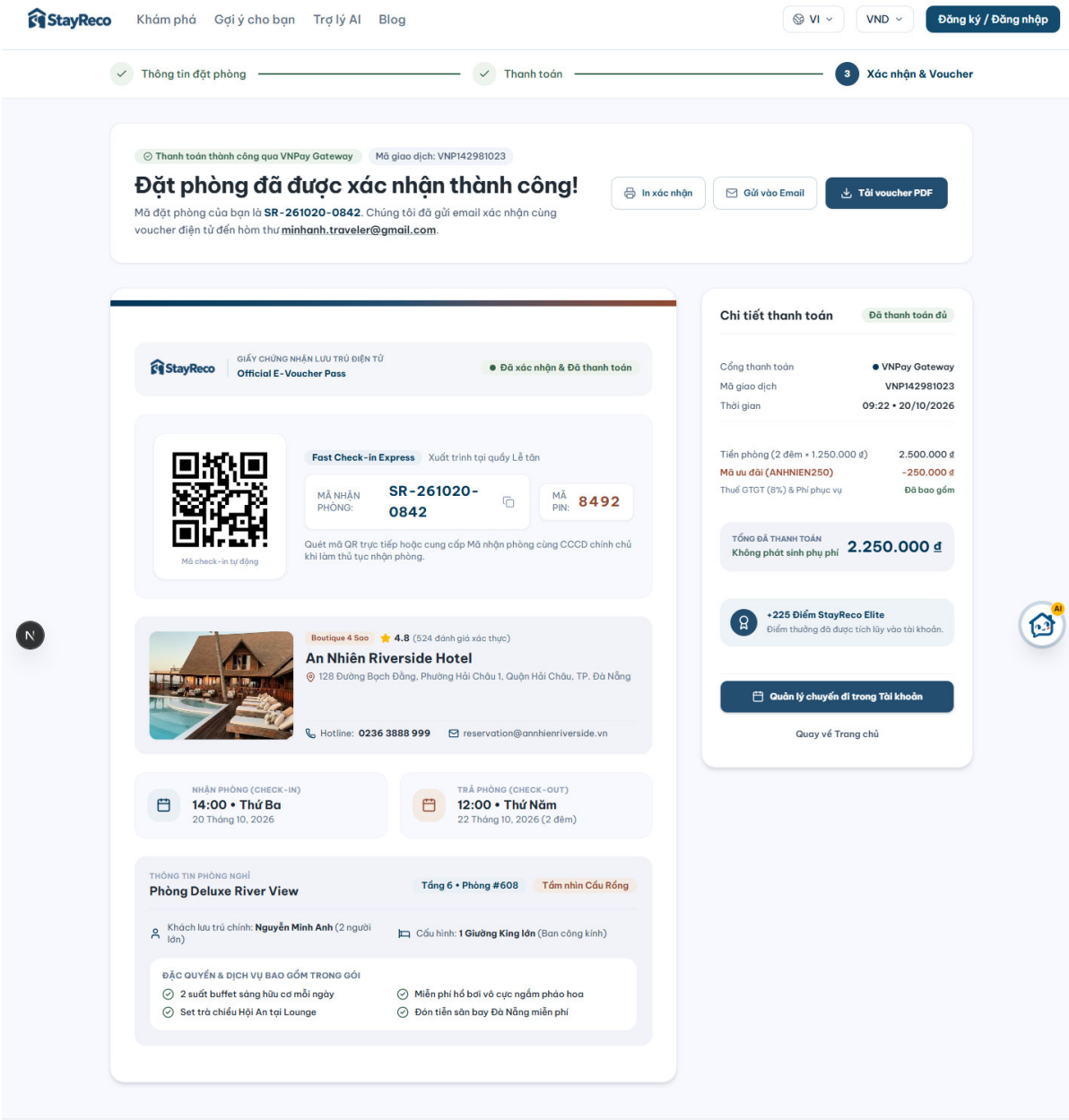


Figure 49. Booking Voucher screen (web)

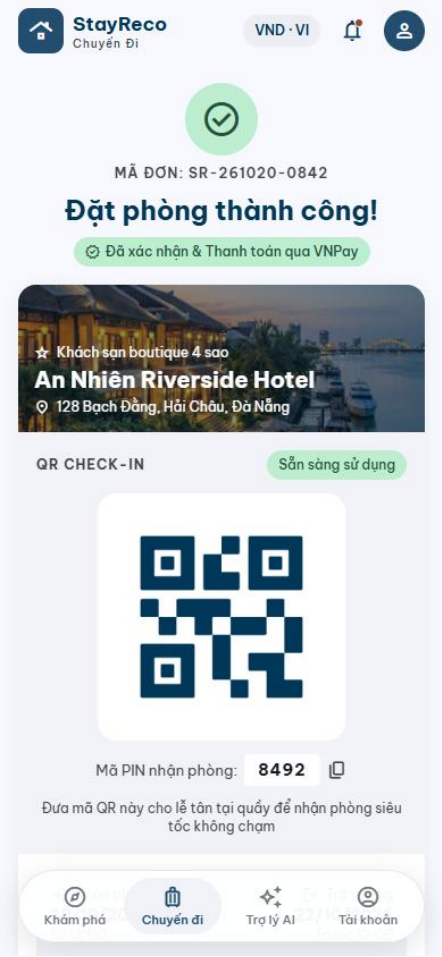


Figure 50. Booking Voucher screen (mobile app)

- **Function details:**
  - The QR code carries a signed token, not the plain booking code.
  - A cancelled booking shows the voucher as invalid.

3.5.4 Booking Hold Expiry Job

- **Function trigger:** Runs every minute.
- **Function description:** Non-screen function. Purpose: release rooms held by bookings whose payment was not completed in time (BR-03).
- **Screen layout:** *not applicable, this function runs without a screen.*
- **Function details:**
  - Bookings still waiting for payment after the hold time are marked expired and their room nights become available again.
  - A transfer that arrives after the booking has expired is recorded and returned to the customer by bank transfer.

3.5.5 Payment Callback Handler

- **Function trigger:** Called by the SePay webhook when a transfer reaches the bank account of the platform.
- **Function description:** Non-screen function. Purpose: receive the SePay webhook, check its API key, match the transfer with a booking by the transfer content and the amount, and update the payment and the booking.
- **Screen layout:** *not applicable, this function runs without a screen.*

- **Function details:**
    - A webhook with a wrong API key, or a transfer whose content or amount matches no booking, is rejected and logged for manual checking.
    - Duplicate notifications for the same transaction are ignored, so a booking is never confirmed twice.
    - On success the booking is confirmed, the hotel share goes to the pending balance of the hotel wallet (BR-22) and the confirmation email is sent.

3.5.6 QR Voucher Generator

- **Function trigger:** Runs when a booking is confirmed or cancelled.
- **Function description:** Non-screen function. Purpose: create the signed QR e-voucher of a confirmed booking and invalidate it when the booking is cancelled.
- **Screen layout:** *not applicable, this function runs without a screen.*
- **Function details:**
    - The token is signed with a server secret; Check-in verifies the signature before opening the booking.

3.5.7 Commission Recording

- **Function trigger:** Runs when a booking is completed (check-out) and when a refund is made.
- **Function description:** Non-screen function. Purpose: record the platform commission of each booking for the revenue reports (BR-13).
- **Screen layout:** *not applicable, this function runs without a screen.*
- **Function details:**
    - The commission rate is the one in force when the booking was made.
    - A refund creates a commission adjustment instead of changing the original record.

3.6 Trip Management

3.6.1 My Bookings

- **Function trigger:** Account menu, "My bookings".
- **Function description:** Actor: Customer. Purpose: see all bookings (UC24). The list has the tabs Upcoming, In-house, Completed and Cancelled, a search by booking code, and the main action of each booking (pay, view voucher, review, cancel).
- **Screen layout: Figure 51 (web) and Figure 52 (mobile app).**

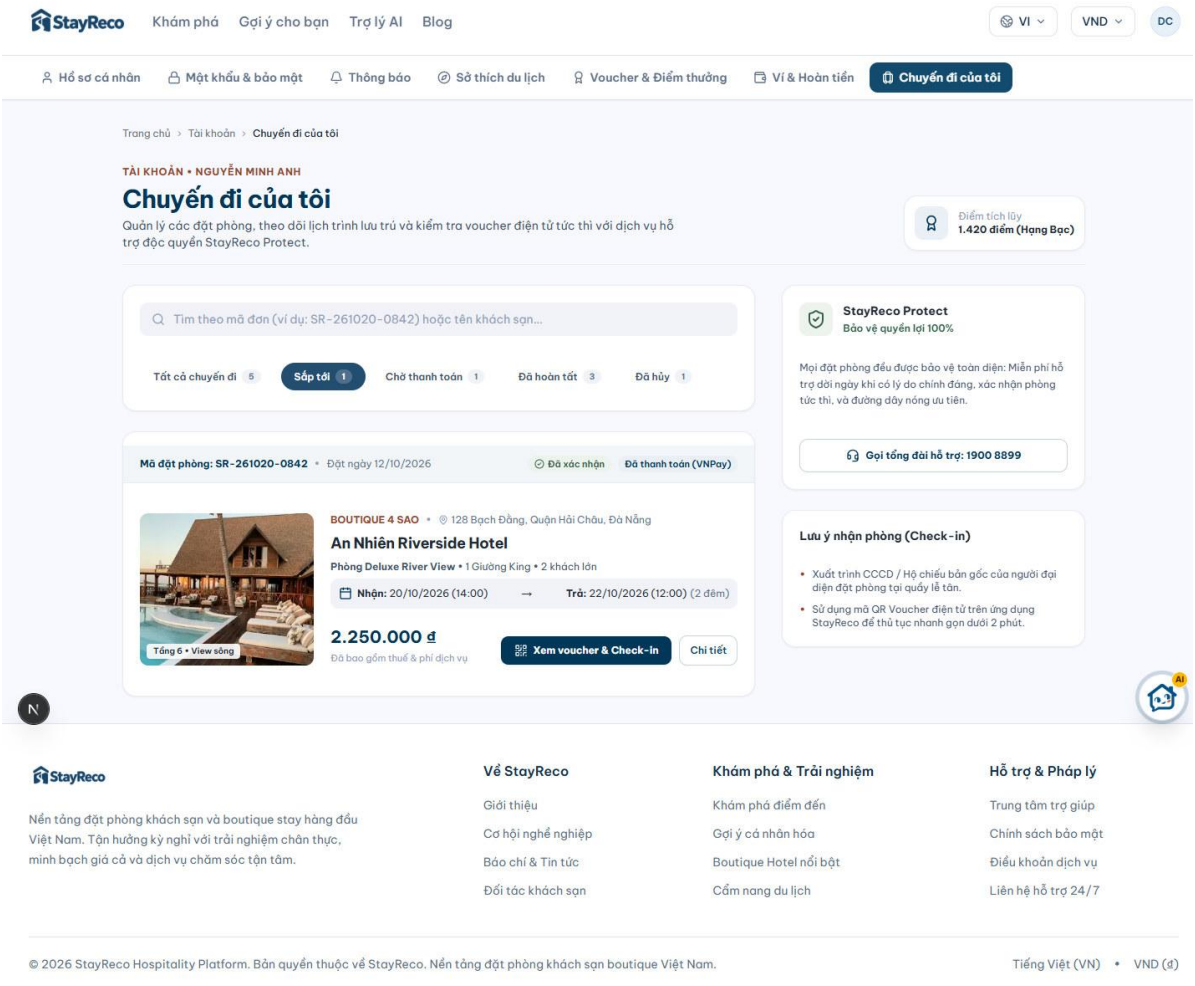


Figure 51. My Bookings screen (web)

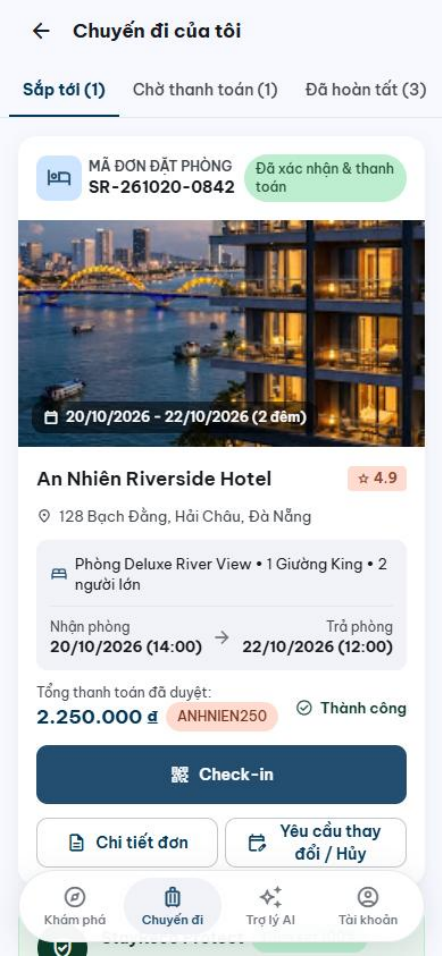


Figure 52. My Bookings screen (mobile app)

- **Function details:**
  - Only the bookings of the signed-in customer are listed.
  - No booking in a tab shows an empty state with a link to Home Page.

3.6.2 Booking Details

- **Function trigger:** Click on a booking in My Bookings, or the link in a notification or email.
- **Function description:** Actor: Customer. Purpose: show one booking in full (UC24, UC30). The page shows booking, payment and refund status, price breakdown, cancellation terms, the timeline of events and the actions allowed for the current status.
- **Screen layout: Figure 53 (web).**

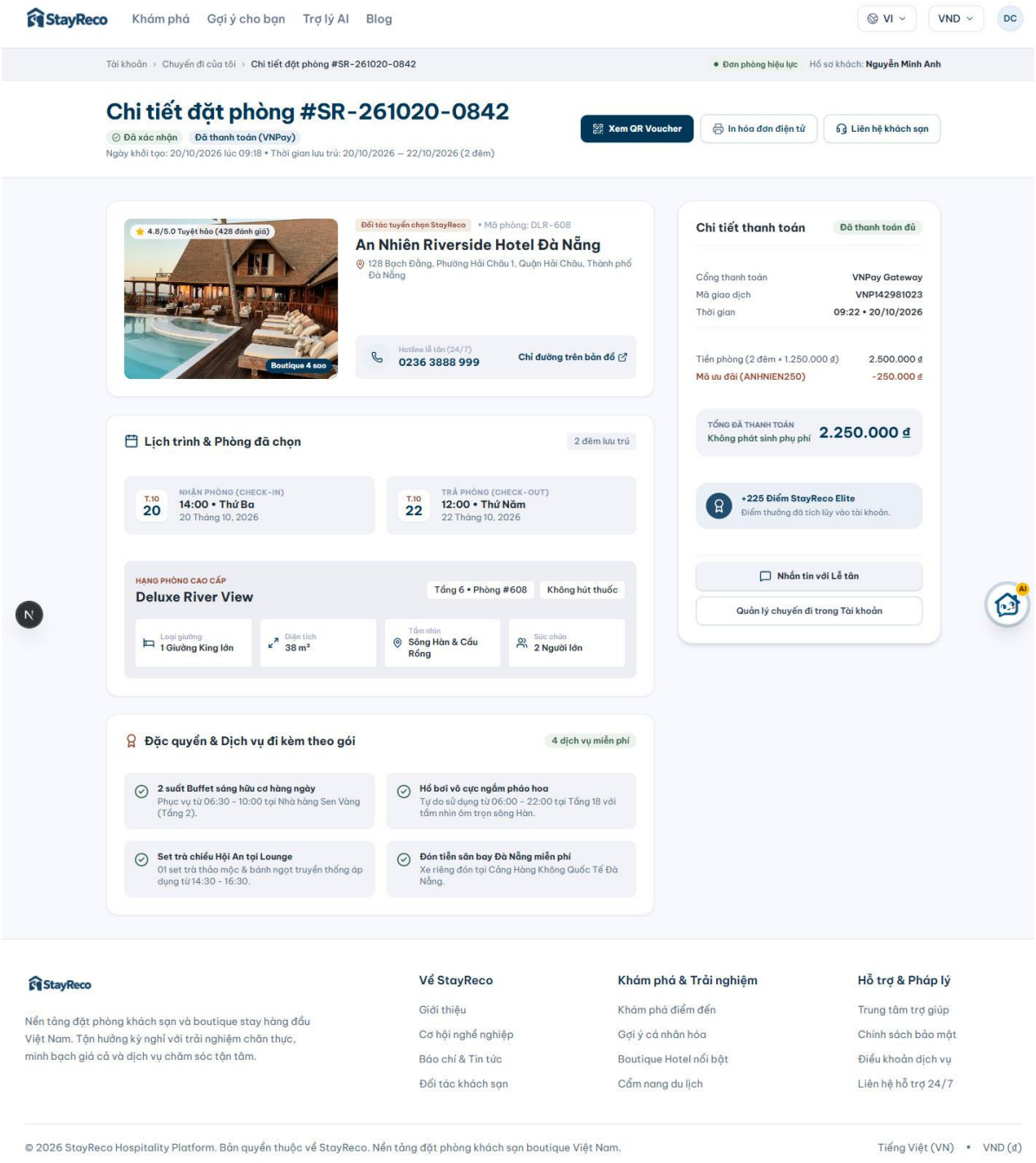


Figure 53. Booking Details screen (web)

- **Function details:**
  - Actions depend on status: a pending booking can be paid, a confirmed booking can be changed or cancelled, a completed stay can be reviewed.
  - The refund status is shown until the refund is completed (UC30).

3.6.3 Change Booking

- **Function trigger:** "Change details" on Booking Details of a confirmed booking.
- **Function description:** Actor: Customer. Purpose: ask the hotel to change details that do not affect the price (UC25). The form has guest name, contact phone, arrival time and special

requests, and a note that new dates, another room type or another number of guests need a new booking.

- **Screen layout: Figure 54 (web).**

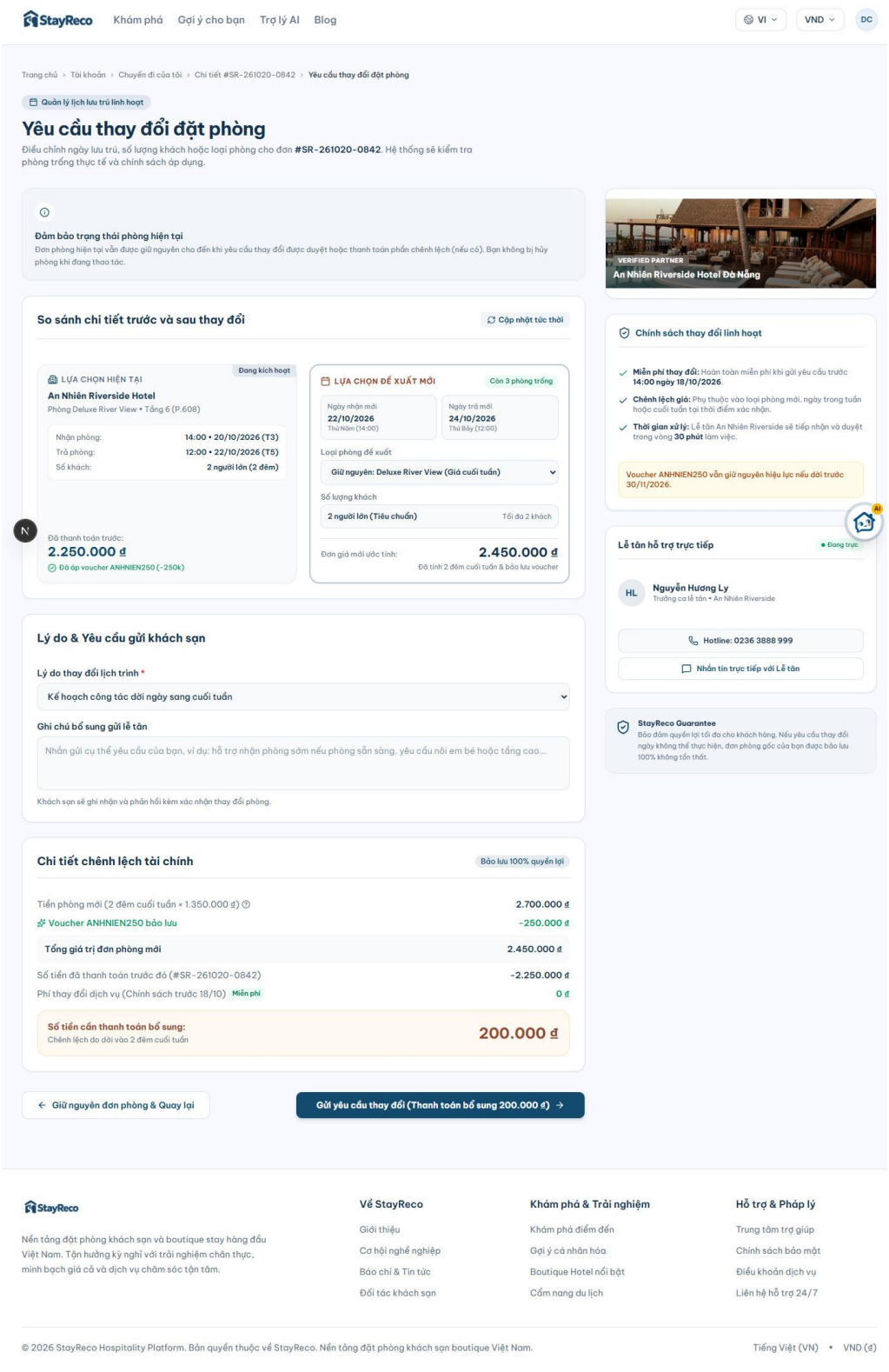


Figure 54. Change Booking screen (web)

- **Function details:**

- The price never changes (BR-09).
- The request takes effect only after the hotel staff accept it (UC43); the customer is notified of the decision.
- A booking that is checked in, completed or cancelled cannot be changed.

3.6.4 Cancel Booking

- **Function trigger:** "Cancel booking" on Booking Details.
- **Function description:** Actor: Customer. Purpose: cancel a booking and see the refund before confirming (UC28, UC29). The screen shows the free-cancellation deadline, the cancellation fee, the refundable amount and a required reason.
- **Screen layout: Figure 55 (web).**

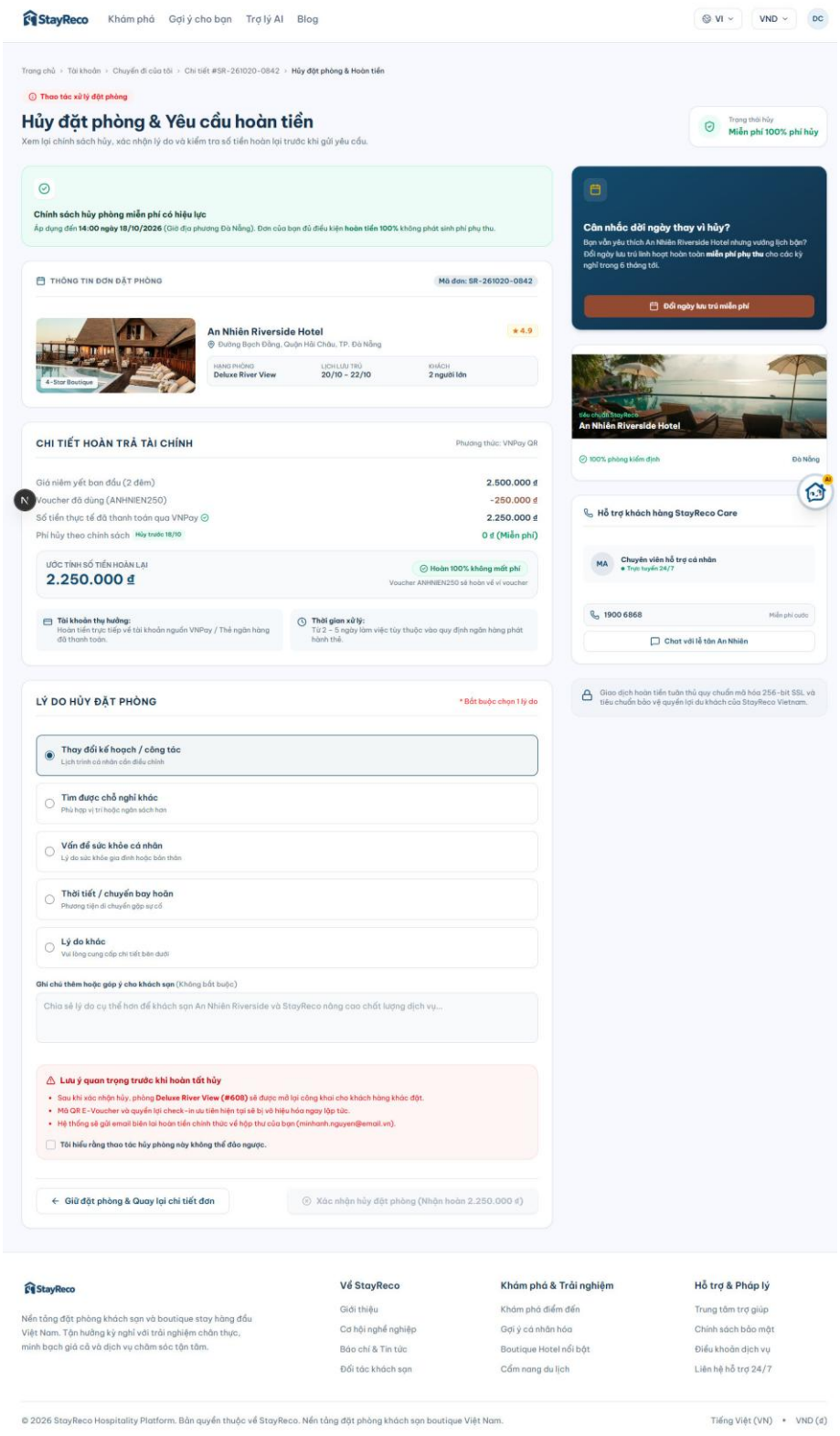


Figure 55. Cancel Booking screen (web)

- **Function details:**
  - The policy applied is the one in force when the booking was made (BR-07); the refundable amount follows BR-08.
  - A confirmation dialog (MSG11) is shown before cancelling. After cancelling MSG18 is shown and a refund request is created when money is due.

- A booking cannot be cancelled after check-in (BR-09).

3.6.5 Stay Review

- **Function trigger:** "Write a review" on a completed booking.
- **Function description:** Actor: Customer. Purpose: rate the hotel after the stay (UC31, UC32). The form has an overall rating, ratings per criterion, a review text and up to 5 photos.
- **Screen layout: Figure 56 (web).**

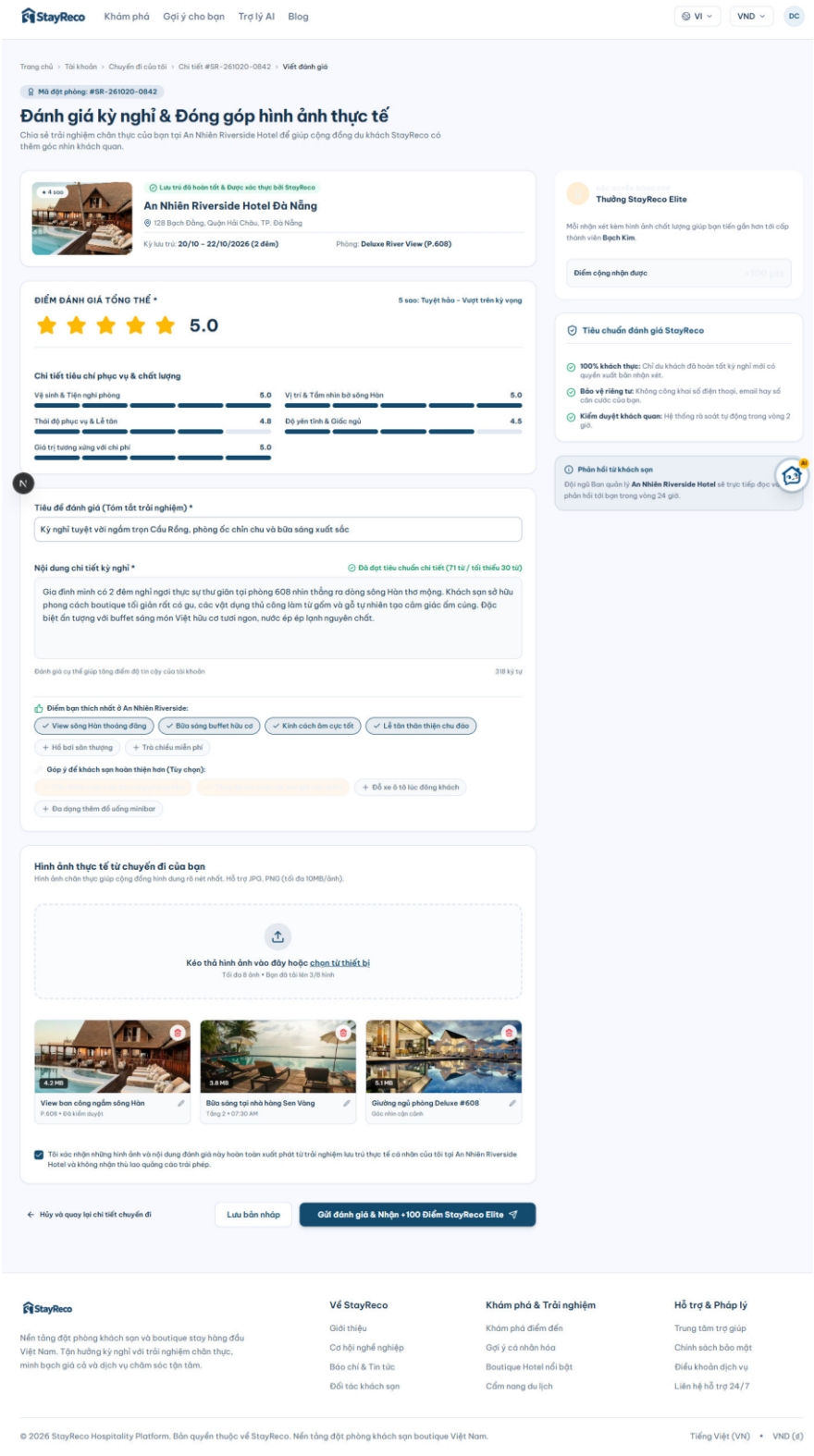


Figure 56. Stay Review screen (web)

- **Function details:**
  - Only one review per completed booking (BR-12); otherwise MSG19 is shown.
  - Review text is required, 20 to 2000 characters; photos follow the upload rules (MSG23).

- The review is published on Hotel Details and used by the recommender; a negative review can raise an alert for the hotel.

3.6.6 Rewards

- **Function trigger:** Account menu, "Points & vouchers".
- **Function description:** Actor: Customer. Purpose: see loyalty points and vouchers per hotel (UC33). The tabs are My Points (balance and history per hotel) and Vouchers (code, value, conditions, expiry).
- **Screen layout: Figure 57 (web).**

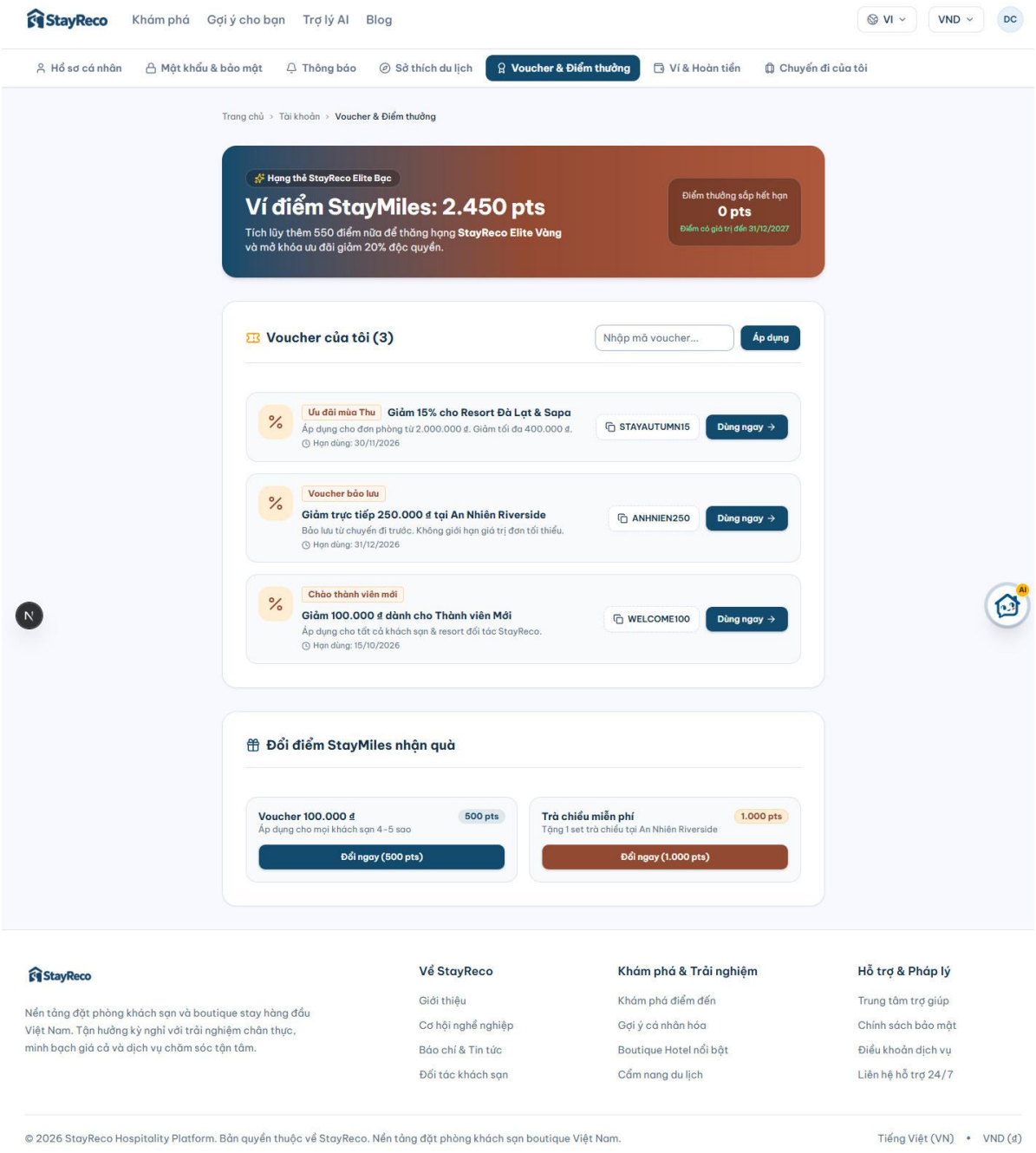


Figure 57. Rewards screen (web)

- **Function details:**

- Points and vouchers can only be used at the hotel that issued them (BR-06).
- Expired vouchers are shown in a separate group and cannot be applied.

3.6.7 Loyalty Point Accrual

- **Function trigger:** Runs when a stay is checked out, and daily for expiry.
- **Function description:** Non-screen function. Purpose: add points after the stay and expire old points and vouchers.
- **Screen layout:** *not applicable, this function runs without a screen.*
- **Function details:**
  - Points are added only after check-out, following the earning rules of the hotel (BR-11).
  - Points used on a booking that is later cancelled are returned to the customer.

3.7 Front Desk

3.7.1 Staff Dashboard

- **Function trigger:** Opened after a staff member logs in.
- **Function description:** Actor: Hotel Staff. Purpose: show the work of the day at the assigned hotel. The dashboard shows arrivals, departures, in-house guests, bookings waiting for review, open refund requests and alerts.
- **Screen layout: Figure 58 (web).**

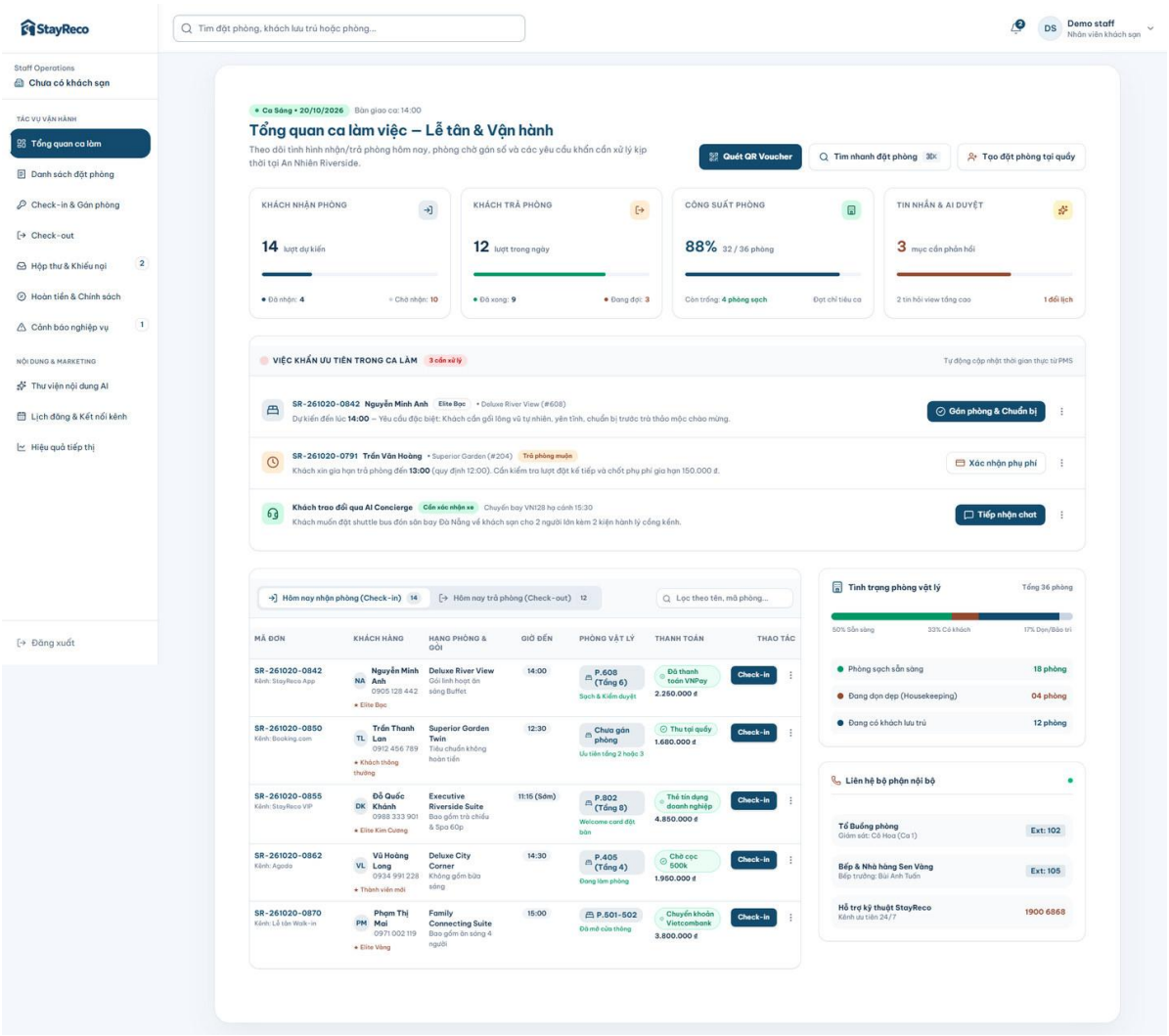


Figure 58. Staff Dashboard screen (web)

- **Function details:**
  - Each block links to the matching list with the same filter.
  - Only blocks for the permission groups granted to the staff member are shown.

3.7.2 Bookings List

- **Function trigger:** Menu "Bookings" (staff), or "Bookings" in the partner workspace.
- **Function description:** Actors: Hotel Staff; Hotel Partner (read-only). Purpose: find and open the bookings of the hotel (UC26, UC57). The table shows booking code, guest, dates, room type, booking status and payment status, with search and filters.
- **Screen layout: Figure 59 (web).**

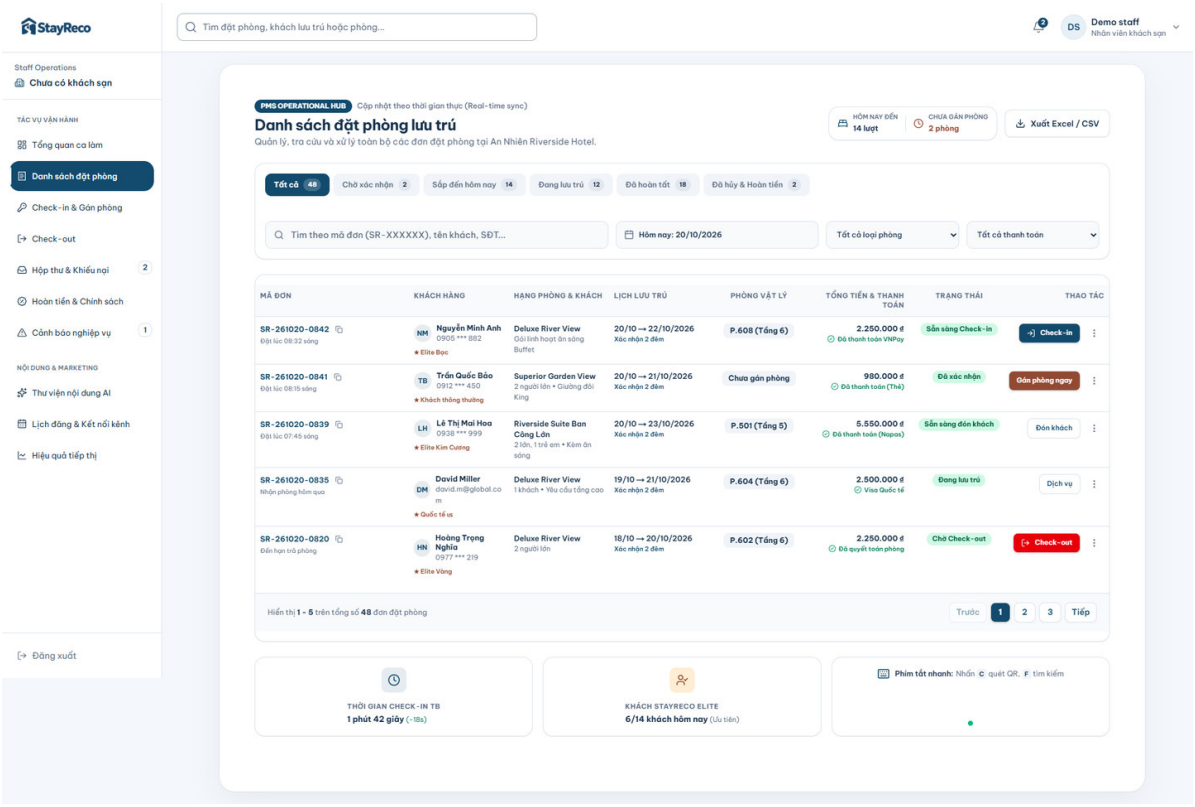


Figure 59. Bookings List screen (web)

- **Function details:**
  - Staff see only the bookings of their hotel; a partner sees the bookings of the selected hotel (BR-21).
  - The partner view has no action buttons.

3.7.3 Booking Review

- **Function trigger:** Click on a booking in Bookings List or on an alert.
- **Function description:** Actors: Hotel Staff; Hotel Partner (read-only). Purpose: review a booking and its change requests (UC27, UC43). The record shows guest, stay, payment status, special requests and change requests, with the actions Accept change and Decline change for staff. A booking is confirmed automatically when its payment is received (BR-04), so staff neither confirm nor decline it.
- **Screen layout: Figure 60 (web).**

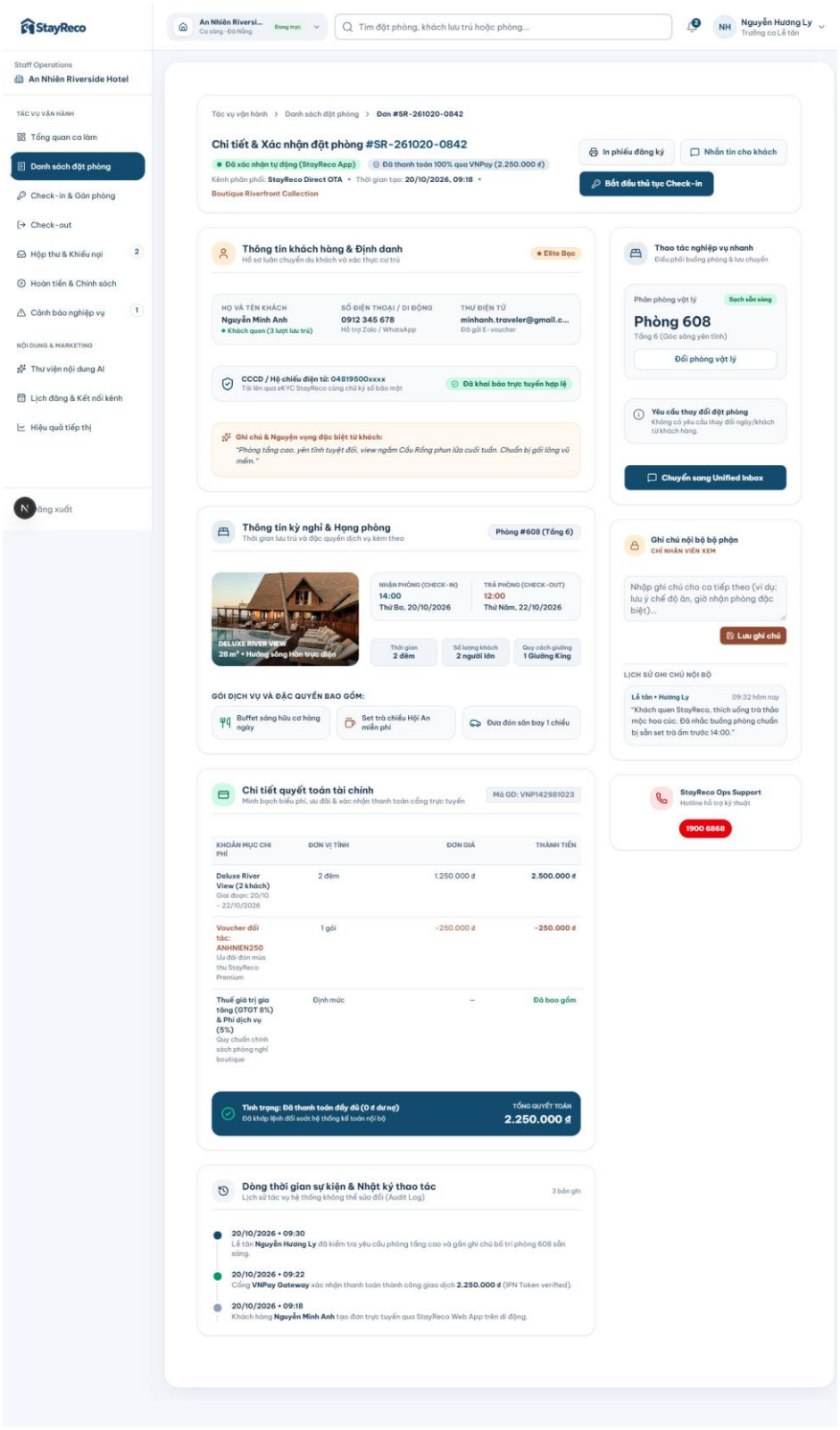


Figure 60. Booking Review screen (web)

- **Function details:**
  - Declining a change request requires a reason, which is sent to the customer.
  - A booking is confirmed only by its payment (BR-04); an unpaid booking stays pending until its hold expires.

3.7.4 Check-in

- **Function trigger:** Menu "Check-in", or "Check in" on a booking arriving today.
- **Function description:** Actor: Hotel Staff. Purpose: check a guest in (UC36, UC37, UC38). The staff member scans the QR voucher or types the booking code, the booking opens, and the screen lists the ready physical rooms of the booked room type to assign.
- **Screen layout: Figure 61 (web).**

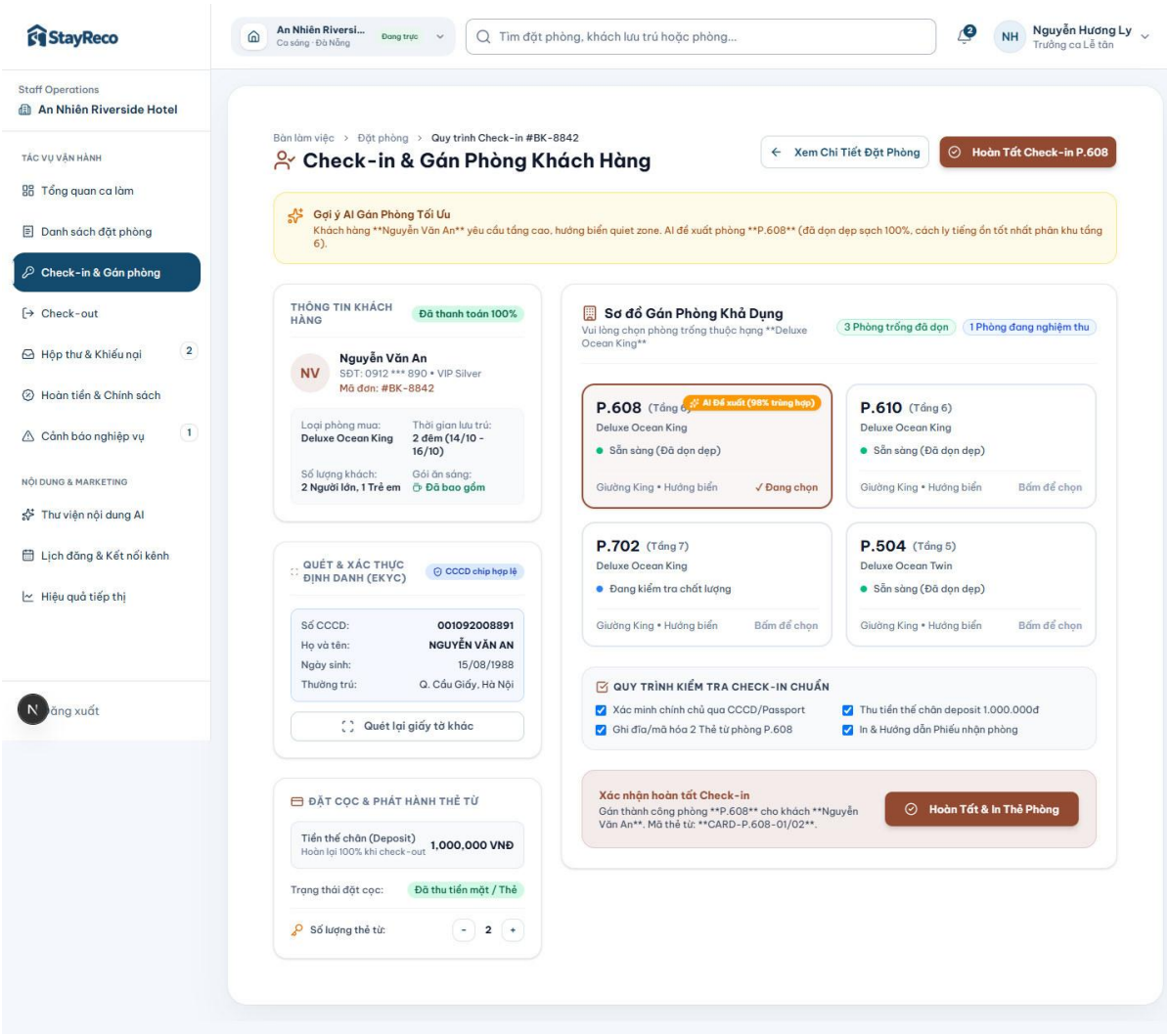


Figure 61. Check-in screen (web)

- **Function details:**
  - The QR token signature and the booking status are verified first; an invalid or cancelled voucher shows MSG20.
  - Only a ready room of the booked room type can be assigned; the room type is never changed silently (BR-10).
  - After check-in the room is marked occupied and the booking becomes in-house.

3.7.5 Check-out

- **Function trigger:** "Check out" on an in-house booking.
- **Function description:** Actor: Hotel Staff. Purpose: close the stay (UC39). The screen summarizes the charges, payments and refunds of the stay and asks for confirmation.
- **Screen layout: Figure 62 (web).**

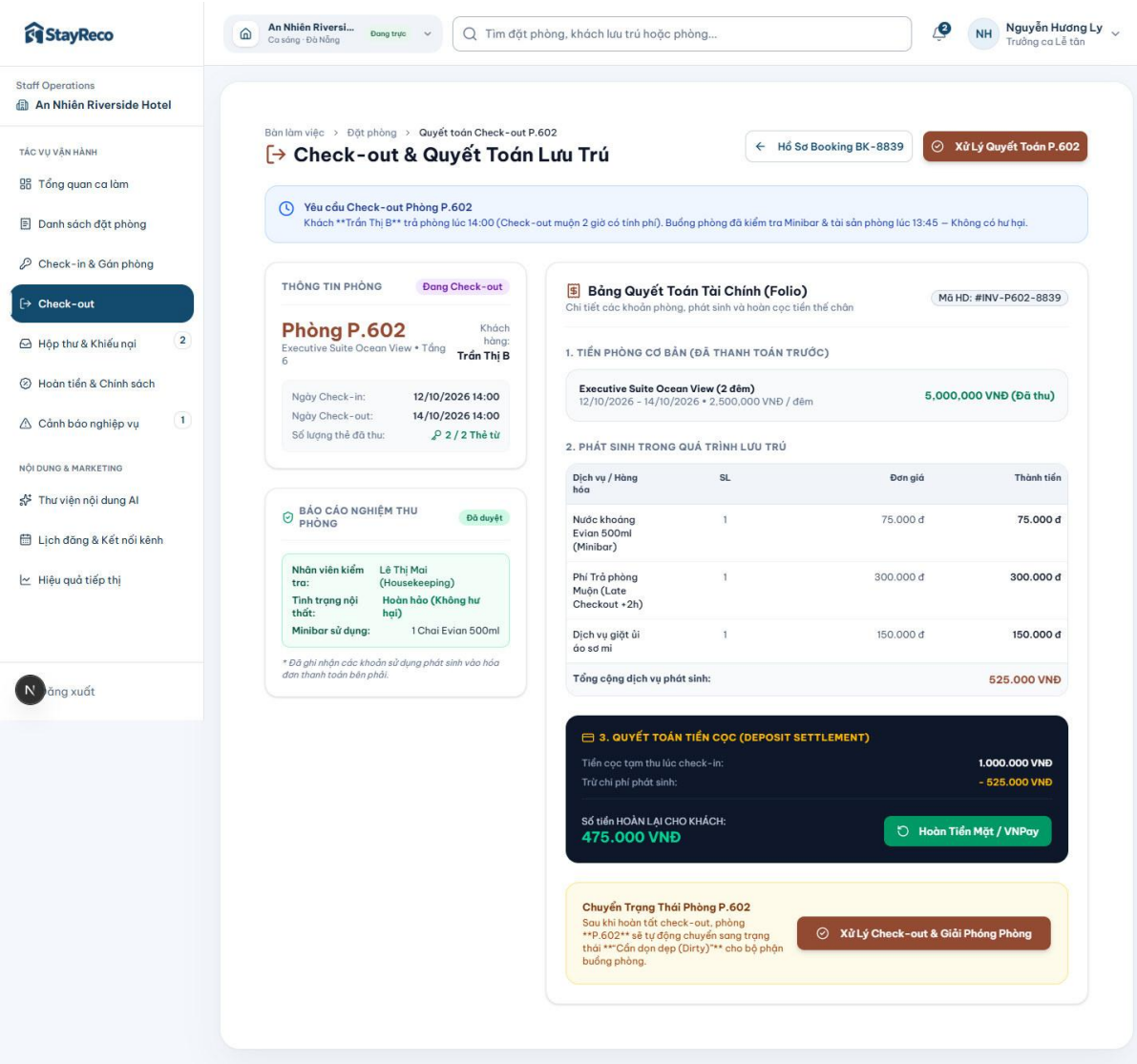


Figure 62. Check-out screen (web)

- **Function details:**
  - After check-out the booking becomes completed, the room goes to cleaning, loyalty points are added (BR-11) and the customer can write a review.

3.8 Customer Care

3.8.1 Unified Inbox

- **Function trigger:** Menu "Inbox".
- **Function description:** Actor: Hotel Staff. Purpose: answer guests in one place (UC40, UC41). Three columns: the conversation list, the message thread with AI and staff messages, and the related booking. A conversation handled by the AI has a "Take over" button.
- **Screen layout: Figure 63 (web).**

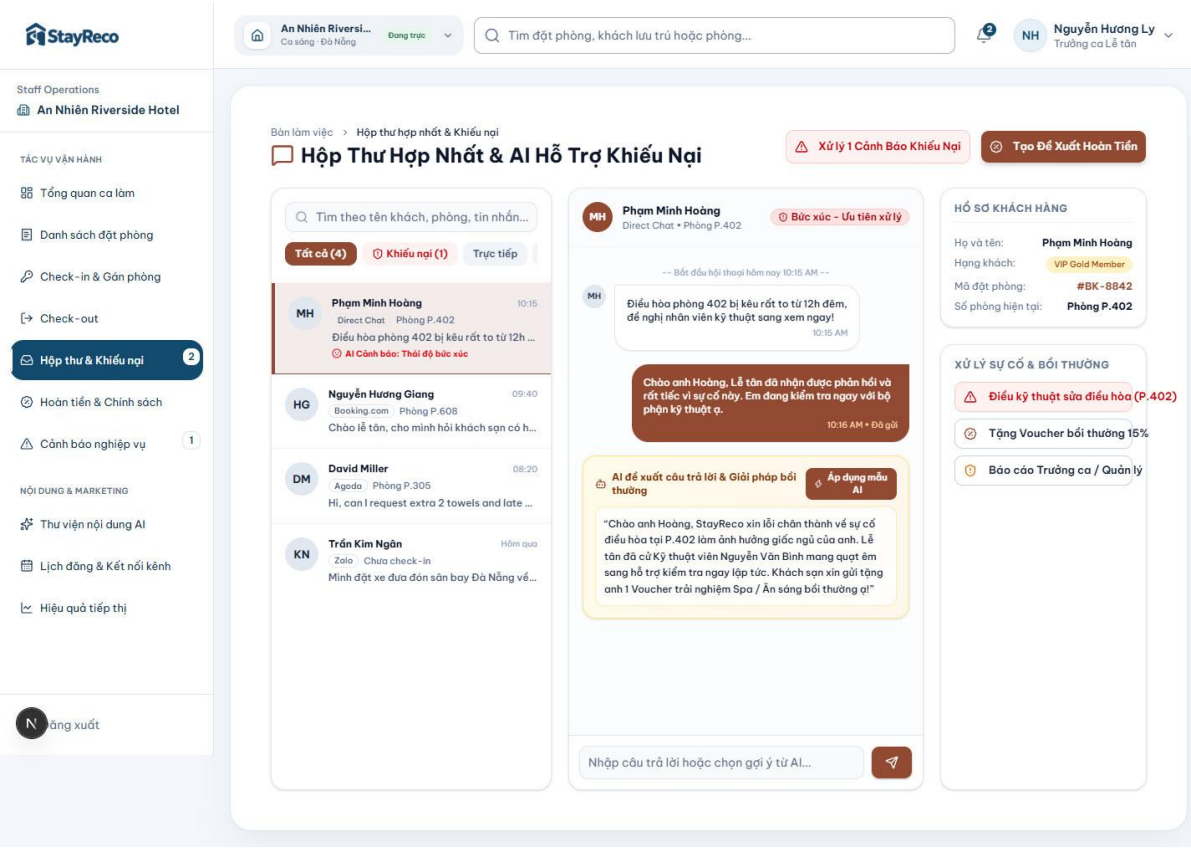


Figure 63. Unified Inbox screen (web)

- **Function details:**
  - Taking over stops the AI in that conversation until the staff member hands it back.
  - Conversations that asked for staff are marked and placed at the top.
  - Staff see only conversations of their hotel.

3.8.2 Refund Requests

- **Function trigger:** Menu "Refunds".
- **Function description:** Actors: Hotel Staff, Hotel Partner. Purpose: see the refund queue (UC34). The table shows booking code, request date, reason, amount and status, with filters by status and date.
- **Screen layout: Figure 64 (web).**

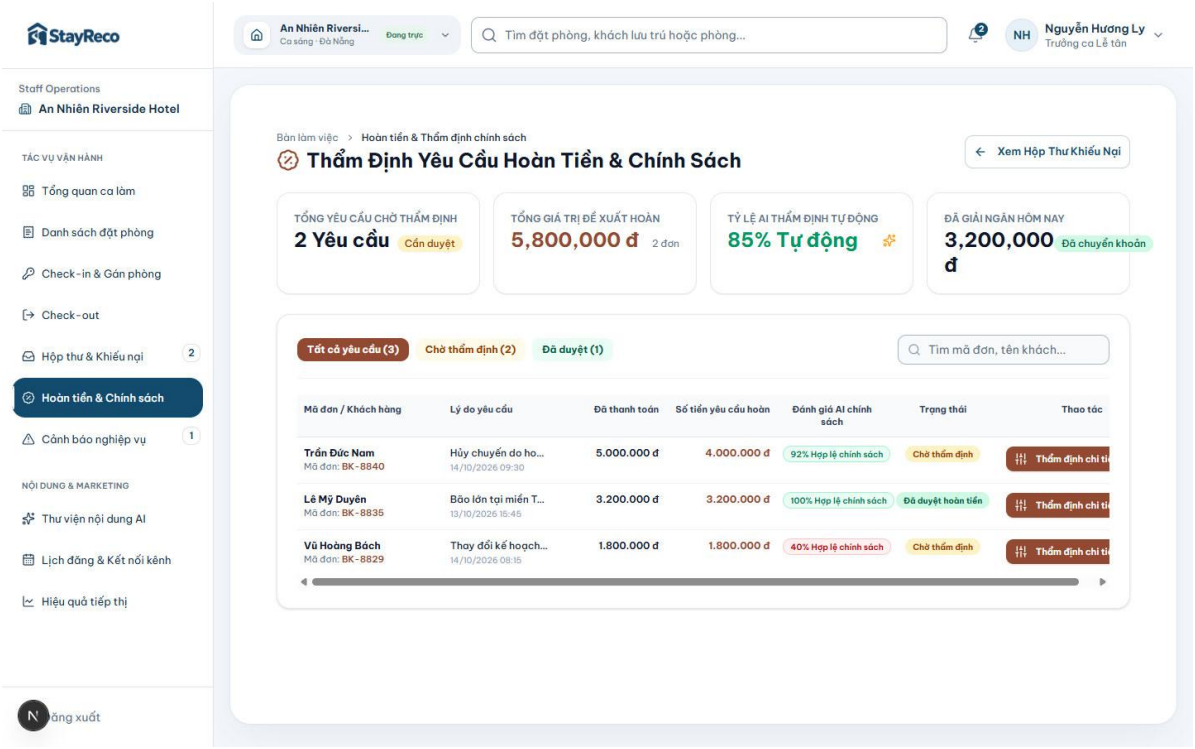


Figure 64. Refund Requests screen (web)

- **Function details:**
  - Requests are created by the system when a cancellation has a refundable amount.

3.8.3 Refund Review

- **Function trigger:** Click on a refund request.
- **Function description:** Actors: Hotel Staff, Hotel Partner. Purpose: check and decide a refund (UC34, UC35). The drawer shows the original payment, the policy applied, the fee and the maximum refundable amount, with the Forward action for staff and the Approve and Reject actions for the hotel partner.
- **Screen layout: Figure 65 (web).**

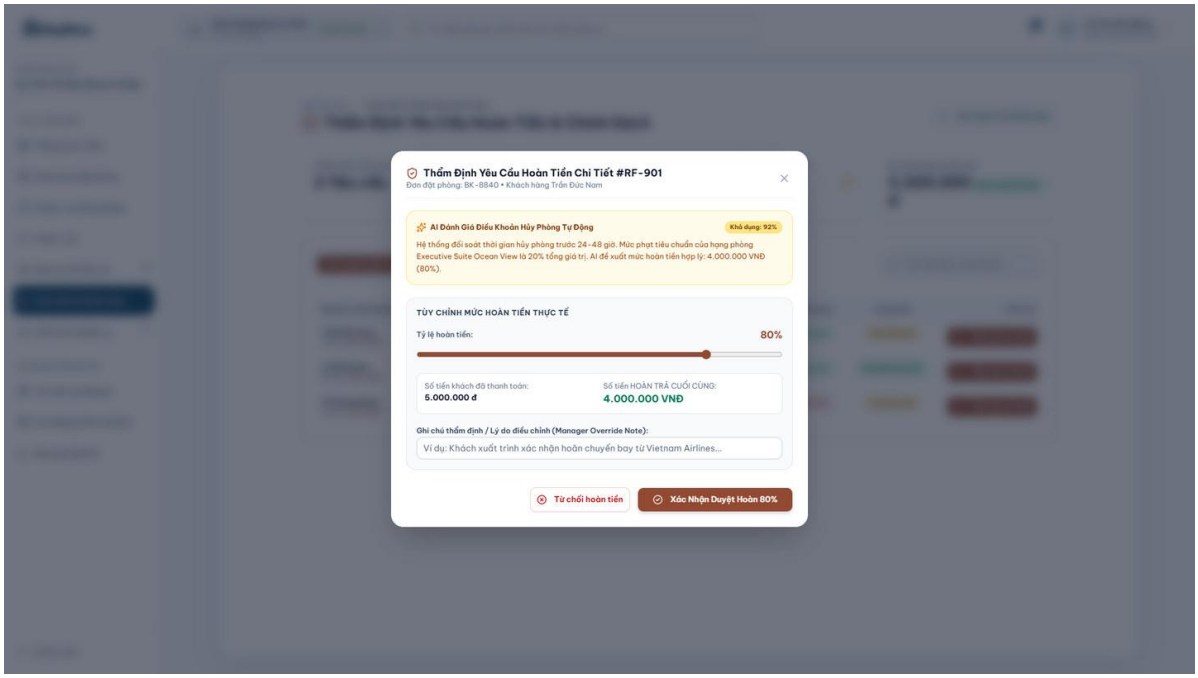


Figure 65. Refund Review screen (web)

- **Function details:**
  - The amount follows BR-08 and cannot exceed the refundable amount (MSG21).
  - Staff check the request and forward it with a note; only the hotel partner approves or rejects. Rejecting is allowed only for the reasons in BR-20 and needs a note; the reason is logged and sent to the customer. A request made before the check-in time and still undecided at that time is approved automatically.
  - An approved refund is moved from the pending balance of the hotel wallet to the customer wallet, and the customer sees the refund status (UC35).

3.8.4 Alerts List

- **Function trigger:** Menu "Alerts", or the alert block on Staff Dashboard.
- **Function description:** Actor: Hotel Staff. Purpose: handle unusual bookings and negative reviews (UC42). The queue shows type, severity, evidence, status and the resolution note.
- **Screen layout: Figure 66 (web).**

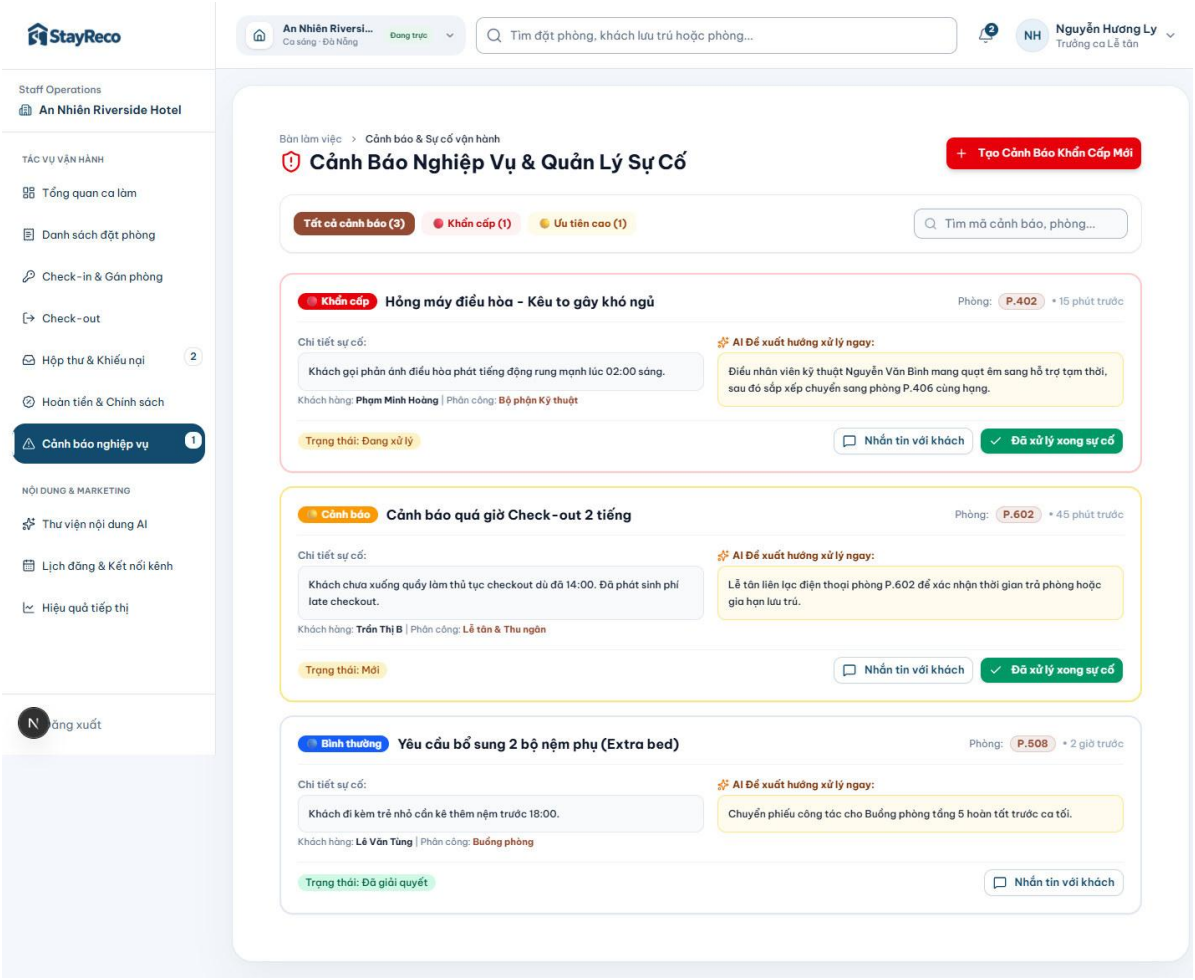


Figure 66. Alerts List screen (web)

- **Function details:**
  - A staff member takes an alert, then closes it with a resolution note.

3.8.5 Alert Detection Job

- **Function trigger:** Runs after each new booking and each new review.
- **Function description:** Non-screen function. Purpose: raise alerts for unusual booking patterns and negative reviews.
- **Screen layout:** *not applicable, this function runs without a screen.*
- **Function details:**
  - Examples: many bookings from one account in a short time, repeated failed payments, a rating below the threshold set for the hotel.

3.9 AI Marketing Support

3.9.1 Content Library

- **Function trigger:** Menu "Marketing".
- **Function description: Actors: Hotel Staff; Hotel Partner (reviews content waiting for approval). Purpose: manage the marketing content of the hotel. The tabs are Drafts, Pending Review, Approved, Rejected, Scheduled and Published; the Rejected tab shows each rejected item with the feedback of the hotel partner.**
- **Screen layout: Figure 67 (web).**

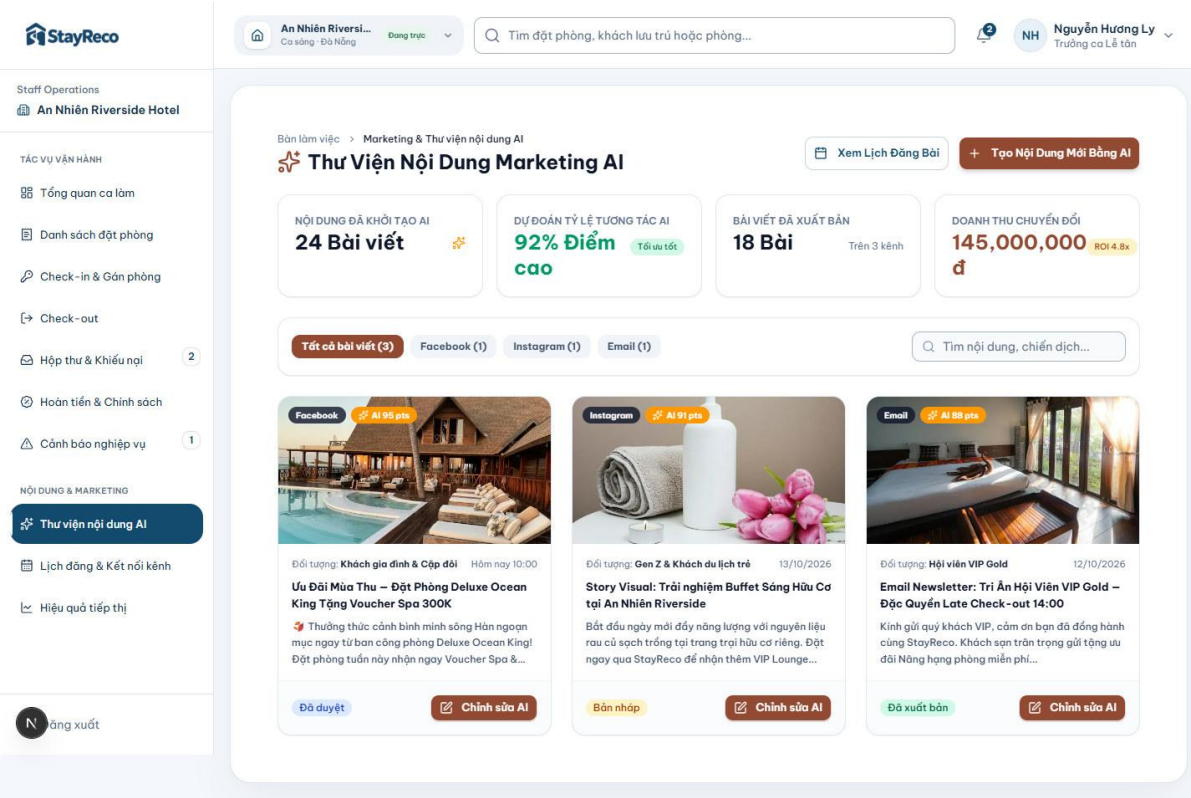


Figure 67. Content Library screen (web)

- **Function details:**
  - Each item shows the type (caption, blog post or promotion), channel, author and last update.

3.9.2 Content Editor

- **Function trigger:** "New content" or "New blog post" in Content Library, or click on a draft.
- **Function description:** Actor: Hotel Staff. Purpose: create content with AI help (UC44, UC45, UC46, UC47). The staff member fills a short brief, the AI writes a draft with the brand settings of the hotel, the staff member edits it, checks the facts, previews it per channel and submits it to the hotel partner for approval (UC47). The hotel partner opens an item from the Pending Review tab in the Content Approval view of this screen, reads it as the staff member wrote it and approves it, or rejects it with feedback.
- **Screen layout: Figure 68 (web).**

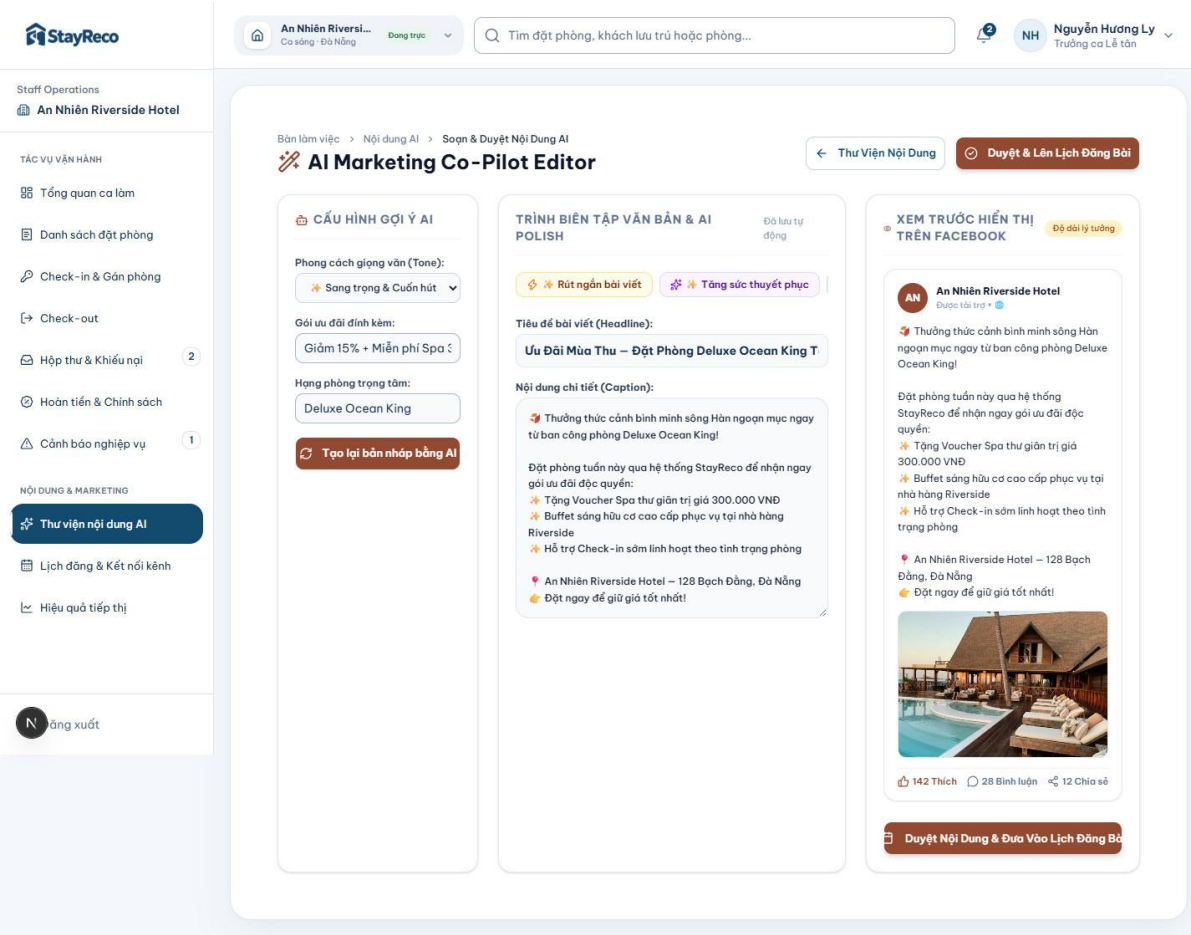


Figure 68. Content Editor screen (web)

- **Function details:**
  - AI content must be approved by the hotel partner before it is scheduled or published; a rejected draft goes back to the staff member with feedback, and editing approved content needs a new approval (BR-14).
  - An approved blog post is published on the hotel blog. A rejected item moves to the Rejected tab with the feedback; the staff member edits it and submits it again.
  - When the AI service is unavailable MSG24 is shown and the staff member can still write by hand.

3.9.3 Publishing Calendar

- **Function trigger:** Menu "Calendar".
- **Function description:** Actor: Hotel Staff. Purpose: schedule approved content and connect social accounts (UC48, UC49). Views by month, week and list; a drawer connects or reconnects supported social media accounts.
- **Screen layout: Figure 69 (web).**

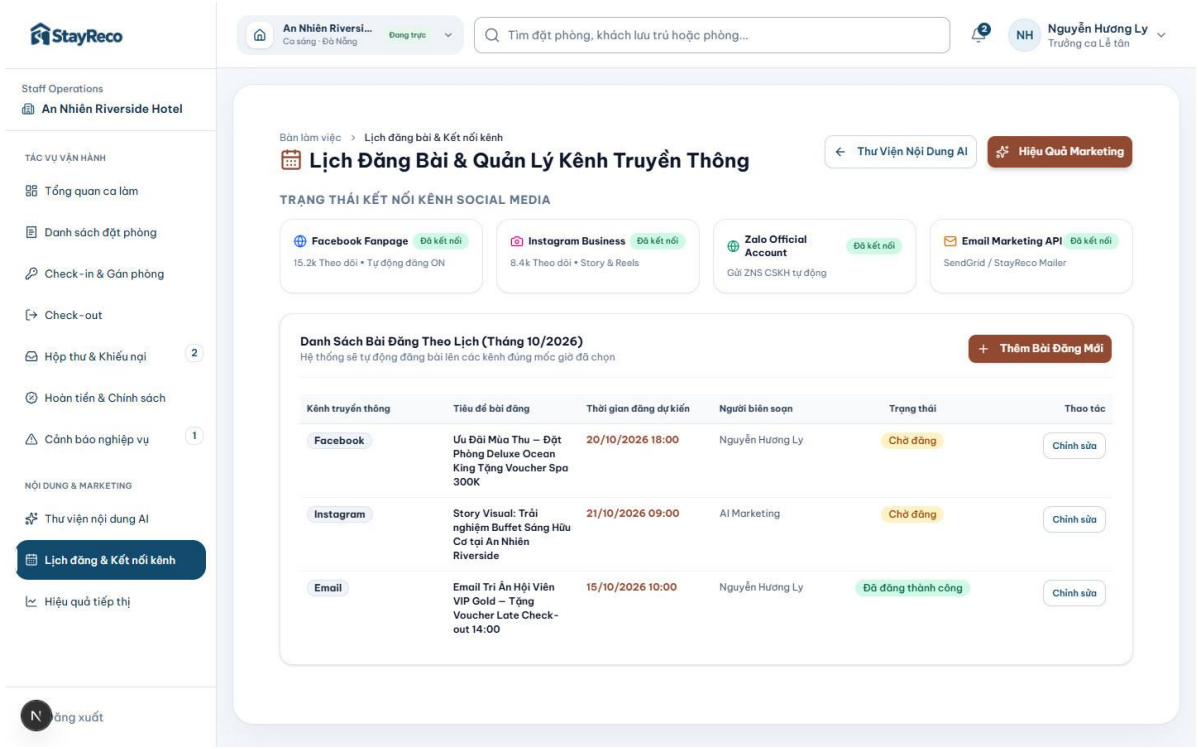


Figure 69. Publishing Calendar screen (web)

- **Function details:**
  - Only approved content can be scheduled (MSG22).
  - An expired social connection is marked and the scheduled post waits until it is reconnected.

3.9.4 Content Performance

- **Function trigger:** Menu "Performance".
- **Function description:** Actor: Hotel Staff. Purpose: see the result of published content (UC50): reach, engagement and attributed bookings, filtered by date and channel.
- **Screen layout: Figure 70 (web).**

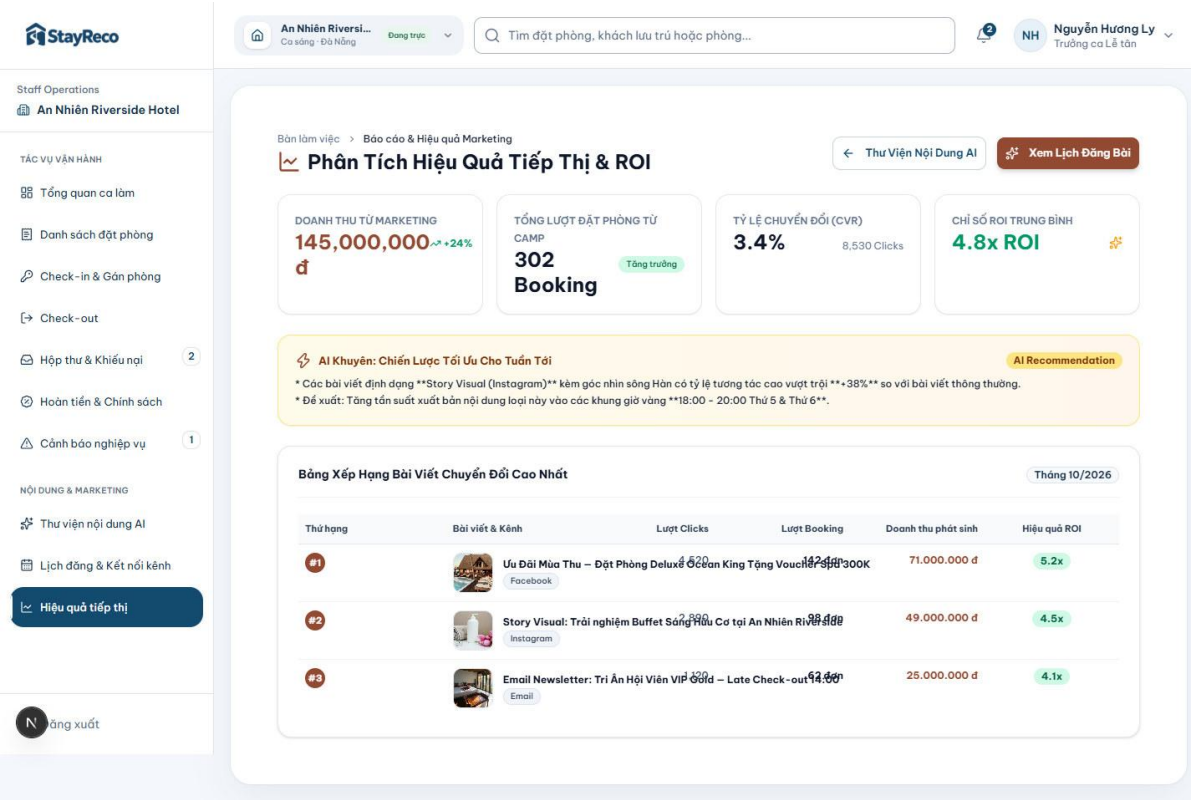


Figure 70. Content Performance screen (web)

- **Function details:**
  - Bookings are attributed to a post through its tracking link.

3.9.5 Content Generation Service

- **Function trigger:** Called by Content Editor.
- **Function description:** Non-screen function. Purpose: write caption, blog and promotion drafts from the brief and the brand settings of the hotel.
- **Screen layout:** *not applicable, this function runs without a screen.*
- **Function details:**
  - Drafts are always saved in Drafts and never published directly.

3.9.6 Scheduled Publisher

- **Function trigger:** Runs every few minutes.
- **Function description:** Non-screen function. Purpose: publish approved content at the scheduled time and collect metrics.
- **Screen layout:** *not applicable, this function runs without a screen.*
- **Function details:**
  - A failed publication is retried and then marked failed with the reason.

3.10 Partner Onboarding

3.10.1 Partner Sign-up

- **Function trigger:** The link "Register as a partner" on User Register or on the footer of the public site.
- **Function description:** Actor: Guest (hotel owner). Purpose: create a partner account (UC02). The form has full name, email, phone number, password, confirm password and company

name (optional). After sign-up the owner enters the email code and then goes straight to Partner Onboarding.

- **Screen layout: Figure 71 (web).**

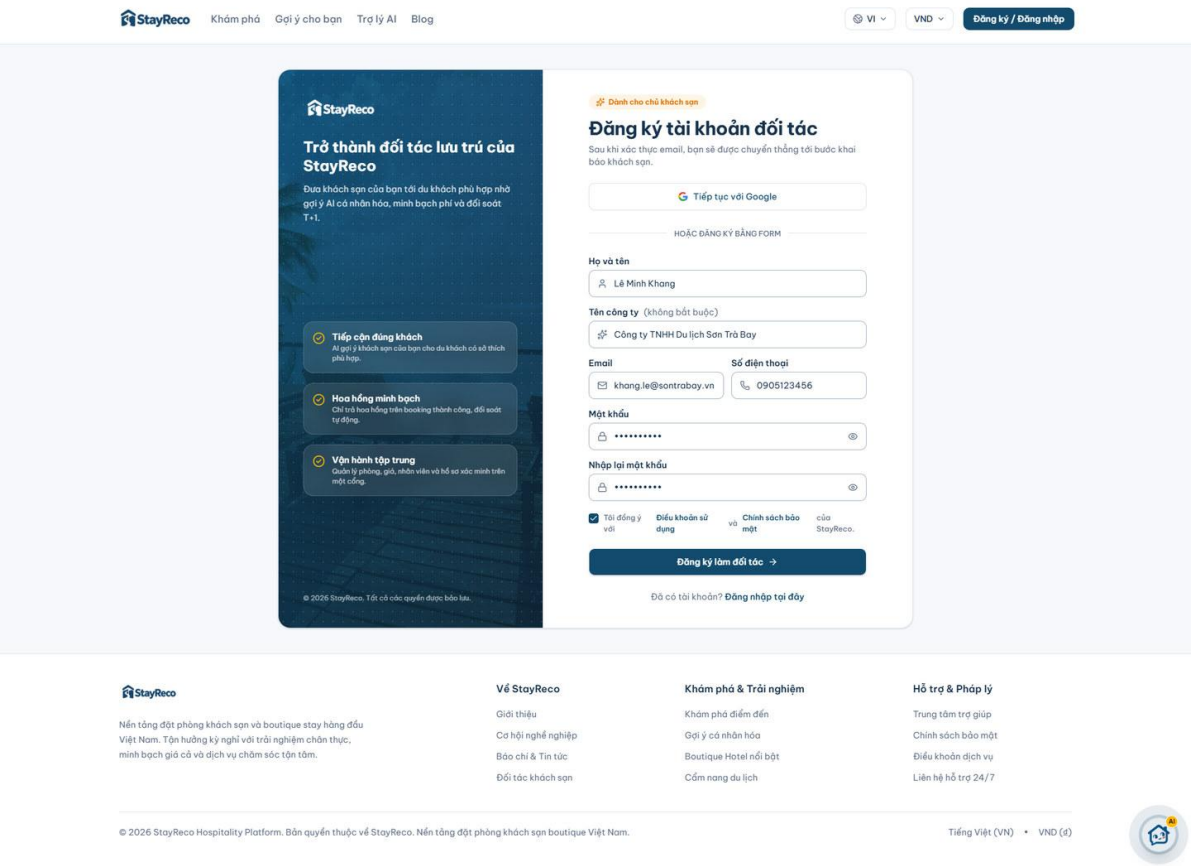


Figure 71. Partner Sign-up screen (web)

- **Function details:**
  - Phone: required, 9 to 15 digits with an optional leading "+". Company name: optional, at most 200 characters.
  - Email and password follow the same rules as User Register (MSG04, MSG07, MSG08).
  - The account can sign in only after the email is verified (BR-19, BR-23).

3.10.2 Partner Onboarding

- **Function trigger:** Opened after the first sign-in of a new partner, or "Add hotel" on My Hotels.
- **Function description:** Actor: Hotel Partner. Purpose: declare a hotel for verification (UC52, UC53, UC55). Four steps: legal entity, accommodation, documents, and confirmation of commitments. The legal entity step holds the legal name, tax code, business type, legal representative (full name, position, citizen ID number), phone, notification email and the beneficiary details (bank, account number, account holder); it is prefilled for later hotels. The accommodation step holds the hotel name, address, contact, star class and short description. A draft is saved automatically.
- **Screen layout: Figure 72 (web).**

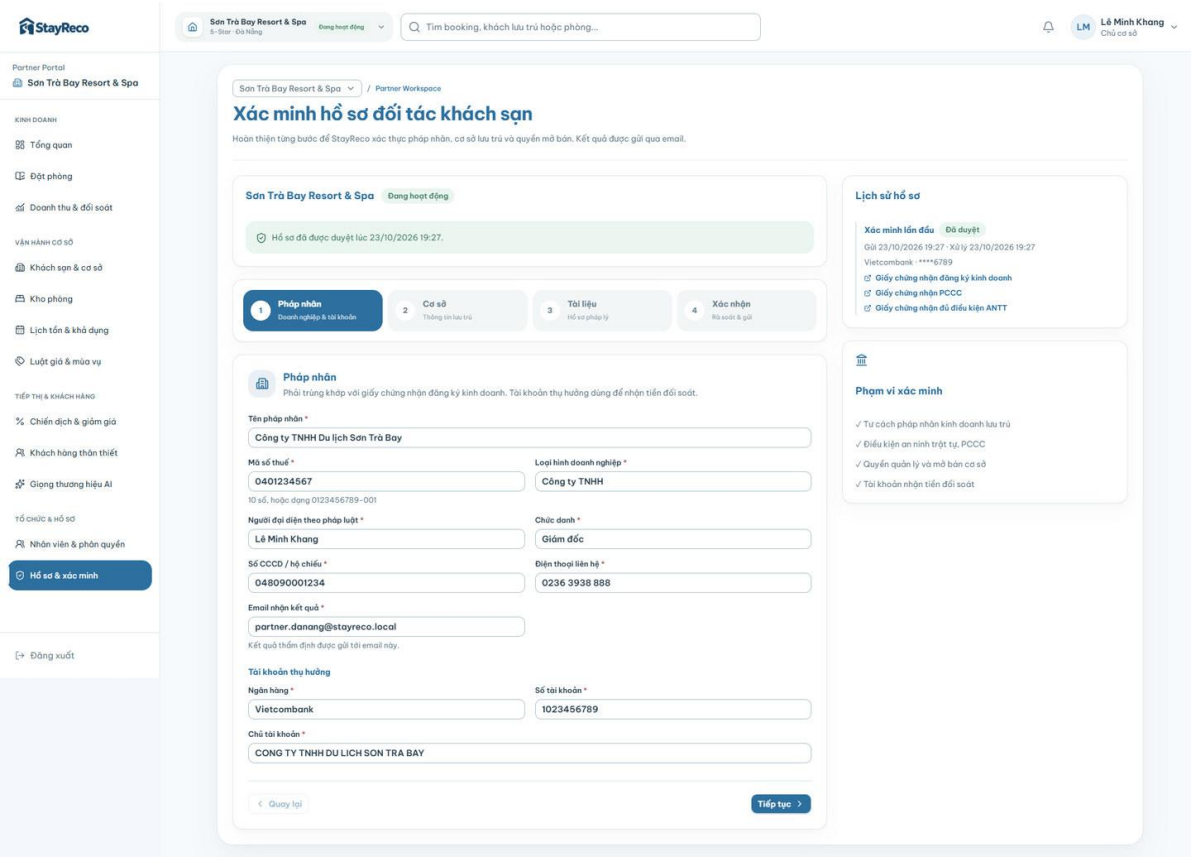


Figure 72. Partner Onboarding screen (web)

- **Function details:**
  - Required fields are checked at each step (MSG02); the partner cannot go to the next step with missing data.
  - Required documents: business license, fire safety (PCCC) certificate and security and order (ANTT) certificate; the star rating certificate and other documents are optional. Files are PDF, JPG or PNG up to 10 MB (MSG23), and only the hotel owner and the platform manager can open them (BR-26).
  - The beneficiary account number is encrypted when stored and shown masked in lists.
  - Submitting sends the hotel to Pending review; the hotel stays hidden from travellers until it is approved (BR-01, BR-19).

3.10.3 Verification Status

- **Function trigger:** Menu "Verification", or the status badge on My Hotels.
- **Function description:** Actor: Hotel Partner. Purpose: follow the review of each hotel (UC54). The screen shows the status (draft, pending review, changes requested, approved, rejected), the timeline, the reasons when changes are requested and the "Resubmit" action; resubmitting creates a new request and keeps the history of earlier ones.
- **Screen layout: Figure 73 (web).**

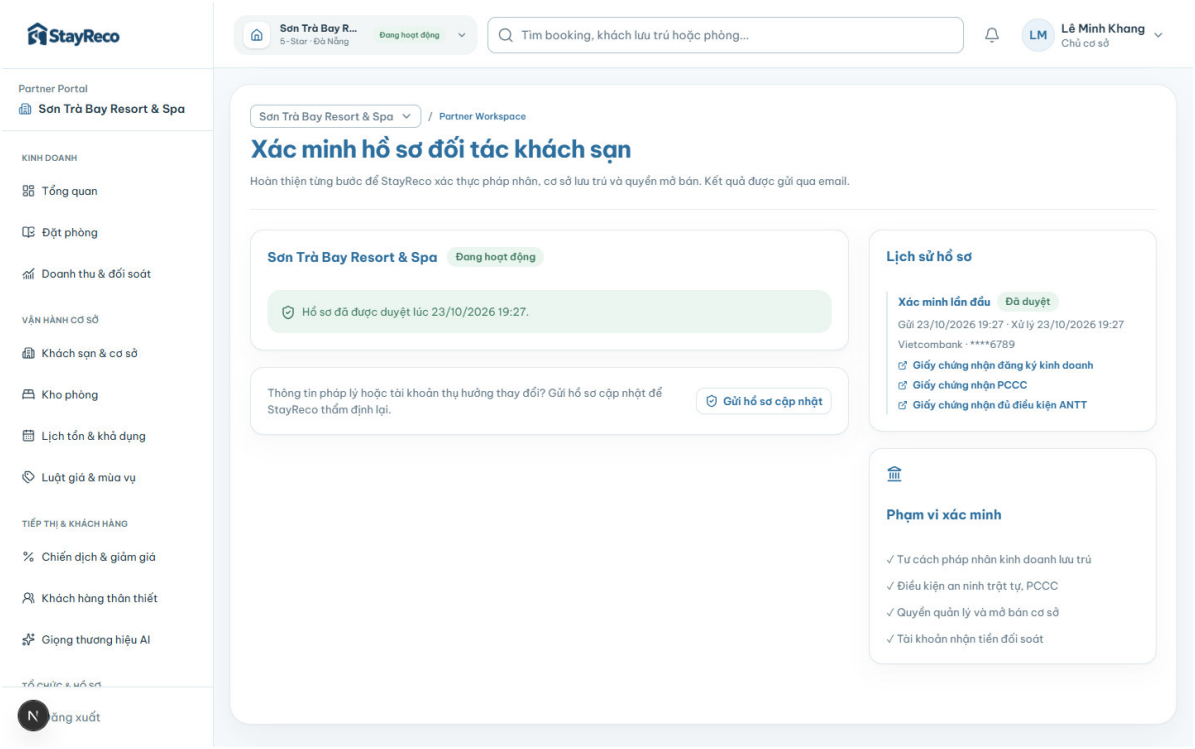


Figure 73. Verification Status screen (web)

- **Function details:**
  - While a hotel is not approved, the partner can only edit its onboarding information and follow this status (MSG27).
  - Each status change sends a notification and an email to the partner.

3.11 Hotel & Room Management

3.11.1 My Hotels

- **Function trigger:** Menu "My hotels", or the hotel switcher on the header of the partner workspace.
- **Function description:** Actor: Hotel Partner. Purpose: list the hotels of the partner and choose the one to work on (UC55). Each card shows the hotel name, verification status and listing status, with the actions "Open" and "Add hotel".
- **Screen layout: Figure 74 (web).**

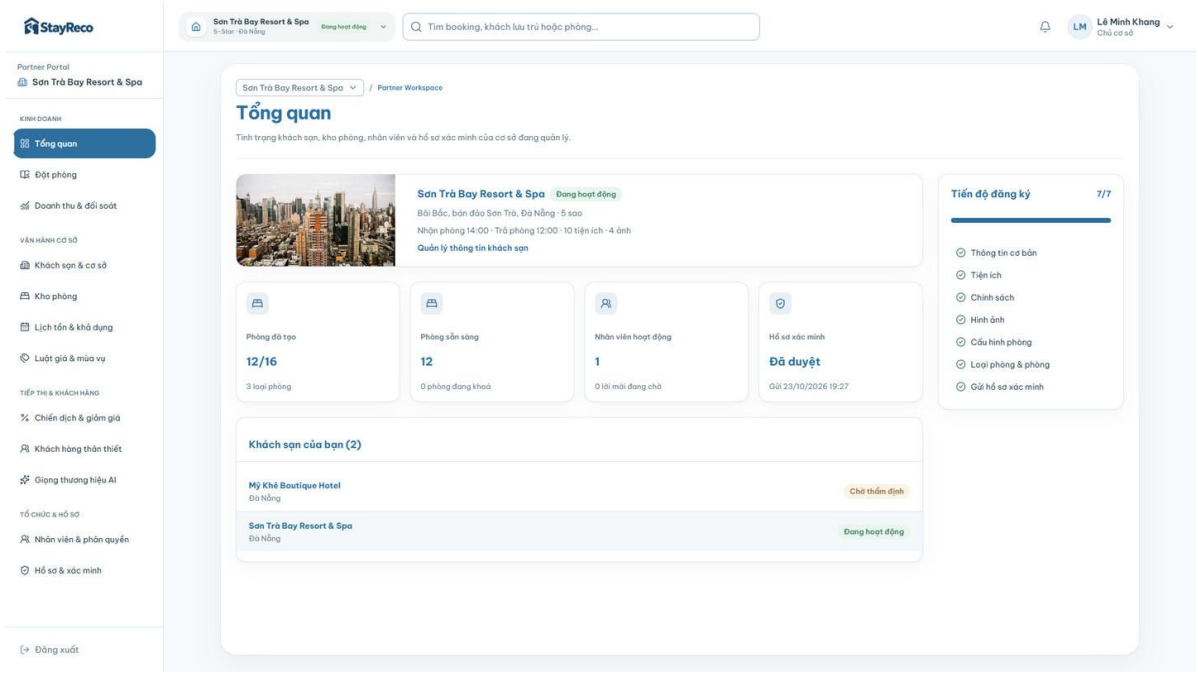


Figure 74. My Hotels screen (web)

- **Function details:**
    - Every screen of the partner workspace works on the selected hotel only (BR-21).
    - "Add hotel" opens Partner Onboarding for a new hotel.

3.11.2 Partner Dashboard

- **Function trigger:** Opened after a partner logs in, or menu "Dashboard".
- **Function description:** Actor: Hotel Partner. Purpose: follow the business of the hotel (UC56). The dashboard shows bookings, occupancy, revenue, booking sources and satisfaction for a chosen period, for the selected hotel or for all hotels together.
- **Screen layout: Figure 75 (web).**

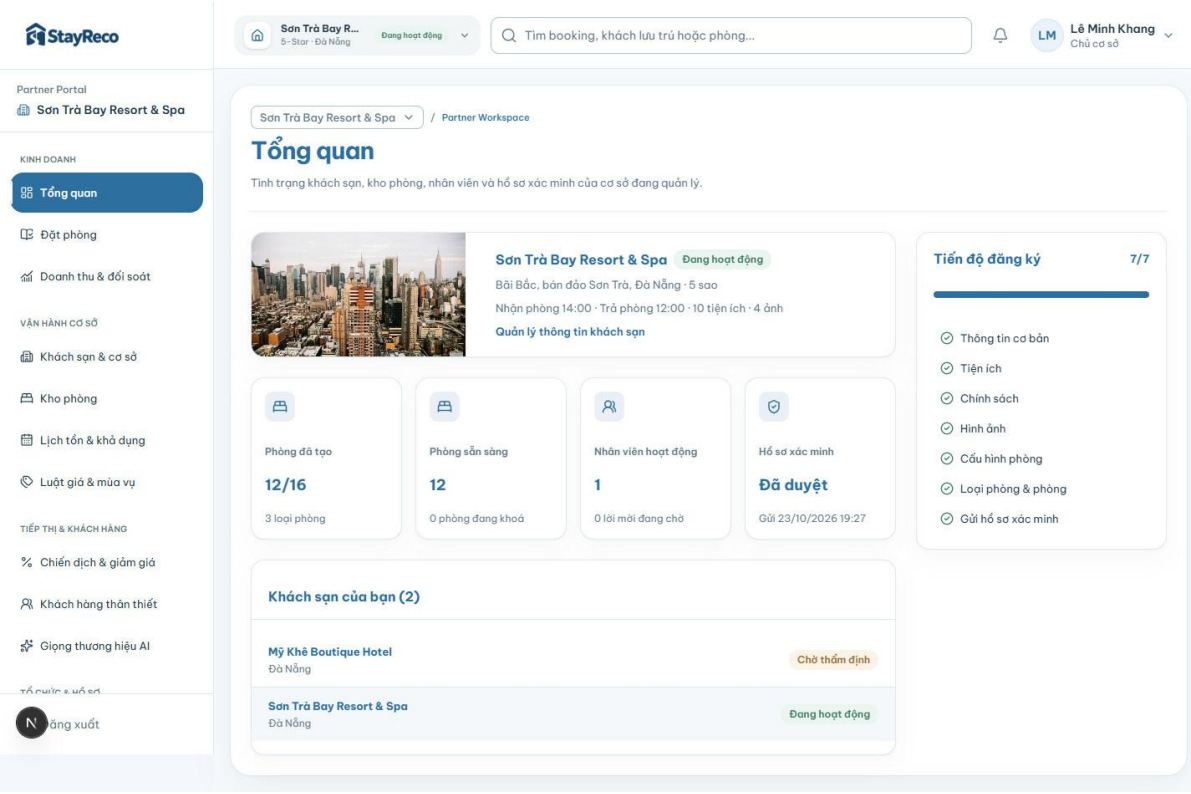


Figure 75. Partner Dashboard screen (web)

- **Function details:**
  - The default period is the current month to date, compared with the same number of days just before it; the end date cannot be before the start date.
  - Revenue is the hotel share after commission.

3.11.3 Hotel Settings

- **Function trigger:** Menu "Hotel".
- **Function description:** Actor: Hotel Partner. Purpose: update what travellers see about the hotel (UC60). Tabs: General Info (name, address, contact, description, amenities), Images (several photos in order, exactly one cover photo) and Policies (check-in and check-out times, cancellation policy, children and pets).
- **Screen layout: Figure 76 (web).**

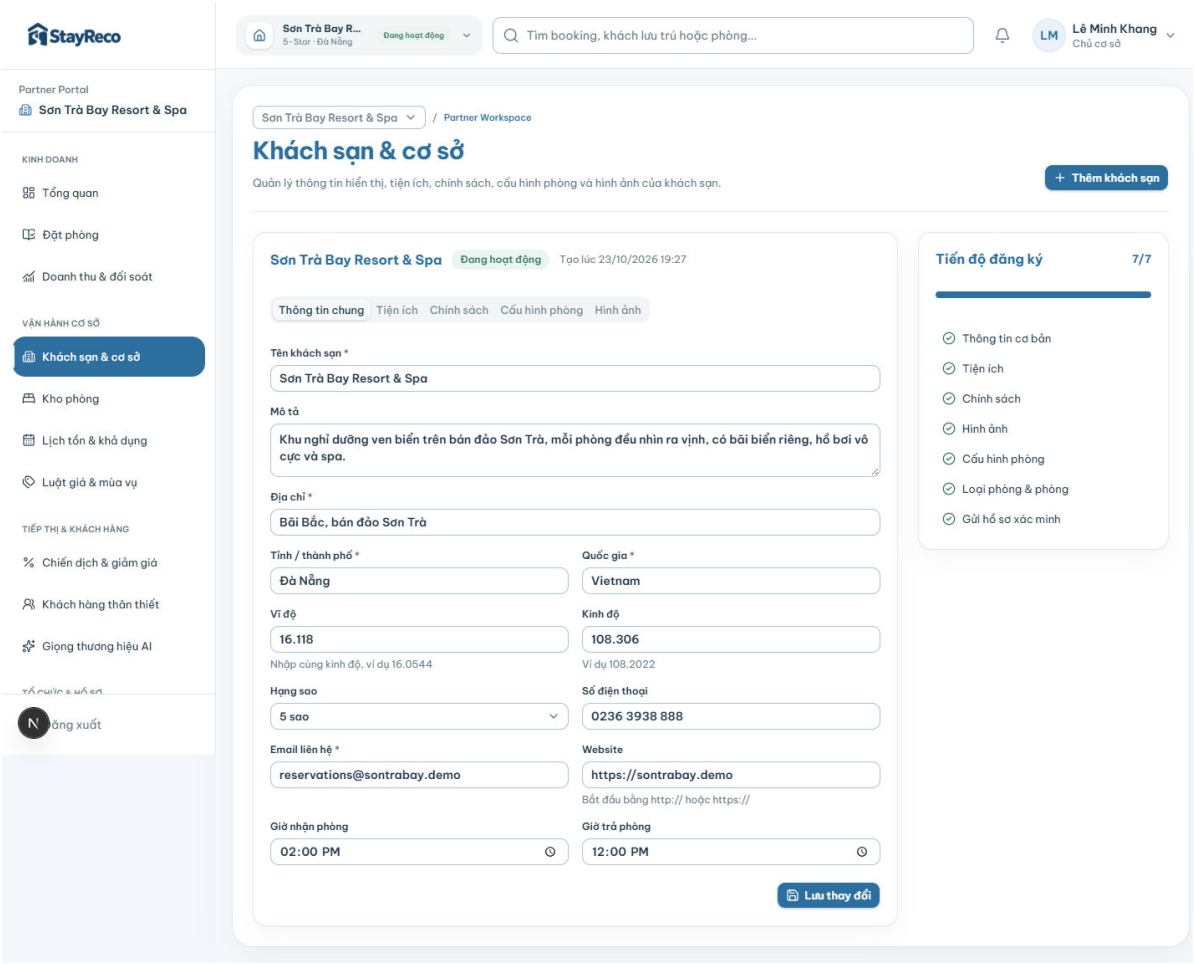


Figure 76. Hotel Settings screen (web)

- **Function details:**
  - Images follow the upload rules (MSG23); they are compressed and thumbnails are created automatically.
  - A new cancellation policy applies only to bookings made after the change (BR-07).
  - Saving shows MSG10.
  - The hotel keeps a room configuration: the declared total number of rooms and the default base price for new room types (BR-27). Editing an active hotel does not need a new approval (BR-19).

3.11.4 Room Inventory

- **Function trigger:** Menu "Rooms".
- **Function description:** Actor: Hotel Partner. Purpose: see room types and physical rooms (UC61, UC62). Two tabs, Room Types and Rooms, with search and status filters and the "Add" action in each tab.
- **Screen layout: Figure 77 (web).**

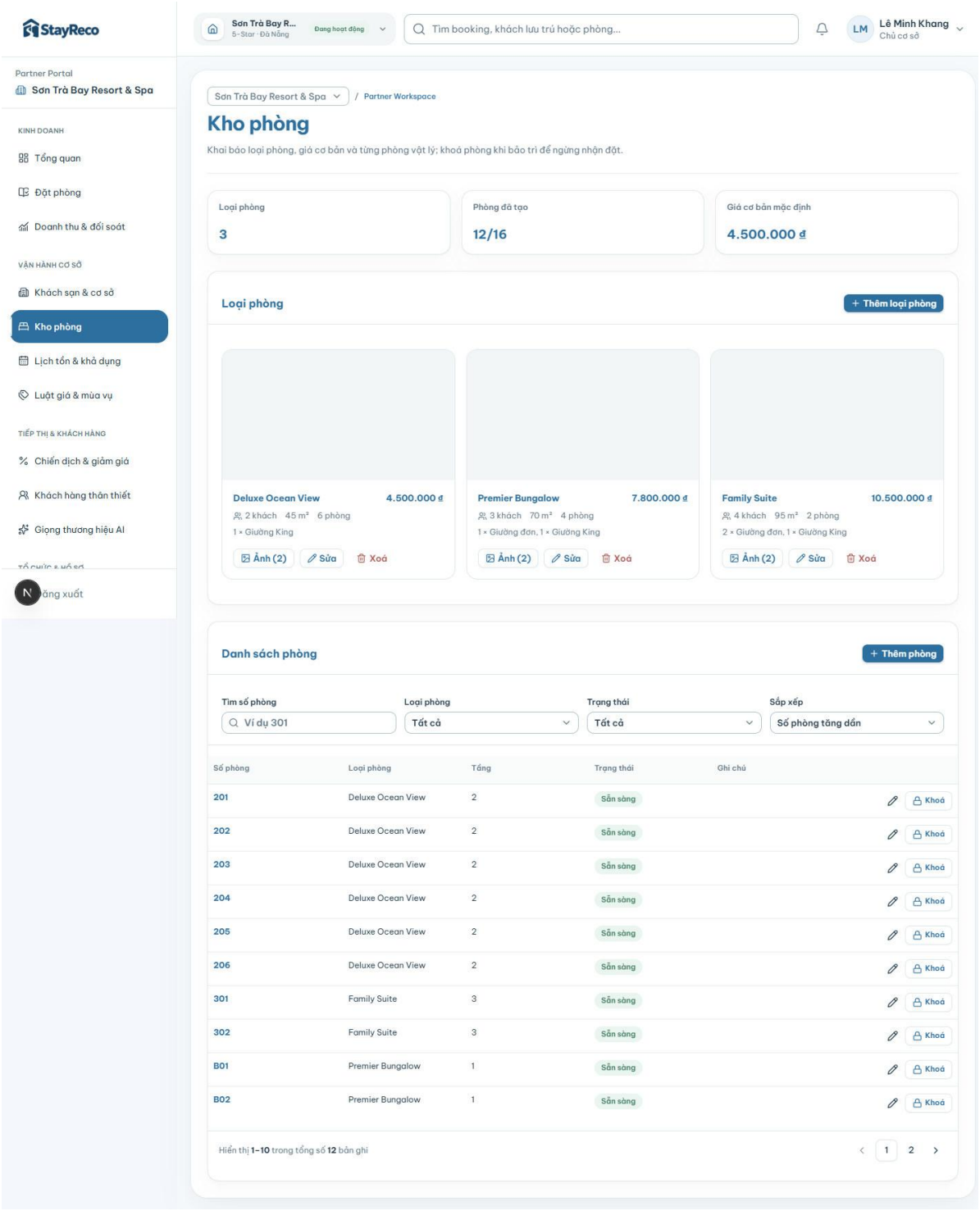


Figure 77. Room Inventory screen (web)

- **Function details:**
  - A room type that still has physical rooms cannot be deleted; it can only be hidden from sale (BR-27).
  - The number of physical rooms cannot exceed the total declared in the room configuration of the hotel.

3.11.5 Room Type Editor

- **Function trigger:** "Add room type" or click on a room type in Room Inventory.

- **Function description:** Actor: Hotel Partner. Purpose: create or edit a room type (UC61). Fields: name, base price, maximum occupancy, area, bed types (several types, each with a quantity), amenities and photos.
- **Screen layout: Figure 78 (web).**

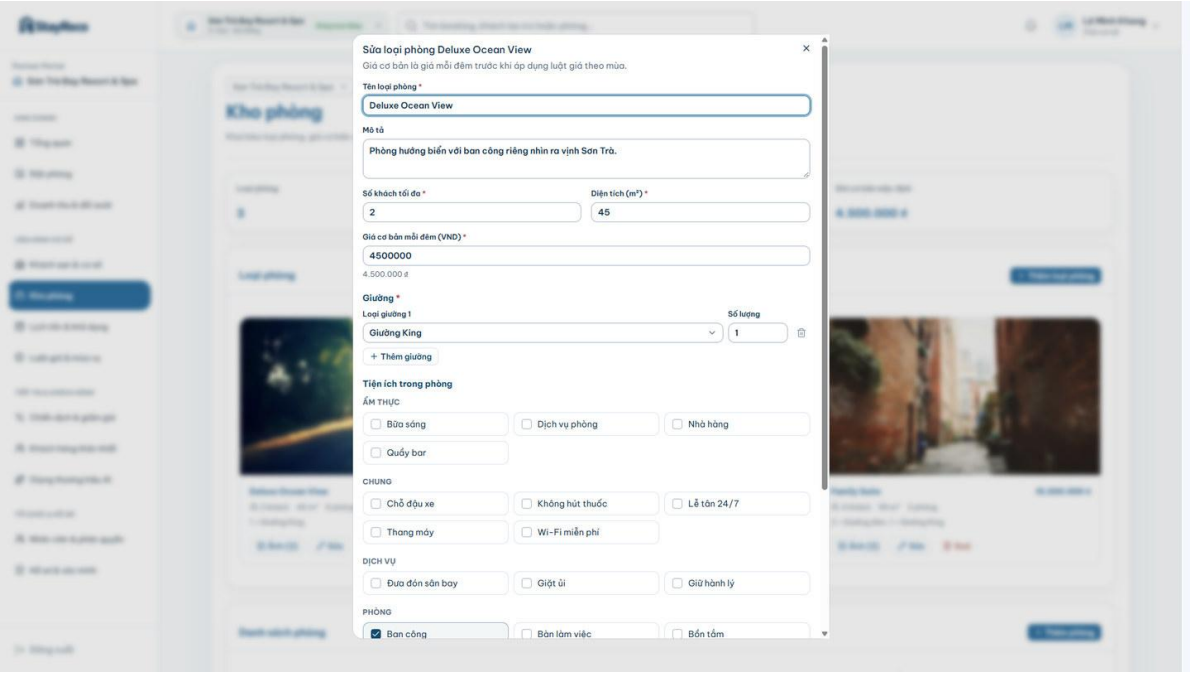


Figure 78. Room Type Editor screen (web)

- **Function details:**
  - Name, occupancy (1 to 20) and base price (greater than 0) are required.
  - A change of base price applies to new bookings only.

3.11.6 Room Editor

- **Function trigger:** "Add room" or click on a room in Room Inventory.
- **Function description:** Actor: Hotel Partner. Purpose: create or edit a physical room (UC62). Fields: room number, floor, room type and status, with the Lock and Unlock actions.
- **Screen layout: Figure 79 (web).**

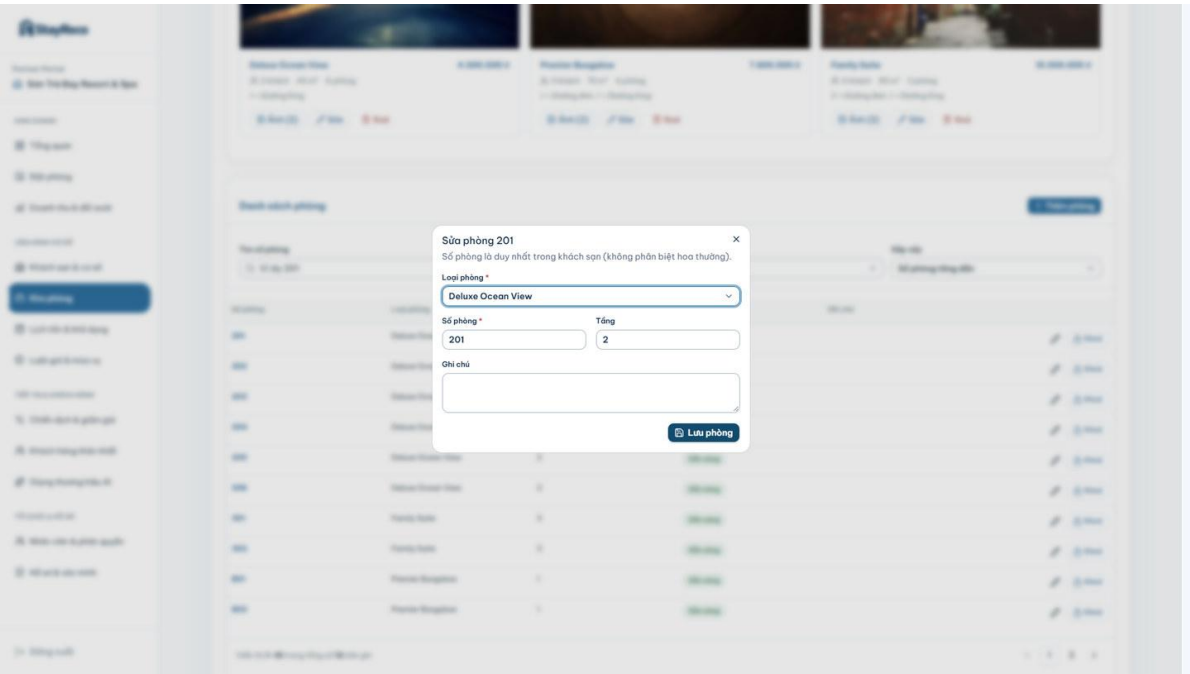


Figure 79. Room Editor screen (web)

- **Function details:**
  - The room number must be unique within the hotel.
  - Only an AVAILABLE room can be booked; MAINTENANCE, CLEANING and OCCUPIED rooms are not ready. Lock moves the room to MAINTENANCE and Unlock returns it to AVAILABLE; a room with a guest cannot be locked (BR-27).

3.11.7 Availability Calendar

- **Function trigger:** Menu "Availability".
- **Function description:** Actor: Hotel Partner. Purpose: manage the rooms open for sale (UC63). A date grid shows available and total rooms per room type; the partner selects dates to open, close or change the number of rooms for sale.
- **Screen layout: Figure 80 (web).**

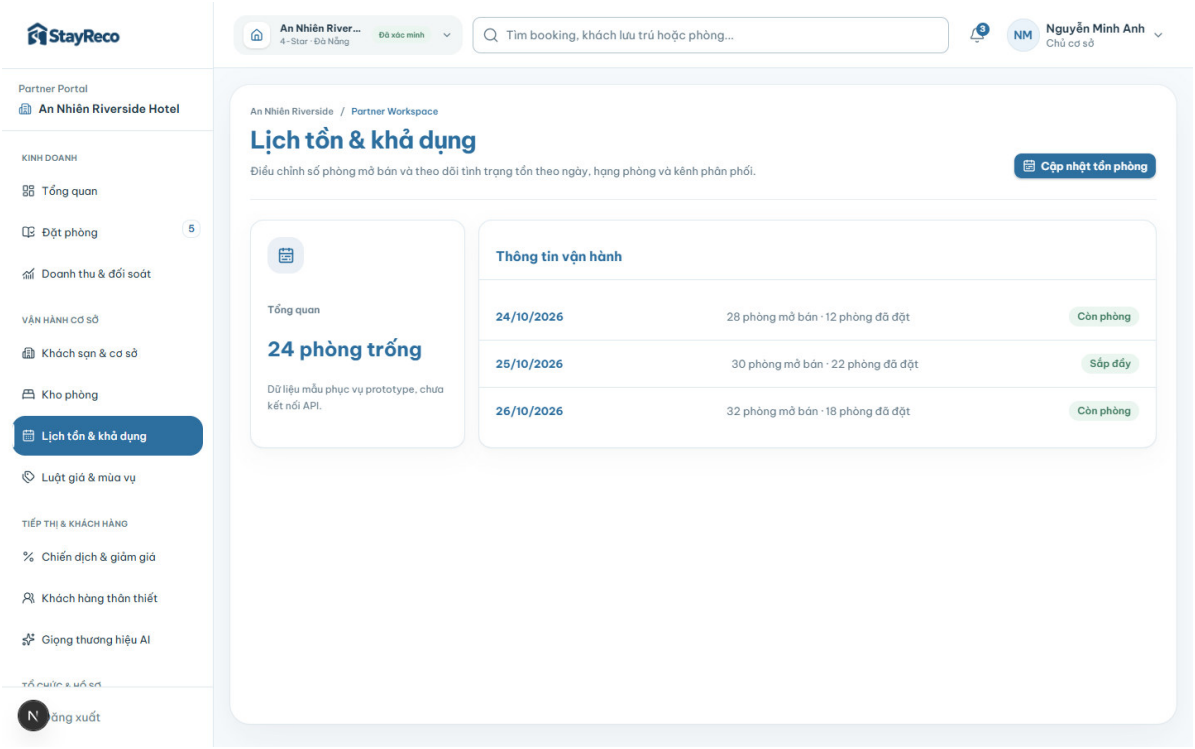


Figure 80. Availability Calendar screen (web)

- **Function details:**
  - The number for sale cannot exceed the number of active rooms of the type or go below the rooms already booked (BR-02).
  - Bookings, cancellations and holds update the grid automatically.

3.12 Pricing & Promotion

3.12.1 Pricing Rules

- **Function trigger:** Menu "Pricing".
- **Function description:** Actor: Hotel Partner. Purpose: set prices by date and by how early guests book (UC64). The table lists base-rate, seasonal and early-booking rules; a drawer edits a rule; a preview shows the resulting price for a room type and a date.
- **Screen layout: Figure 81 (web).**

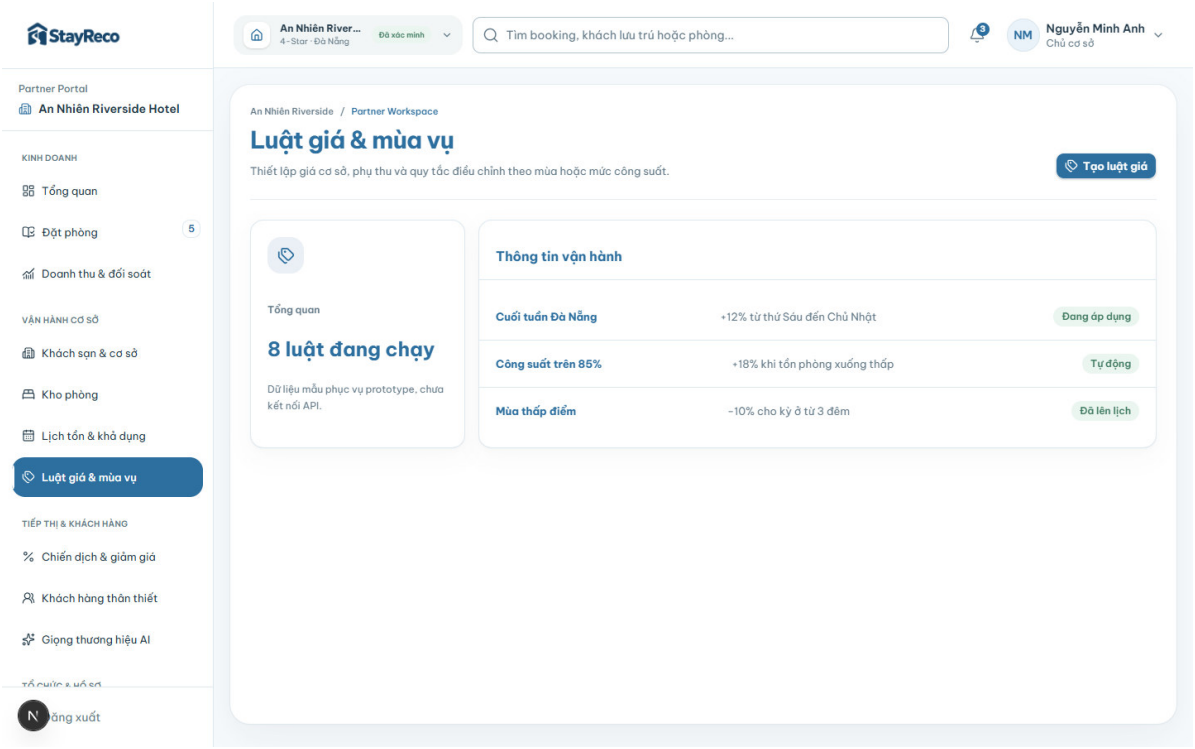


Figure 81. Pricing Rules screen (web)

- **Function details:**
  - A rule needs room types, a date range and an adjustment (fixed amount or percentage).
  - When rules overlap, the rule with the higher priority is applied and the preview shows which rule was used.
  - Prices already include VAT and service charge (BR-05).

3.12.2 Promotions

- **Function trigger:** Menu "Promotions".
- **Function description:** Actor: Hotel Partner. Purpose: create campaigns and discount codes (UC65). Fields: name, code, validity, eligible room types, discount type and value, usage limits, and an example of the resulting price.
- **Screen layout: Figure 82 (web).**

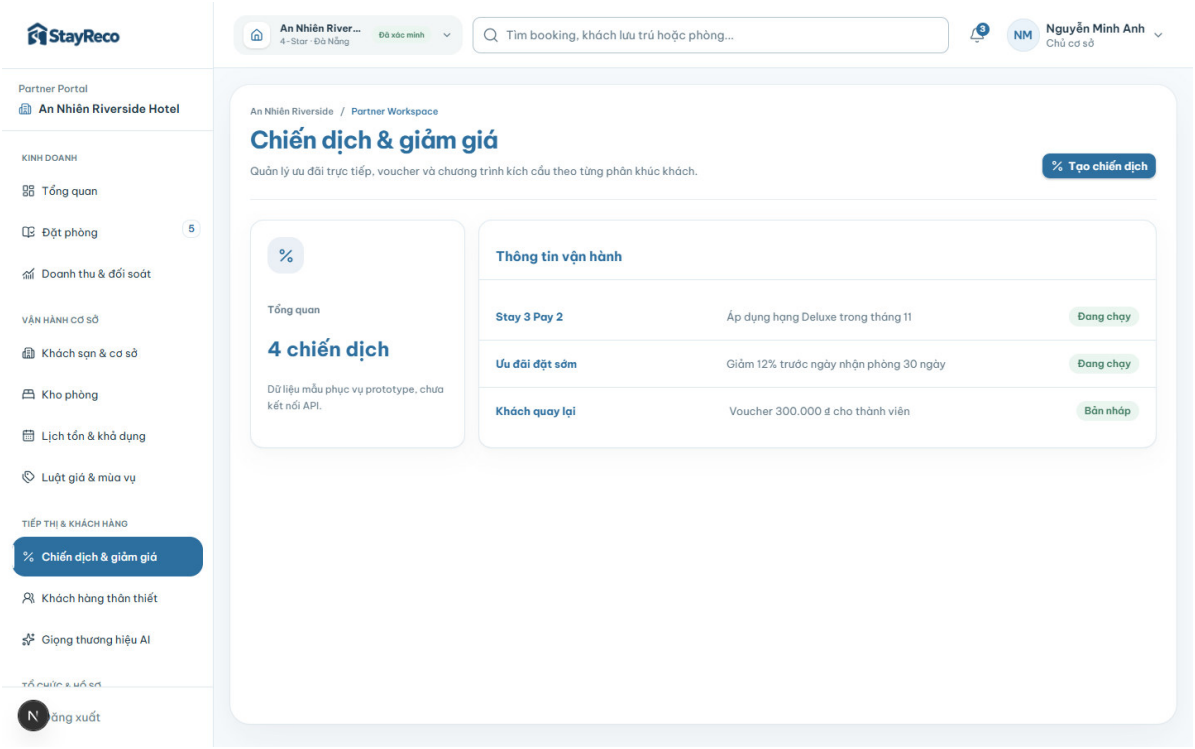


Figure 82. Promotions screen (web)

- **Function details:**
  - The code is unique within the hotel; the end date is after the start date; a percentage discount is between 1 and 100.
  - A booking can use only one code together with points (BR-06).

3.12.3 Loyalty Program

- **Function trigger:** Menu "Loyalty".
- **Function description:** Actor: Hotel Partner. Purpose: configure the loyalty program of the hotel (UC66). Tabs: Earning Rules (points per amount spent), Redemption (value of a point, maximum share of a booking payable with points, expiry) and Vouchers (issue vouchers with value, conditions and expiry).
- **Screen layout: Figure 83 (web).**

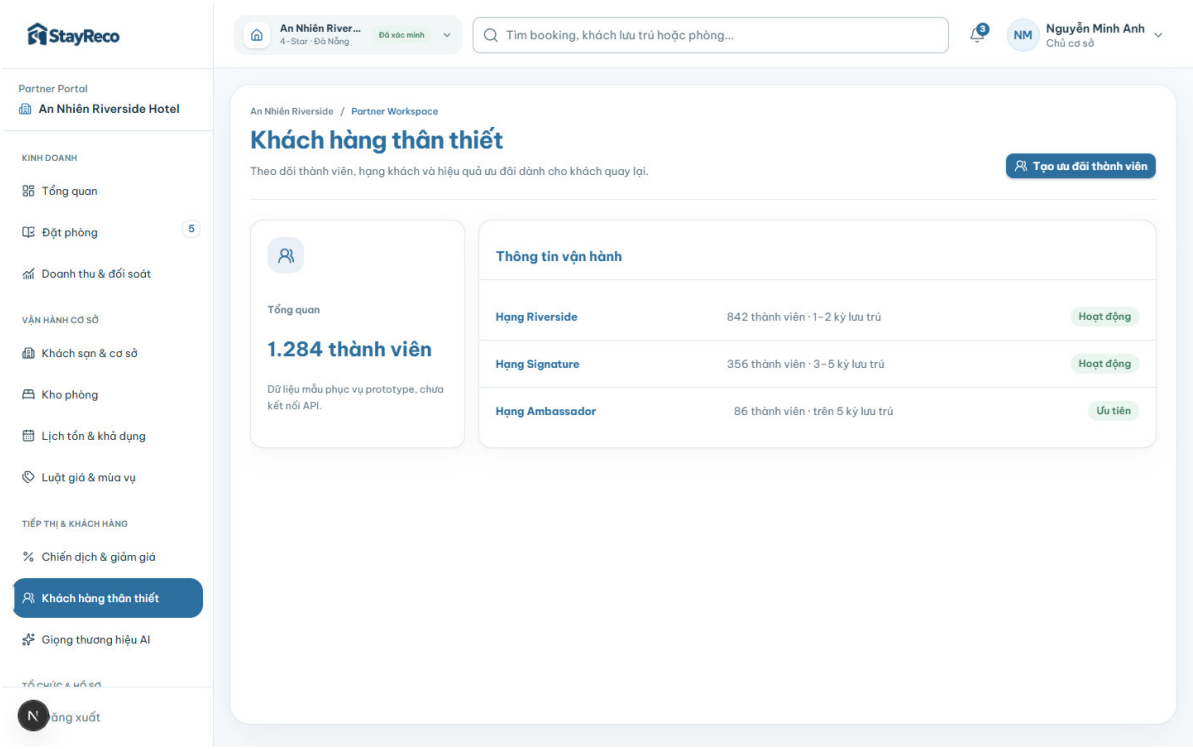


Figure 83. Loyalty Program screen (web)

- **Function details:**
  - Points are earned only after check-out (BR-11) and can be used only at this hotel (BR-06).
  - Changing the rules does not change points already earned.

3.13 Partner Administration

3.13.1 Staff List

- **Function trigger:** Menu "Staff".
- **Function description:** Actor: Hotel Partner. Purpose: manage the staff accounts of the selected hotel (UC67). The list shows name, email, position, status and the actions "Invite staff" and "Deactivate".
- **Screen layout: Figure 84 (web).**

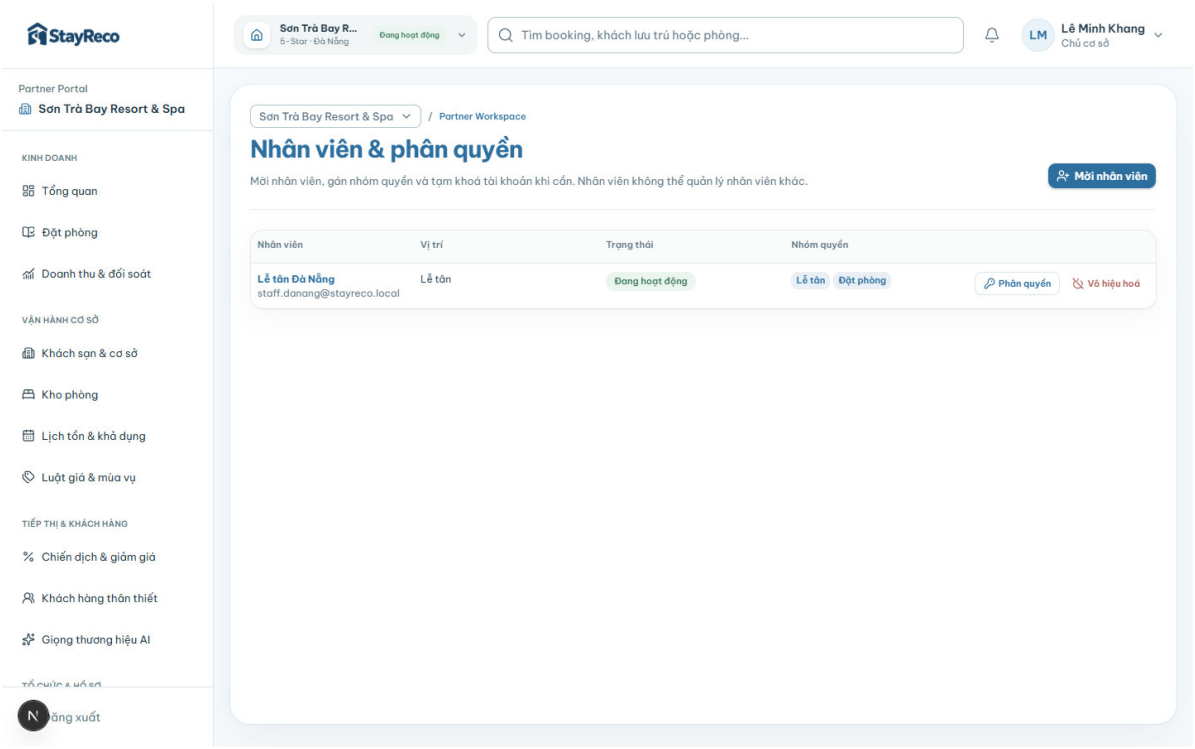


Figure 84. Staff List screen (web)

- **Function details:**
  - Inviting asks for the email, the hotel and the position, then sends an email with a link to set the password, valid for 72 hours; the account stays invited until the link is used.
  - A staff account belongs to exactly one hotel (BR-21). A partner cannot see or manage staff of another partner (BR-16).
  - Deactivating asks for confirmation (MSG11) and signs the staff member out at once.

3.13.2 Staff Permissions

- **Function trigger:** "Permissions" on a row of Staff List.
- **Function description:** Actor: Hotel Partner. Purpose: grant permission groups to a staff member (UC68). The drawer has five groups: Booking, Front desk, Customer care, Refund and Marketing.
- **Screen layout: Figure 85 (web).**

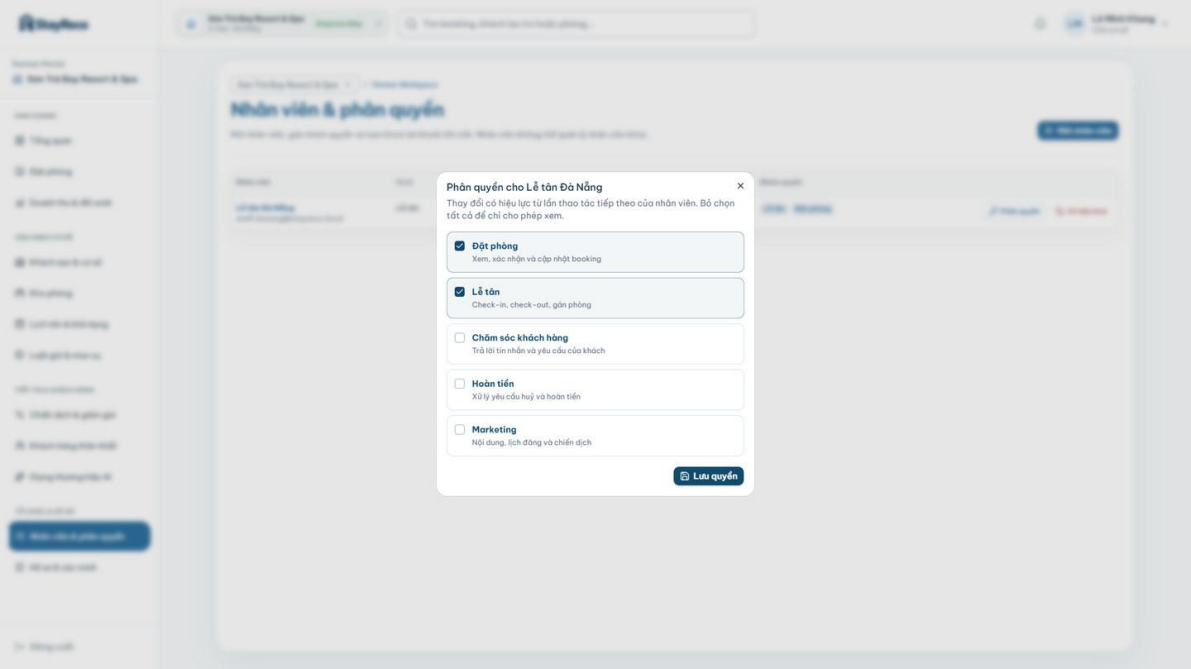


Figure 85. Staff Permissions screen (web)

- **Function details:**
  - A partner cannot grant Admin or Platform Manager permissions (BR-16).
  - A staff member without any group can only view the data of the hotel. The system has no Accountant role (BR-28).
  - Changes apply from the next request of the staff member.

3.13.3 AI Brand Settings

- **Function trigger:** Menu "AI brand".
- **Function description:** Actor: Hotel Partner. Purpose: set how the AI writes for the hotel (UC51). Fields: brand description, tone, words to use and to avoid, content templates, and a preview of sample output.
- **Screen layout: Figure 86 (web).**

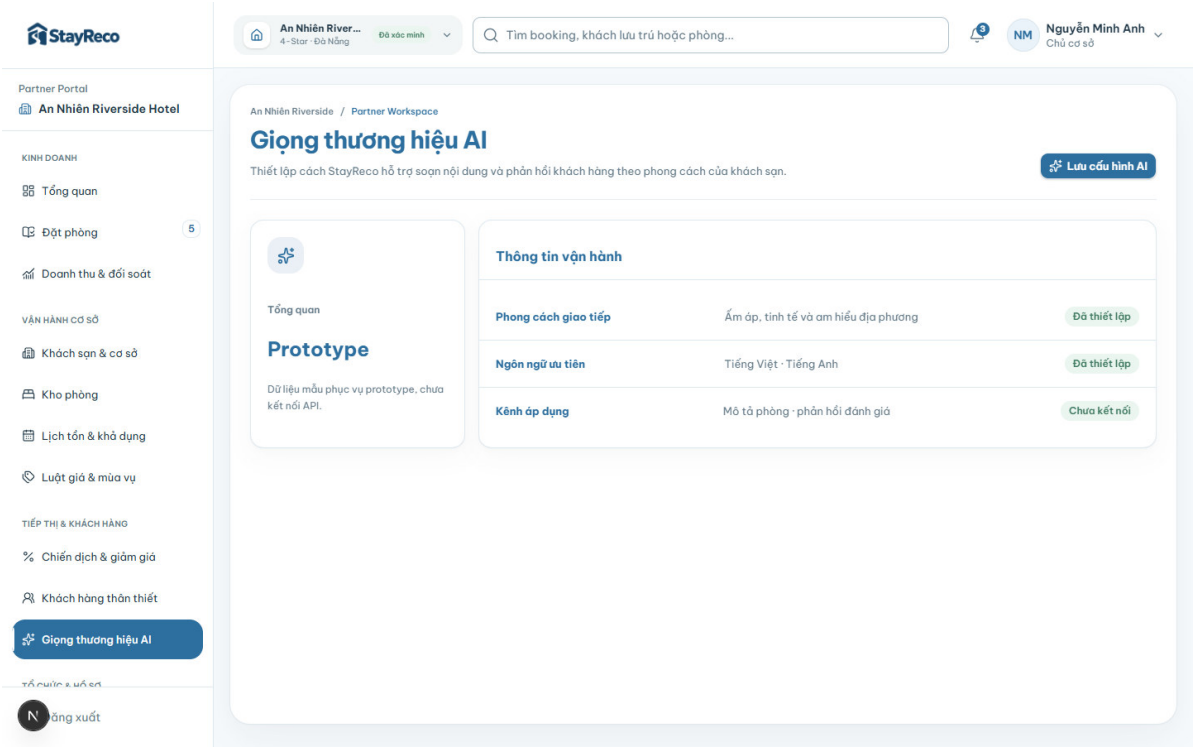


Figure 86. AI Brand Settings screen (web)

- **Function details:**
  - Saving shows MSG10; the new settings are used for the next draft.

3.13.4 Hotel Wallet

- **Function trigger:** Menu "Wallet".
- **Function description:** Actor: Hotel Partner. Purpose: see the money of the selected hotel (UC58). The screen shows the pending and available balance and the list of wallet transactions (booking payments, commission, refunds and withdrawals) with date and type filters, and the "Withdraw" button.
- **Screen layout: Figure 87 (web).**

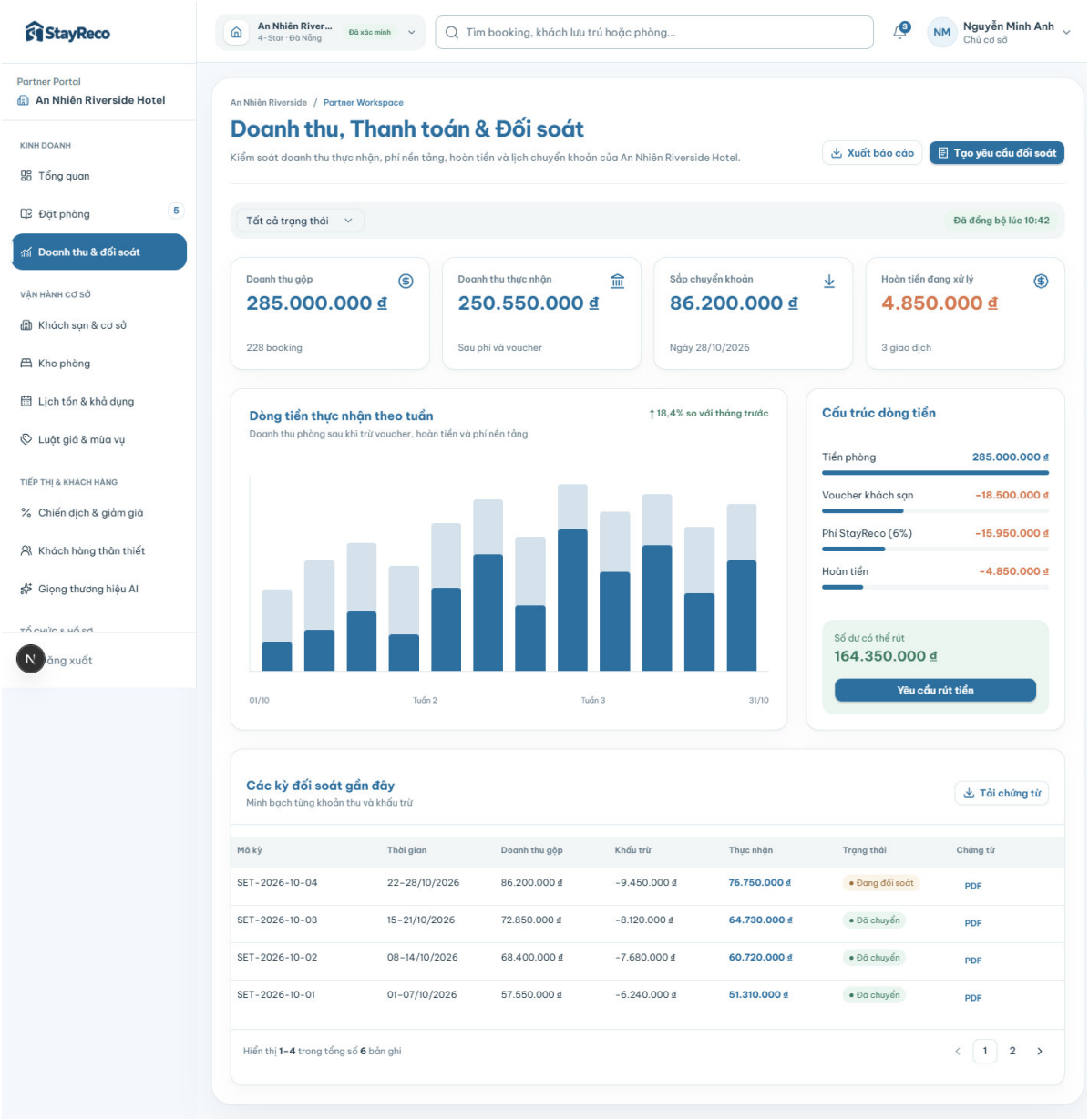


Figure 87. Hotel Wallet screen (web)

- **Function details:**
  - The hotel share of a paid booking is pending until the guest checks out and the hold period has passed, then it becomes available (BR-22).

3.13.5 Withdrawal Request

- **Function trigger:** "Withdraw" on Hotel Wallet.
- **Function description:** Actor: Hotel Partner. Purpose: withdraw money to the personal bank account of the partner (UC59). The pop-up shows the available balance, the amount field and the beneficiary account declared in the legal entity step (shown masked), followed by the status of earlier requests.
- **Screen layout: Figure 88 (web).**

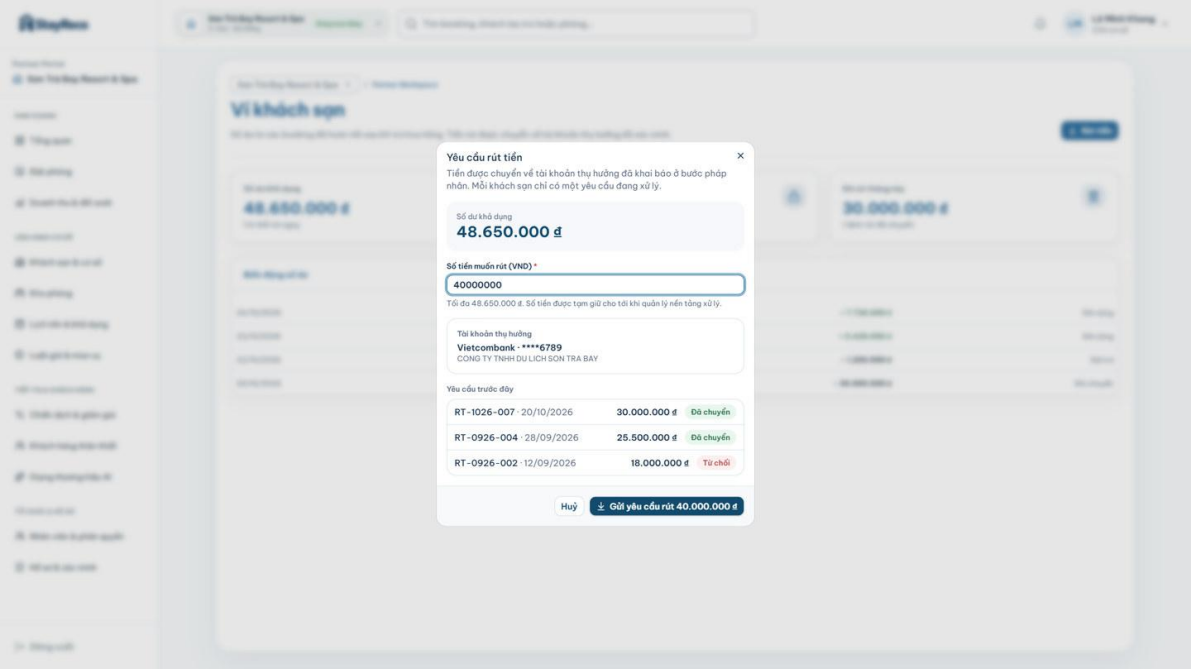


Figure 88. Withdrawal Request screen (web)

- **Function details:**
  - The amount must be greater than 0 and not more than the available balance (MSG30).
  - Each hotel can have only one open request; the amount is held until the platform manager processes it (BR-22).
  - The request is paid to the saved bank account only; changing the bank account needs a new verification.

3.14 Platform Operations

3.14.1 Manager Dashboard

- **Function trigger:** Opened after a platform manager logs in.
- **Function description:** Actor: Platform Manager. Purpose: follow the platform (UC69). The dashboard shows pending applications, active partners, bookings, commission revenue, customer growth and a short action queue (applications and withdrawals waiting).
- **Screen layout: Figure 89 (web).**

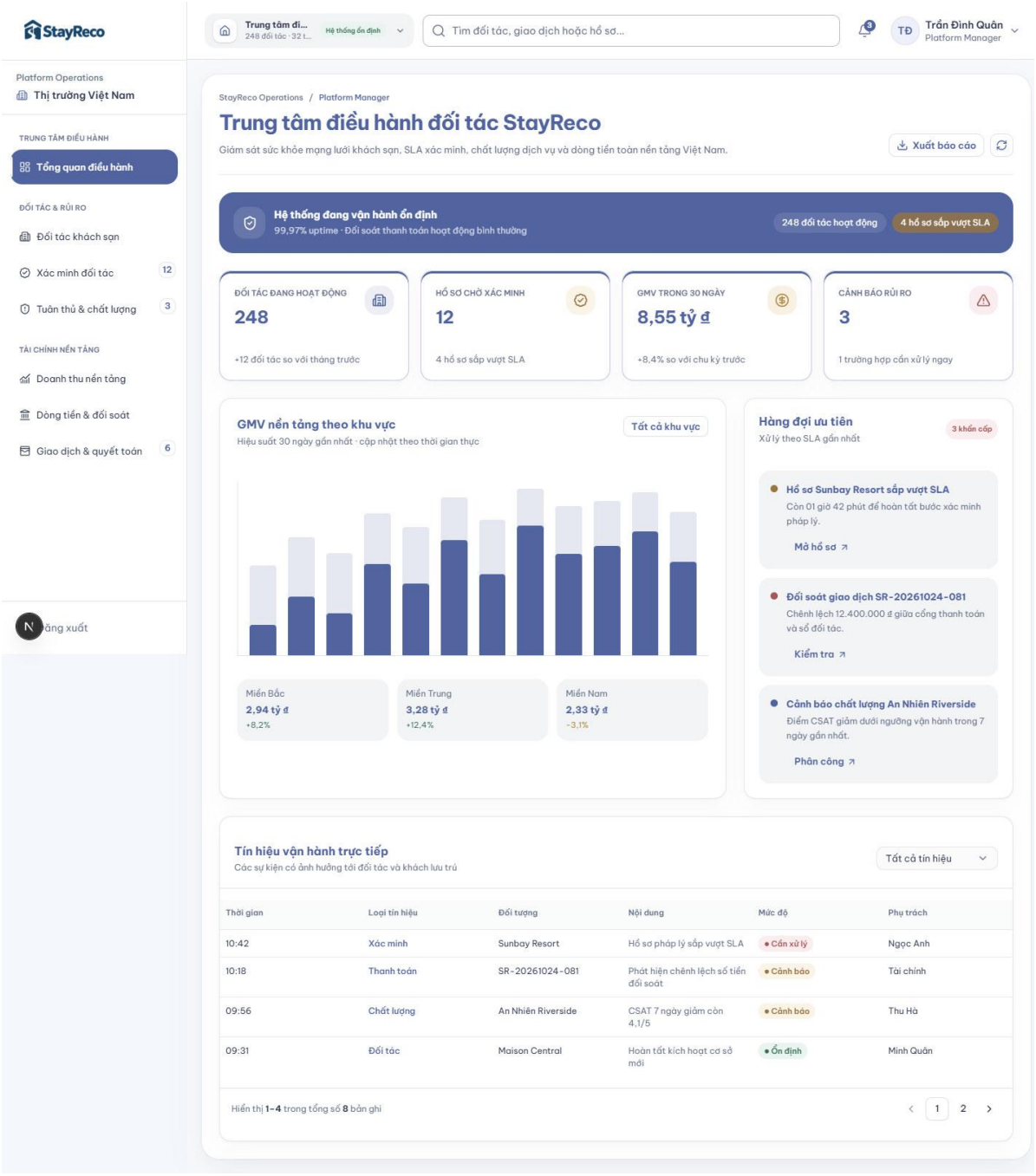


Figure 89. Manager Dashboard screen (web)

- **Function details:**
  - Each block links to the matching list.

3.14.2 Verification Queue

- **Function trigger:** Menu "Verifications".
- **Function description:** Actor: Platform Manager. Purpose: list hotel applications (UC70). Search by hotel or partner, filters by status and submission date.
- **Screen layout: Figure 90 (web).**

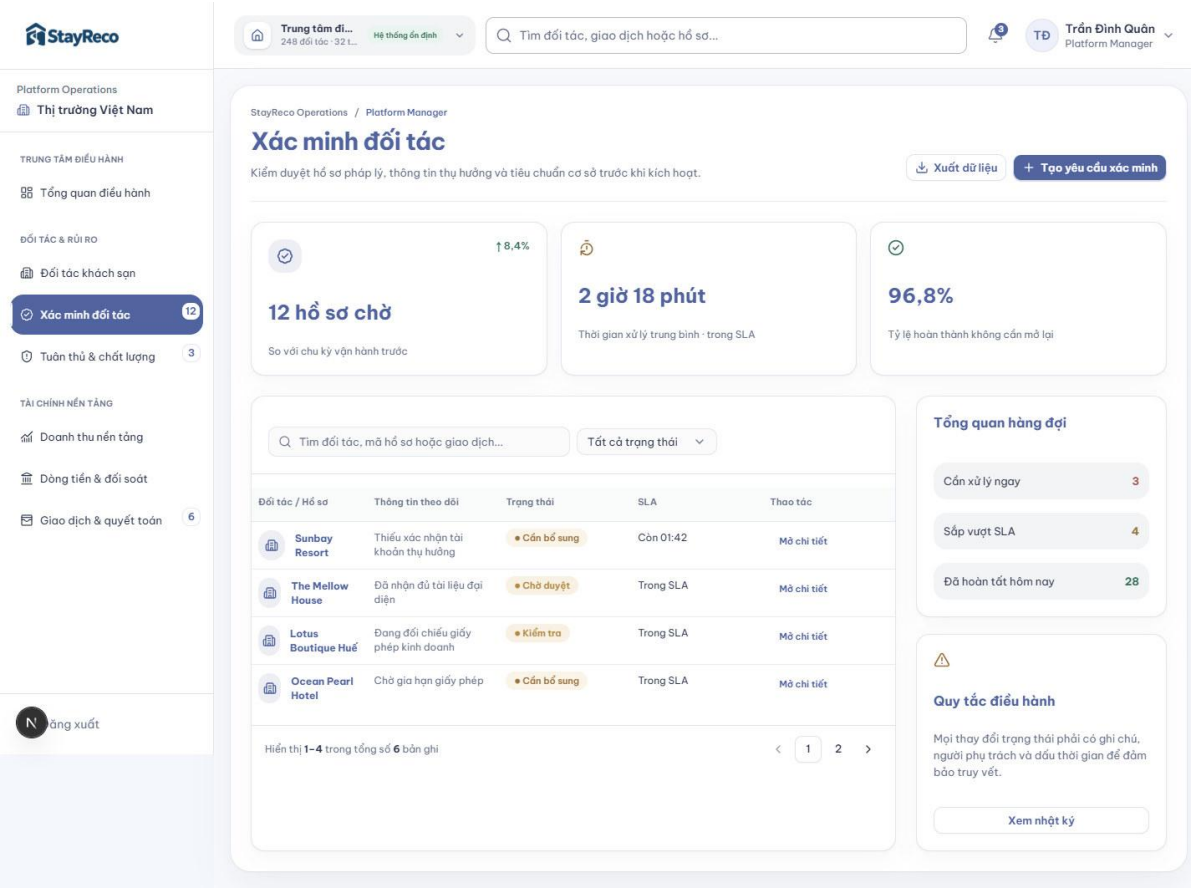


Figure 90. Verification Queue screen (web)

- **Function details:**
    - The oldest pending application is shown first.

3.14.3 Verification Review

- **Function trigger:** Click on an application in Verification Queue.
- **Function description:** Actor: Platform Manager. Purpose: decide the application of one hotel (UC70). A document viewer and a review panel with a checklist and the decisions Approve, Request changes and Reject.
- **Screen layout: Figure 91 (web).**

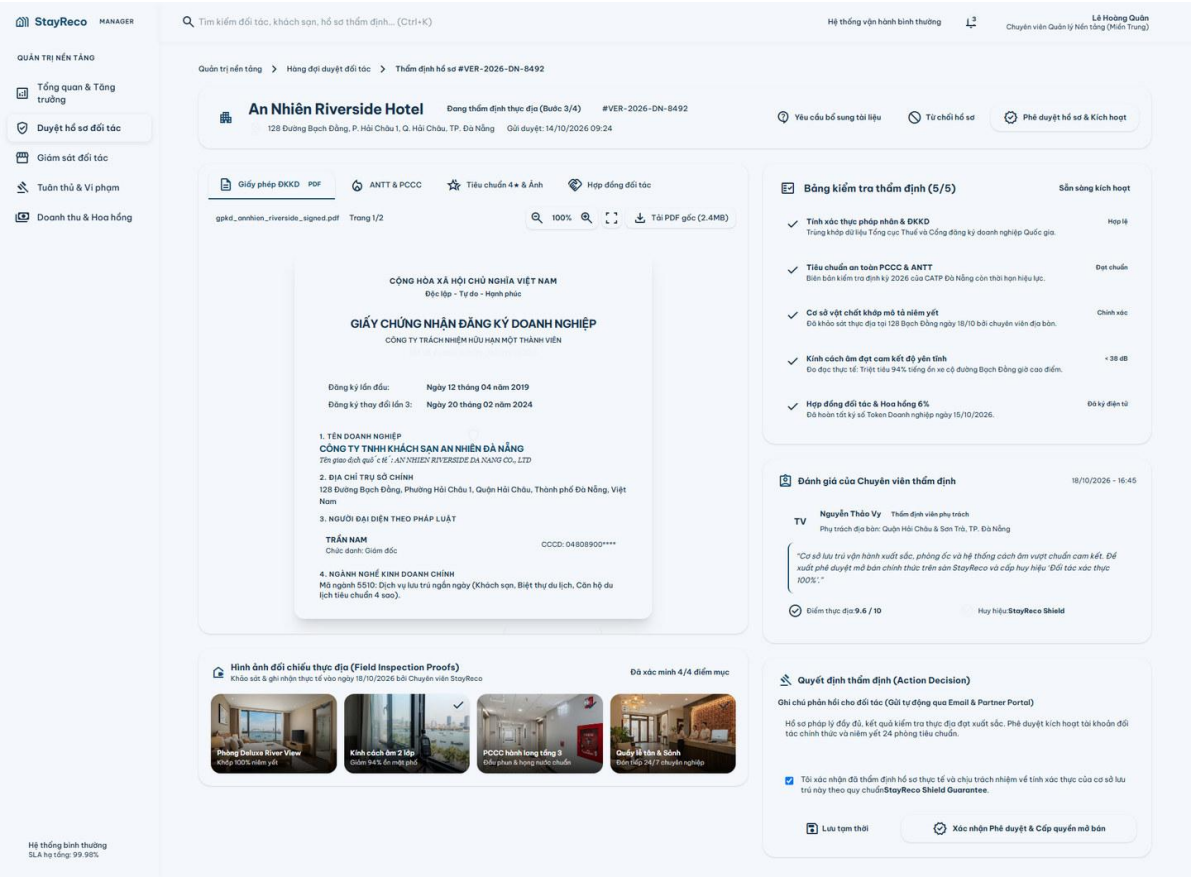


Figure 91. Verification Review screen (web)

- **Function details:**
  - Request changes and Reject require a reason, which is sent to the partner.
  - Approving makes the hotel visible to travellers (BR-01).
  - Approval is given per hotel; when the first hotel of a partner is approved, the partner becomes approved (BR-19).

3.14.4 Partners List

- **Function trigger:** Menu "Partners".
- **Function description:** Actor: Platform Manager. Purpose: monitor partner hotels (UC71). The table shows verification and listing state, bookings, cancellation rate and satisfaction, with links to Partner Revenue Details and Compliance Cases.
- **Screen layout: Figure 92 (web).**

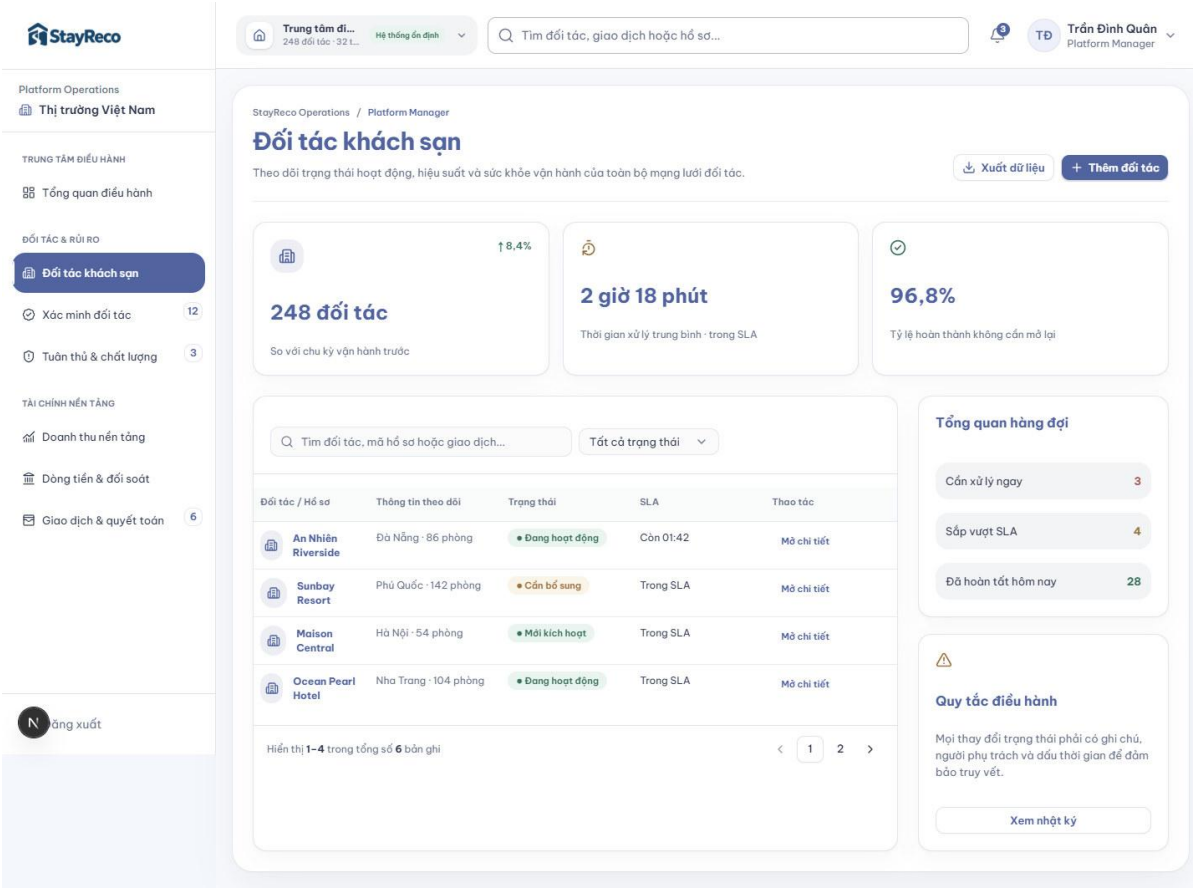


Figure 92. Partners List screen (web)

- **Function details:**
  - The list can be sorted by cancellation rate and satisfaction to spot weak hotels.
  - The manager can ask an active hotel to verify again, with a reason; the hotel then goes through the review again (BR-19).

3.14.5 Compliance Cases

- **Function trigger:** "Open case" on Partners List, or menu "Compliance".
- **Function description:** Actor: Platform Manager. Purpose: handle policy violations (UC72, UC73). A case holds the evidence and findings, and the manager can send a suspension request to the admin.
- **Screen layout: Figure 93 (web).**

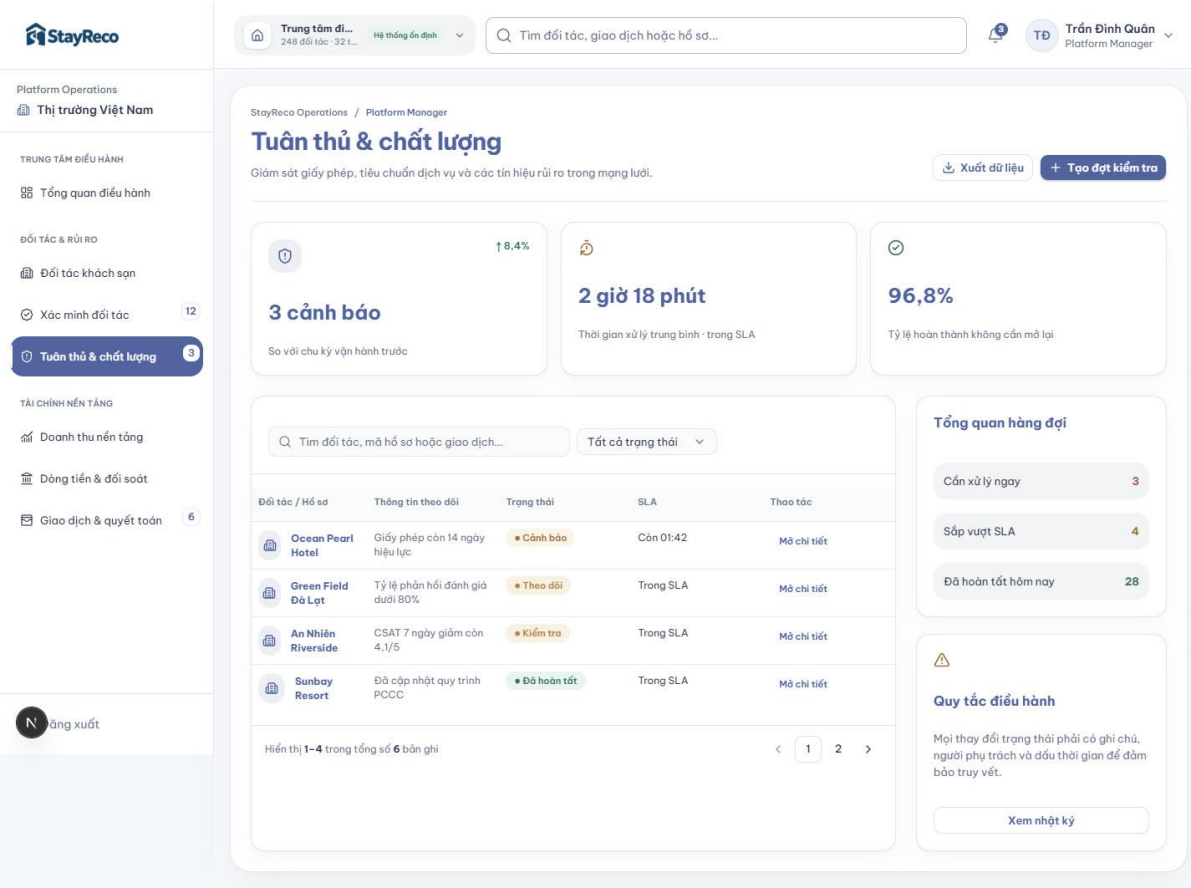


Figure 93. Compliance Cases screen (web)

- **Function details:**
  - The platform manager can only request a suspension; the admin decides (BR-15).

3.15 Platform Revenue

3.15.1 Platform Revenue

- **Function trigger:** Menu "Revenue".
- **Function description:** Actor: Platform Manager. Purpose: follow platform revenue (UC75, UC76). The screen shows commission earned, refund adjustments and gross booking value with a trend chart, a drill-down table and the "Export" action.
- **Screen layout: Figure 94 (web).**

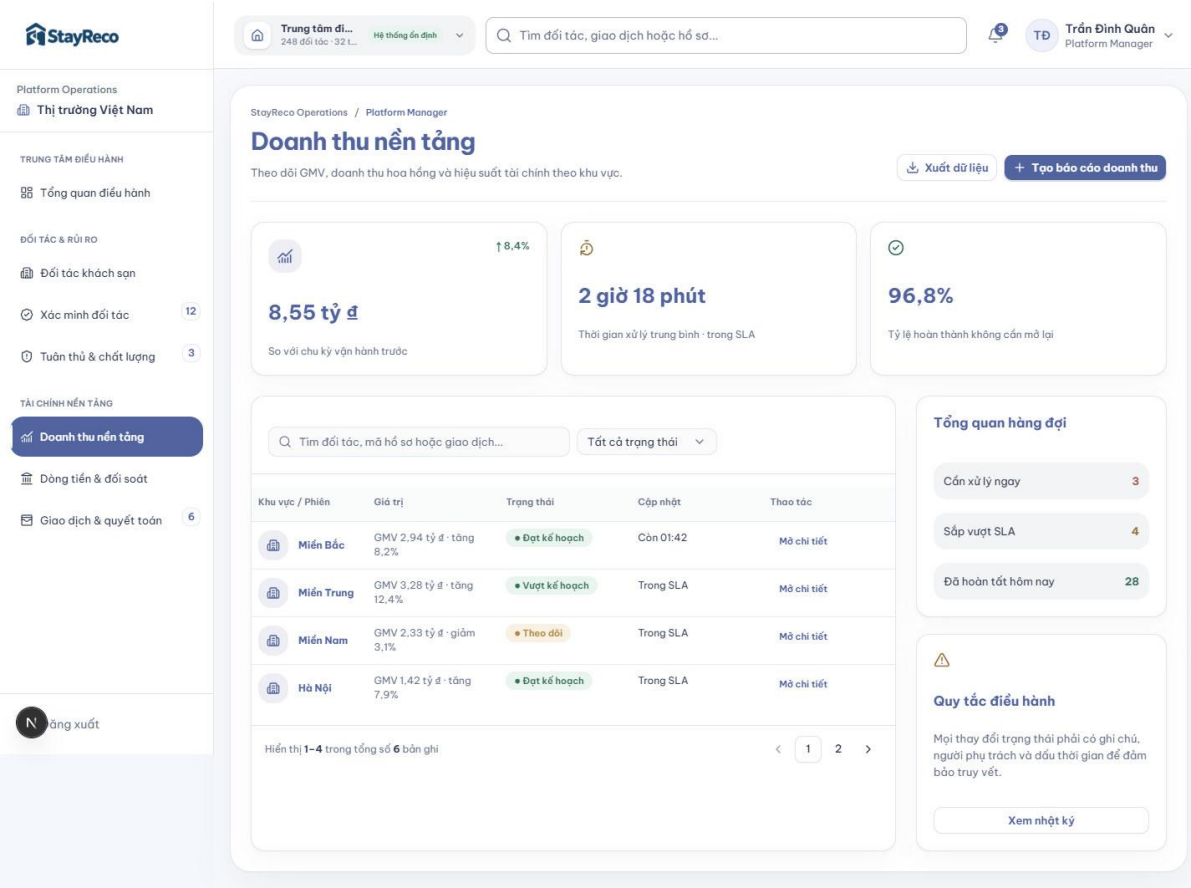


Figure 94. Platform Revenue screen (web)

- **Function details:**
    - Figures come from the commission records (BR-13); export produces an Excel file of the filtered data.

3.15.2 Partner Revenue Details

- **Function trigger:** Click on a hotel in Platform Revenue or Partners List.
- **Function description:** Actor: Platform Manager. Purpose: see the bookings and money flow of one hotel in the selected period.
- **Screen layout: Figure 95 (web).**

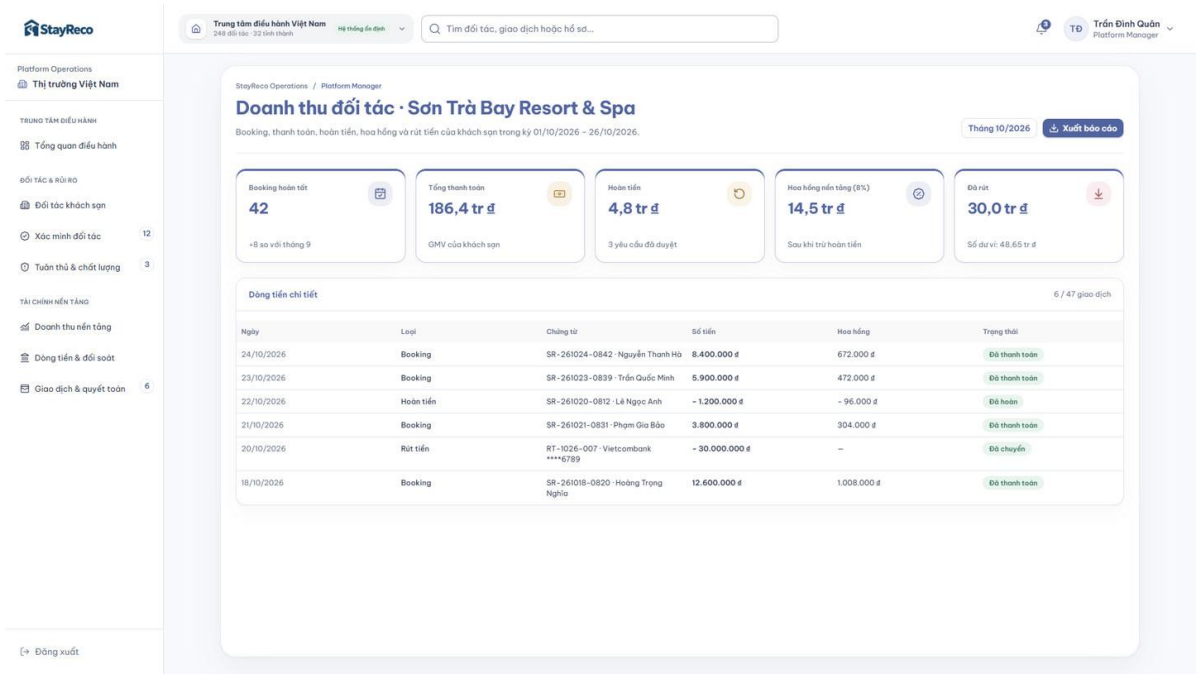


Figure 95. Partner Revenue Details screen (web)

- **Function details:**
    - The screen shows bookings, payments, refunds, commission and withdrawals of the hotel.

3.15.3 Commission Transactions

- **Function trigger:** Menu "Commission".
- **Function description:** Actor: Platform Manager. Purpose: check commission per booking: rate, amount, adjustment and status.
- **Screen layout: Figure 96 (web).**

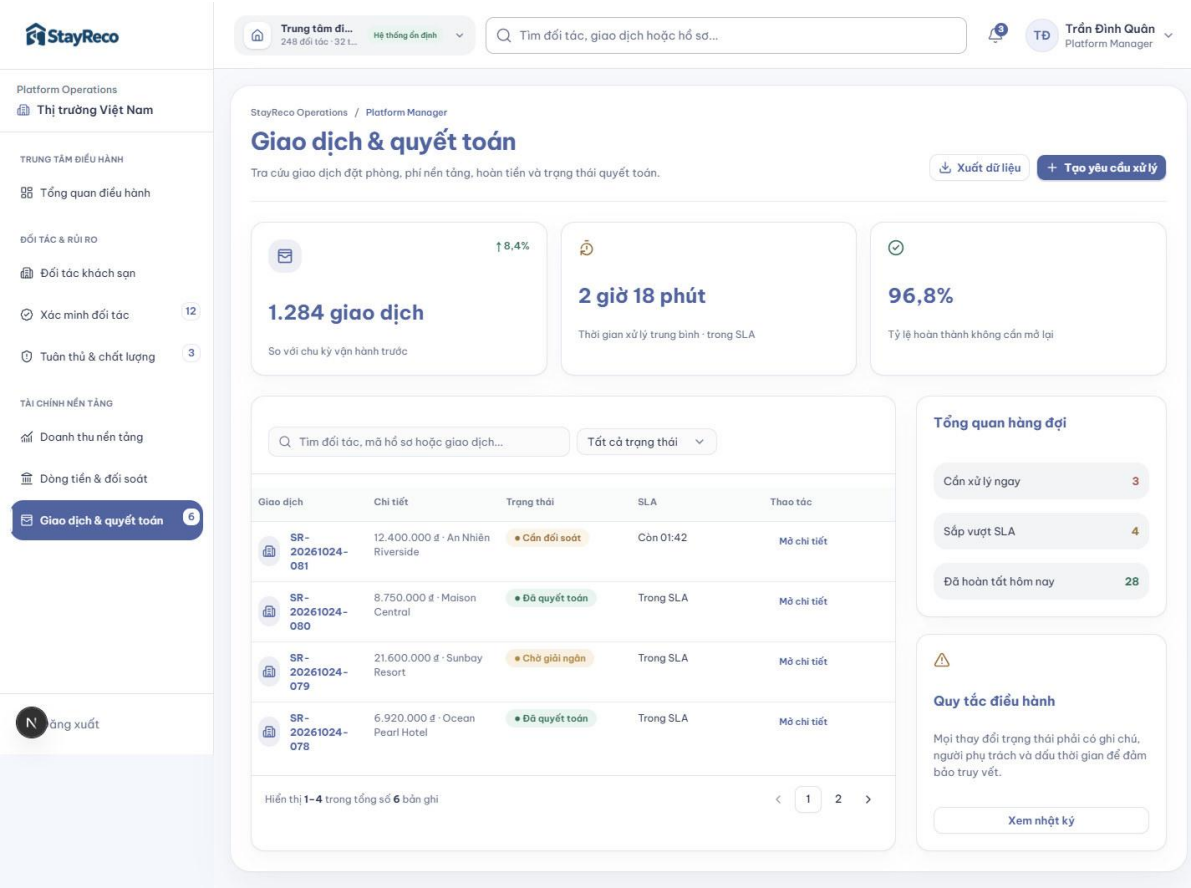


Figure 96. Commission Transactions screen (web)

- **Function details:**
    - Adjustments created by refunds are shown next to the original record.

3.15.4 Cash Flow Report

- **Function trigger:** Menu "Cash flow".
- **Function description:** Actor: Platform Manager. Purpose: split the money received through the gateway into hotel share, platform commission and refunds for reconciliation.
- **Screen layout: Figure 97 (web).**

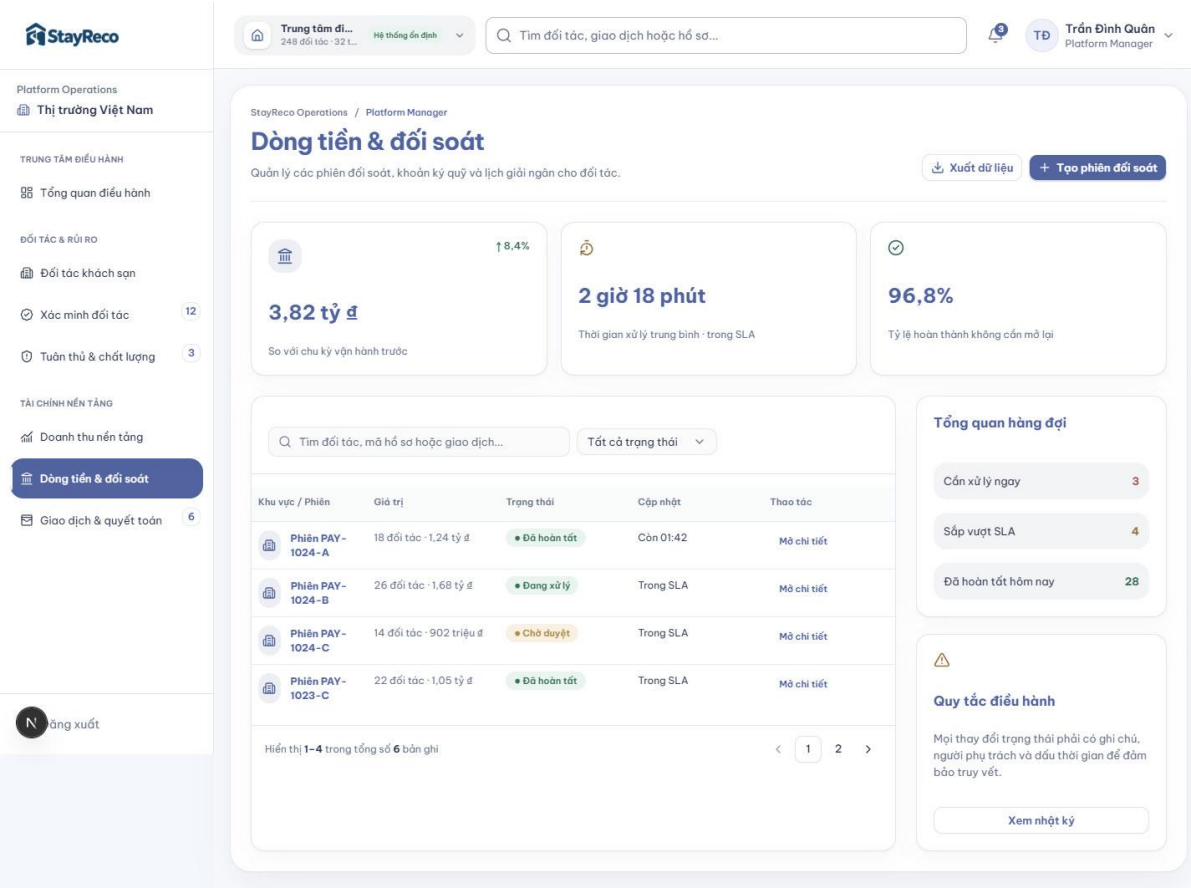


Figure 97. Cash Flow Report screen (web)

- **Function details:**
  - The total of the three parts must equal the amount received; a difference is highlighted.

3.15.5 Withdrawal Requests

- **Function trigger:** Menu "Withdrawals", or the action queue on Manager Dashboard.
- **Function description:** Actor: Platform Manager. Purpose: process partner withdrawals (UC74). The queue shows hotel, amount, bank account and status; a drawer confirms the transfer with its reference or rejects the request with a reason.
- **Screen layout: Figure 98 (web).**

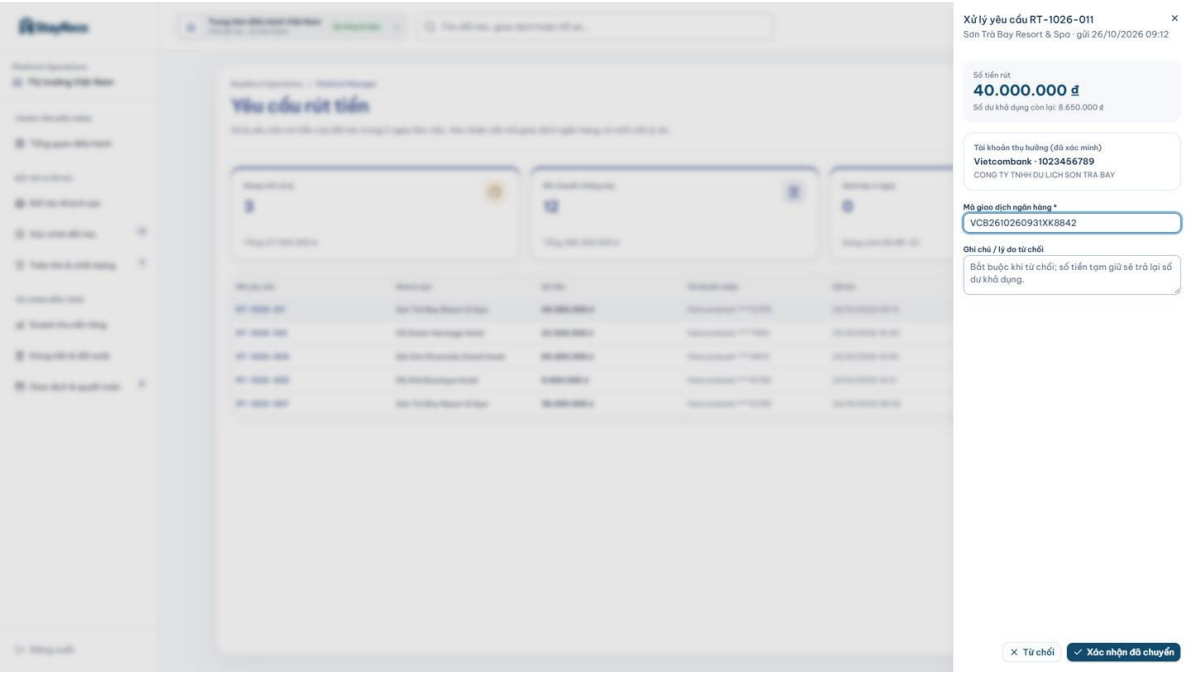


Figure 98. Withdrawal Requests screen (web)

- **Function details:**
  - Requests are processed within 2 working days (BR-22).
  - Confirming requires the bank transfer reference; rejecting requires a reason and returns the held amount to the available balance.
  - The partner is notified of the result.

3.16 System Administration

3.16.1 Admin Dashboard

- **Function trigger:** Opened after an admin logs in.
- **Function description:** Actor: Admin. Purpose: watch the system (UC77). The dashboard shows account statistics, pending account decisions, payment issues and recent security events.
- **Screen layout: Figure 99 (web).**

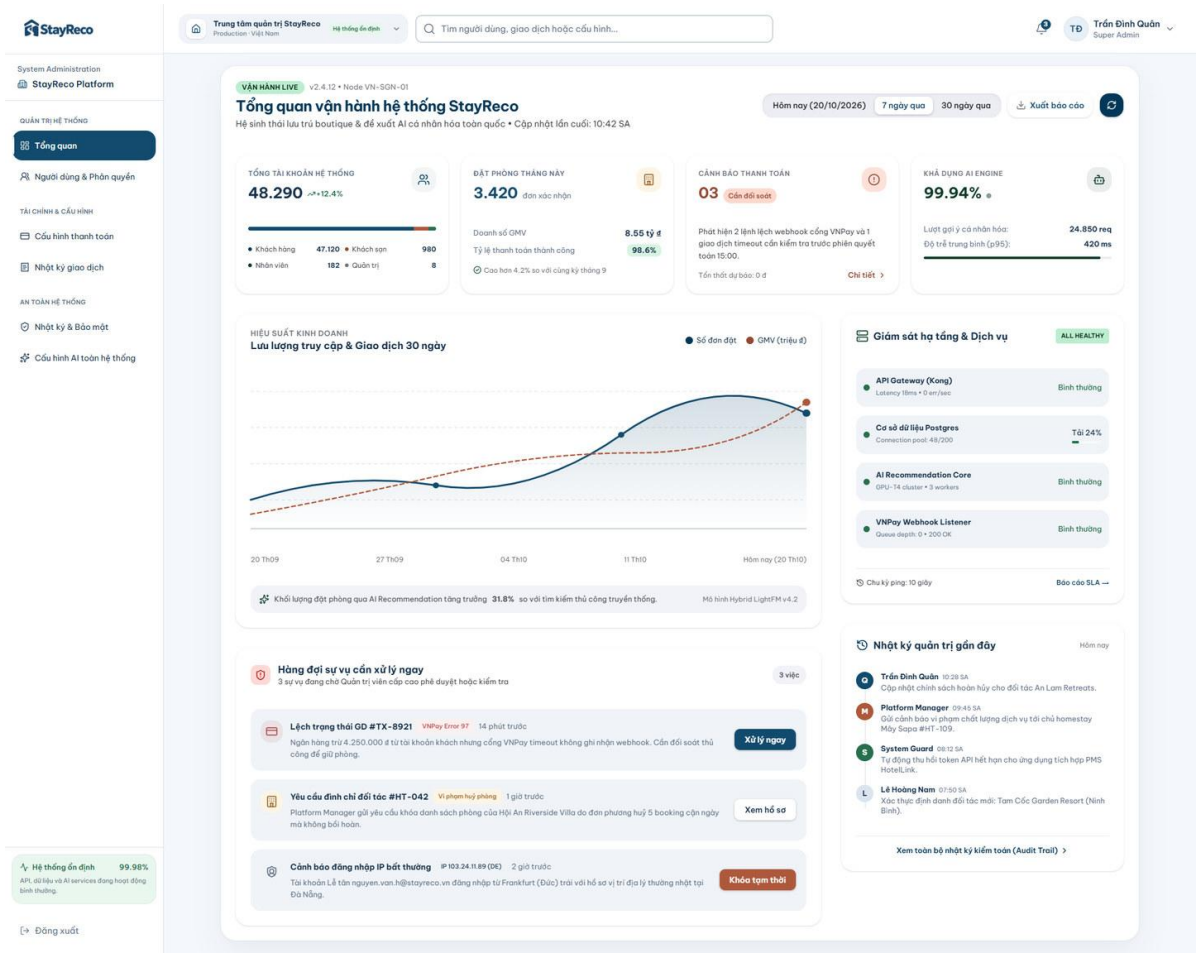


Figure 99. Admin Dashboard screen (web)

- **Function details:**
    - Each block links to the matching list.

3.16.2 Users List

- **Function trigger:** Menu "Users".
- **Function description:** Actor: Admin. Purpose: find any account (UC82). The table shows name, email, role, organization, status and created date, with search and filters.
- **Screen layout: Figure 100 (web).**

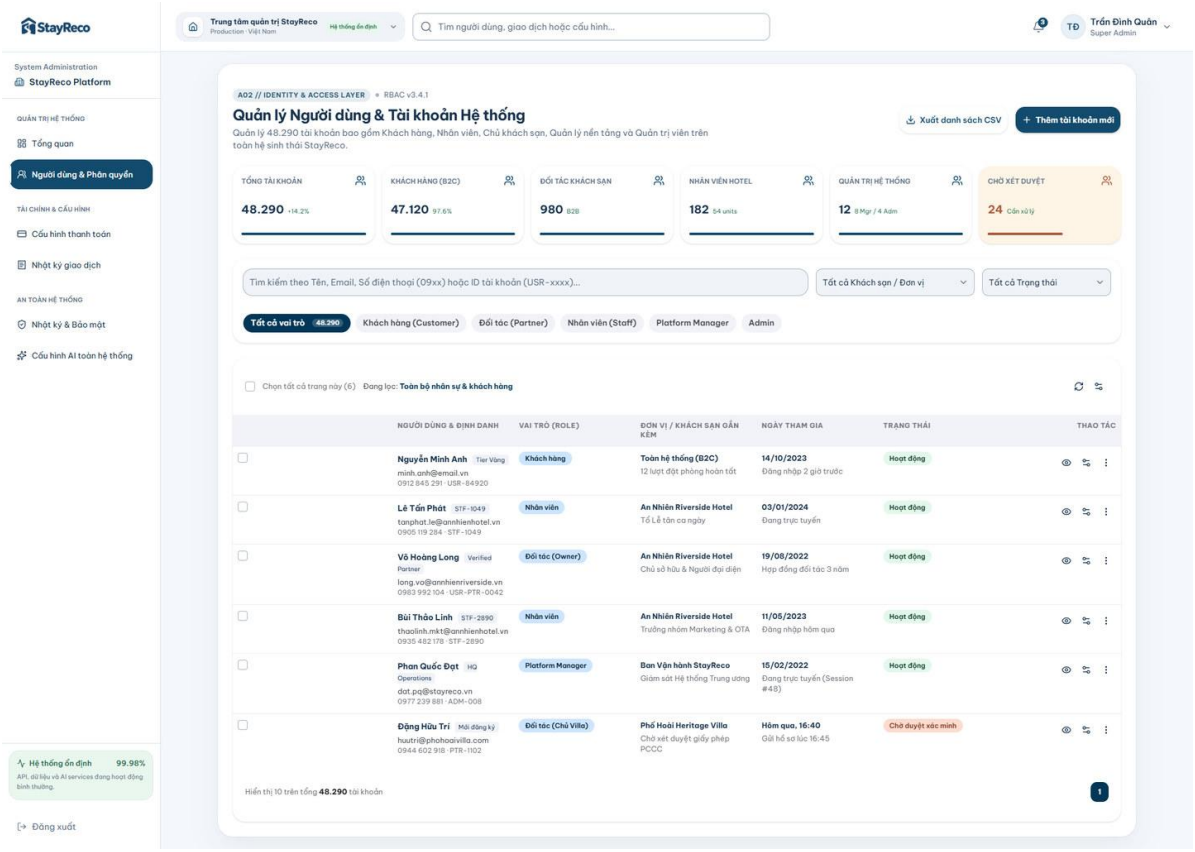


Figure 100. Users List screen (web)

- **Function details:**
    - The admin can invite a platform manager from this screen (email link valid for 72 hours).

3.16.3 User Details

- **Function trigger:** Click on an account in Users List.
- **Function description:** Actor: Admin. Purpose: manage one account (UC82, UC83). The screen shows the account summary, the audit timeline, a linked suspension request if any, and the activate, deactivate or decide-suspension actions.
- **Screen layout: Figure 101 (web).**

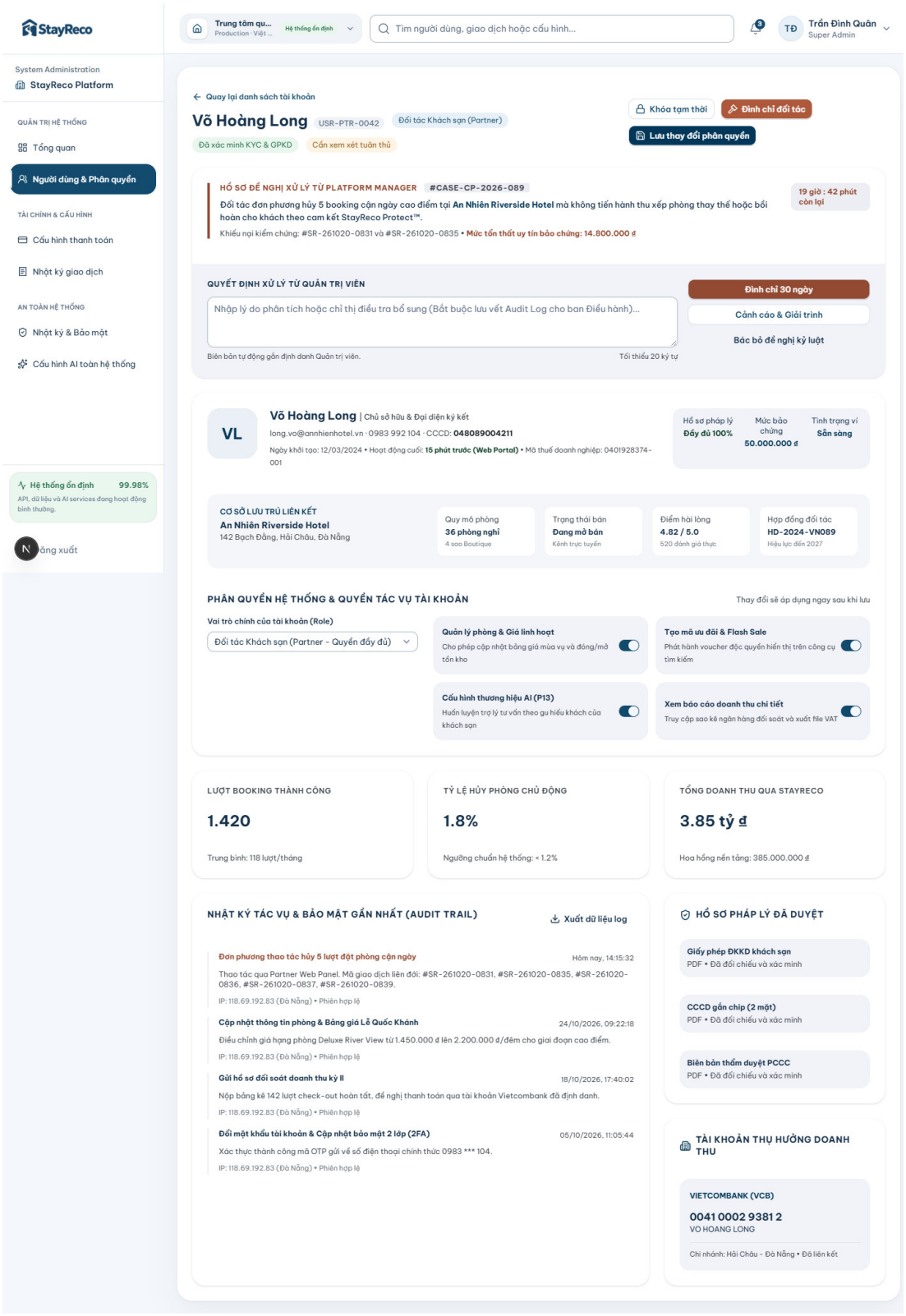


Figure 101. User Details screen (web)

- **Function details:**
  - Deactivating or suspending asks for confirmation (MSG11) and signs the account out of all devices.
  - The decision on a suspension request needs a reason and is logged.

3.16.4 Payment Settings

- **Function trigger:** Menu "Payments".
- **Function description:** Actor: Admin. Purpose: configure the SePay gateway (UC78): the bank account that receives payments, the API key and the webhook URL (shown masked), and a connection test.
- **Screen layout: Figure 102 (web).**

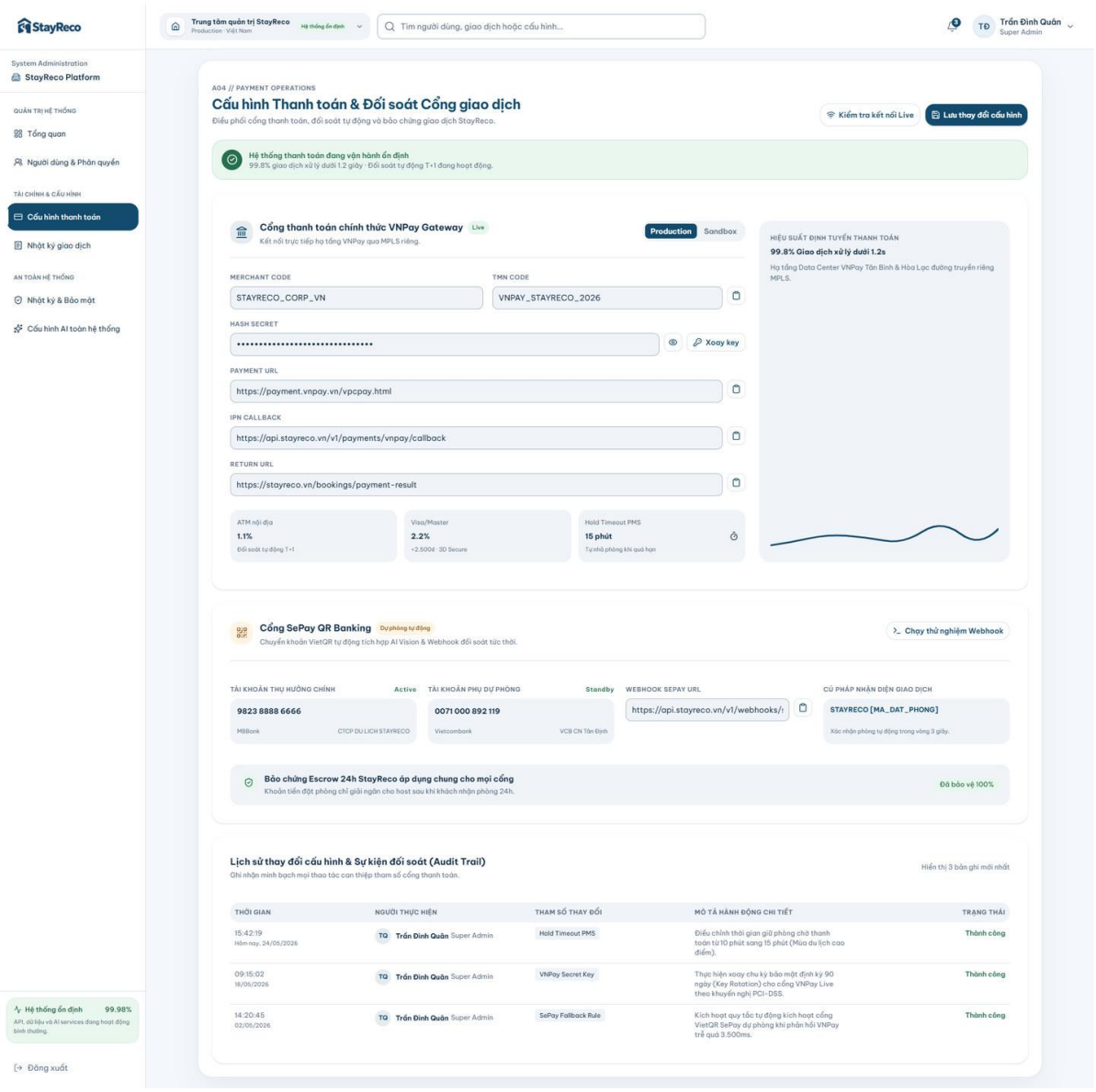


Figure 102. Payment Settings screen (web)

- **Function details:**
  - Credentials are stored encrypted and never shown in full again after saving.
  - Changes are written to the audit log.

3.16.5 Transactions List

- **Function trigger:** Menu "Transactions".
- **Function description:** Actor: Admin. Purpose: check payments and refunds (UC79), with filters by date, status, booking code and transaction ID, and export.
- **Screen layout: Figure 103 (web).**

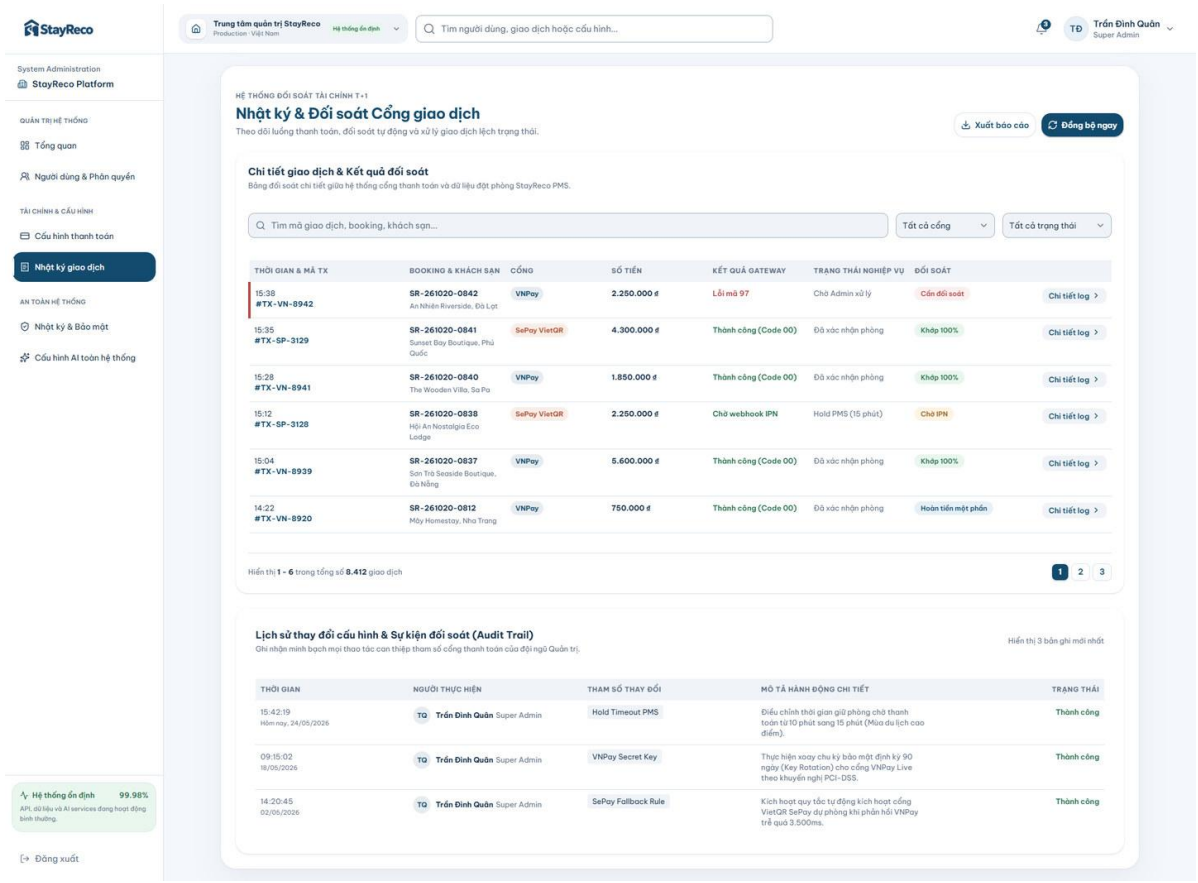


Figure 103. Transactions List screen (web)

- **Function details:**
    - Export produces an Excel file of the filtered data.

3.16.6 Transaction Details

- **Function trigger:** Click on a transaction.
- **Function description:** Actor: Admin. Purpose: inspect one transaction: linked booking, expected and confirmed amounts, event history and a status check with the gateway.
- **Screen layout: Figure 104 (web).**

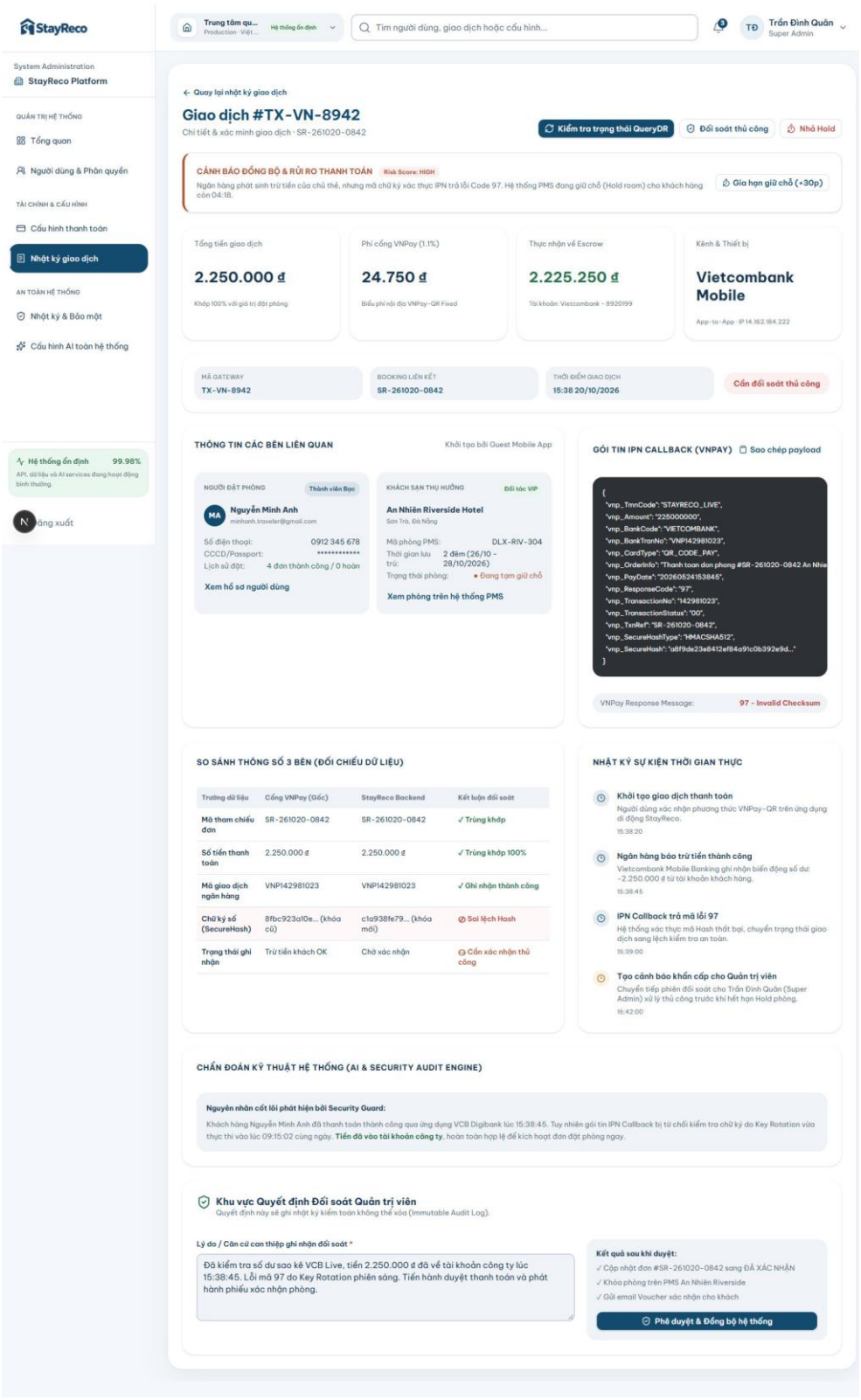


Figure 104. Transaction Details screen (web)

- **Function details:**
    - A mismatch between expected and confirmed amounts is highlighted.

3.16.7 System Logs

- **Function trigger:** Menu "Logs".

- **Function description:** Actor: Admin. Purpose: read the audit log and security events (UC80), filtered by time, actor, action and severity.
- **Screen layout: Figure 105 (web).**

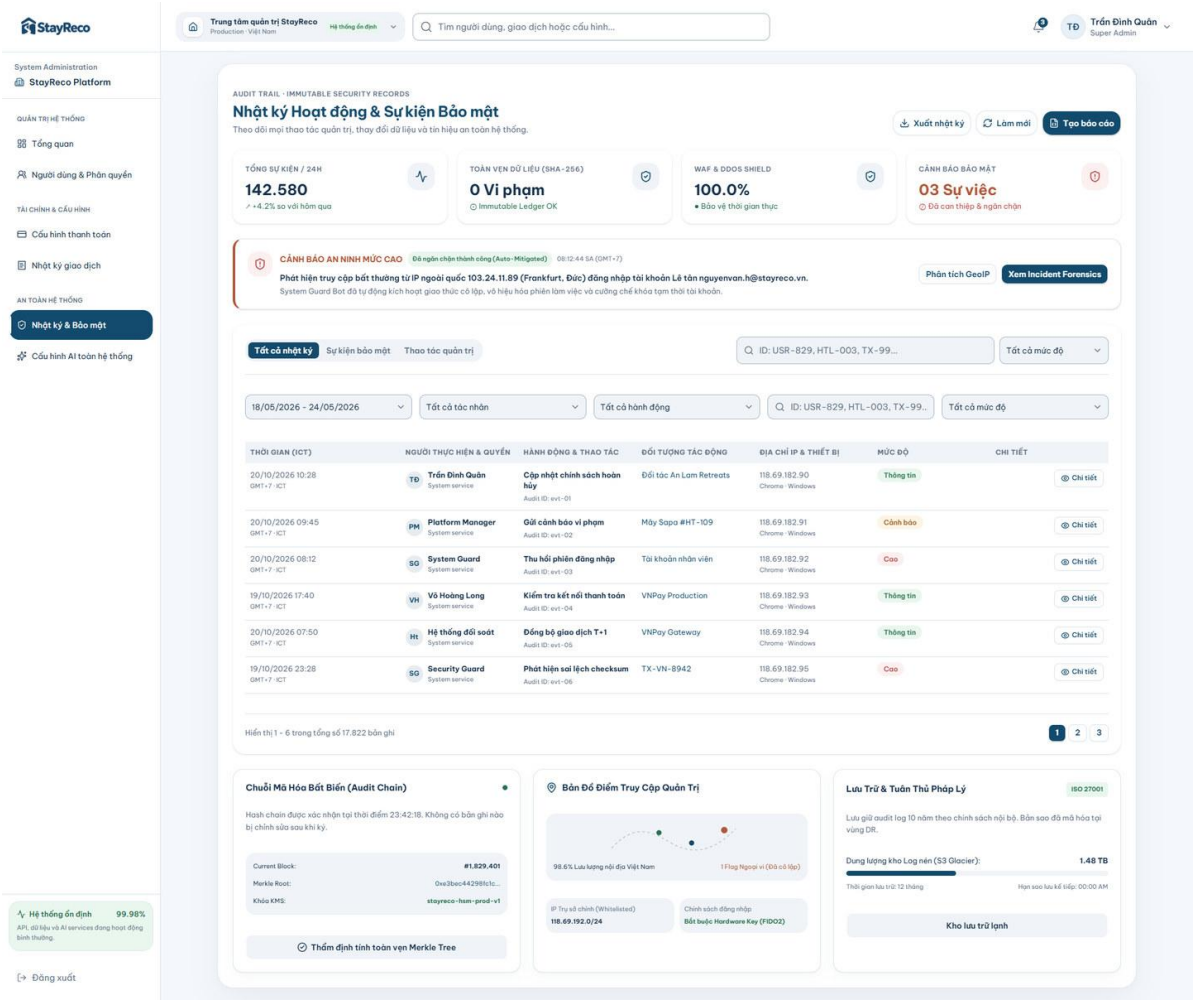


Figure 105. System Logs screen (web)

- **Function details:**
  - Logs are read-only and cannot be deleted from the screen.

3.16.8 System AI Settings

- **Function trigger:** Menu "AI settings".
- **Function description:** Actor: Admin. Purpose: configure the AI of the platform (UC81): model, API credential (masked), usage limits and safety filters.
- **Screen layout: Figure 106 (web).**

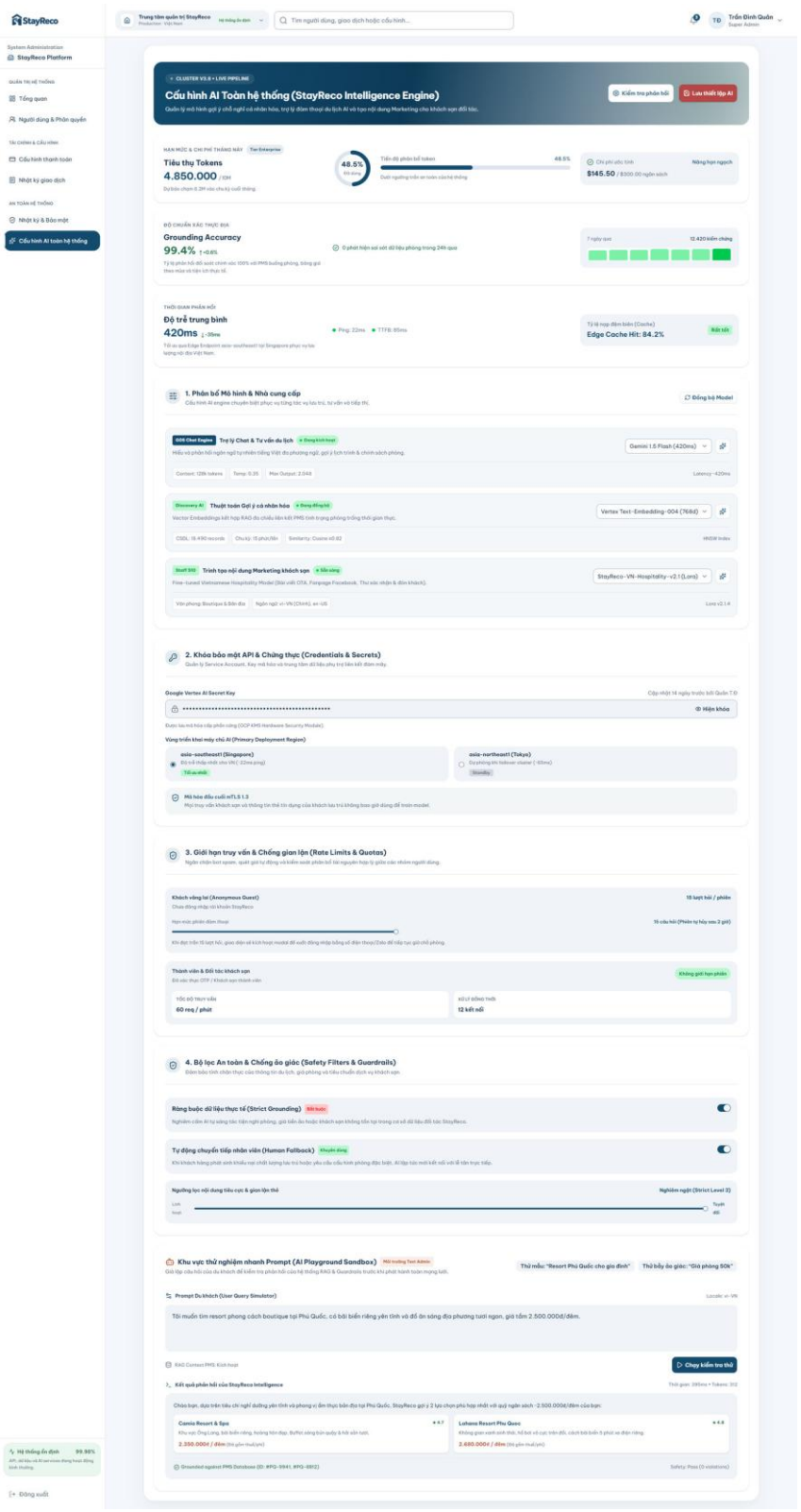


Figure 106. System AI Settings screen (web)

- **Function details:**
  - A limit reached makes the AI functions show MSG24 until the next period.

3.16.9 Audit Logging

- **Function trigger:** Runs on every create, update, approval and security-relevant action.

- **Function description:** Non-screen function. Purpose: keep a record of who did what and when.
- **Screen layout:** *not applicable, this function runs without a screen.*
- **Function details:**
    - Each record stores the actor, action, target, time and IP address.
    - Failed logins, account locks, permission changes, refund decisions and withdrawal decisions are always logged.

4. Non-Functional Requirements

4.1 External Interfaces

- **User interfaces:** A responsive web application (Next.js with React 19, TypeScript, TailwindCSS) for all six roles, and a React Native (Expo) mobile application for Guest, Customer and Hotel Staff on Android and iOS. The interface is in Vietnamese and English.
- **Hardware interfaces:** No special hardware. The front desk scans QR vouchers with the camera of a phone, tablet or laptop; a manual booking-code search is always available.
- **Software interfaces:** SePay for payments (VietQR bank transfer into the account of the platform, confirmed by webhook); OpenAI API for chat and content drafts; PostgreSQL 16 with pgvector; Cloudinary for hotel images, room type images and verification documents (documents are stored in private mode); Firebase Cloud Messaging for push notifications; Nodemailer over SMTP for email; the APIs of the supported social media platforms for publishing.
- **Communication interfaces:** All clients call the NestJS RESTful API over HTTPS with JSON bodies, and every response uses the common envelope in section 5.2. The API calls the Python FastAPI AI service inside the private network. Payment results arrive through the SePay webhook, checked with its API key.

4.2 Quality Attributes

The quality attributes below are the targets the team designs and tests against.

4.2.1 Usability

- A first-time customer can find a hotel, book and pay in no more than 5 minutes without guidance.
- A new staff member can look up a booking and complete a check-in after a 30-minute introduction.
- Screens work from 360 px mobile width to desktop, required fields are marked with *, and every destructive action asks for confirmation.

4.2.2 Reliability

- Availability of at least 99% during the demonstration and user acceptance test period.
- Booking, payment, check-in and check-out are transactional: a failure leaves no half-updated record, and the same room can never be sold twice for the same night.
- The database is backed up daily and can be restored within 4 hours.
- A critical defect is one that loses data, charges a wrong amount or blocks booking or payment; the release for a review or the defence must have no open critical defect.

4.2.3 Performance

- 95% of API requests for search, detail and booking respond within 2 seconds with 100 concurrent users.
- The first reply of the AI assistant starts within 5 seconds; a full answer is returned within 15 seconds.

- A recommendation list is returned within 1 second from precomputed scores.

4.2.4 Security

- Passwords are hashed; authentication uses JWT with expiry, and every request is checked by role.
- All traffic uses HTTPS; secrets and API keys are stored encrypted and shown masked on screens.
- No card data is stored; a payment is confirmed only by the SePay webhook, checked with its API key and matched with the booking.
- After 5 failed logins in a row the account is locked for 15 minutes (the lock is not an account status). Login is limited to 5 requests per minute and email sending to 3 requests per 15 minutes (BR-24).
- The access token is valid for 15 minutes. The refresh token is valid for 30 days, is rotated on every use and is revoked at logout; users can see their active sessions and sign out of one device or of all devices.
- The beneficiary account number of a partner is encrypted when stored and shown masked; verification documents are stored privately on Cloudinary and only the hotel owner and the platform manager can open them.
- Staff and platform manager accounts are created only by invitation; the invitation link is valid for 72 hours and the invited person sets the password (BR-25).
- The QR voucher carries a signed token instead of the plain booking code. At check-in the system checks the signature and the booking status before it opens the booking.
- Uploaded files are checked on both the client and the server against the allowed types (JPG, PNG or WebP for images; PDF, JPG or PNG for documents) and the size limit (10 MB for verification documents); MSG23 shows the limits.

4.2.5 Multi-tenancy & Data Privacy

- Every hotel-scoped record carries the hotel (tenant) identifier and every query is filtered by it; a partner or staff member can never read or change the data of another hotel.
- Personal data is handled according to Decree 13/2023/ND-CP and GDPR principles: only the data needed by a role is shown, and users can view and update their own data.

4.2.6 Maintainability & Compatibility

- The code follows a modular, layered structure with consistent naming; business logic is covered by unit tests.
- The web application supports the latest versions of Chrome, Edge, Safari and Firefox; the mobile application supports Android 10+ and iOS 15+.

5. Requirement Appendix

This appendix lists the business rules, the common requirements and the messages used by all functions of StayReco.

5.1 Business Rules

The rules below apply to every function that touches the related data.

ID	Rule Definition
BR-01	A hotel is shown to travellers only after its verification request is approved by the platform manager.

BR-02	Availability is counted per night. A room booked for the nights from the check-in date to the day before the check-out date cannot be sold again for those nights.
BR-03	When a customer confirms a booking, the room is held for a limited time (default 15 minutes) while the payment is completed; after that the hold is released automatically.
BR-04	A booking becomes confirmed only after the payment gateway reports a successful payment. No screen can mark a booking as paid by hand.
BR-05	The price shown at checkout is the final price: it already includes VAT and the service charge of the hotel, and it is paid in full when the booking is made. Final price = effective base price (pricing rules for the stay dates and for how early the booking is made) − promotion or voucher discount − value of the loyalty points used. The point value and the maximum share that can be paid with points follow the loyalty program of the hotel. The final price is never below zero and does not change after booking.
BR-06	A booking can use at most one discount code, either a promotion code or a voucher, together with loyalty points. Discounts are applied in this order: base price, then the promotion or voucher, then the points. Vouchers and points belong to one hotel and can only be used for bookings of that hotel.
BR-07	The cancellation policy that applies to a booking is the policy in force at the time of booking; later policy changes do not affect existing bookings.
BR-08	Refund amount = paid amount − cancellation fee − amounts already refunded, and never more than the paid amount.
BR-09	Because a booking is paid in full when it is made, its price never changes afterwards. A customer can ask to change details that do not affect the price (guest name, contact details, arrival time, special requests); the change takes effect after the hotel accepts it. New dates, another room type or another number of guests are handled by cancelling under the cancellation policy and making a new booking at the current price. A booking cannot be cancelled after check-in.
BR-10	At check-in the staff assigns a ready physical room of the booked room type; the room type is never changed silently.
BR-11	Loyalty points are added only after check-out is completed, following the program of the hotel.
BR-12	A customer can review a booking only once and only after the stay is completed.
BR-13	The platform commission is calculated on each completed booking and on each no-show booking at the commission rate in force at booking time. The default rate is 8% and the admin can set it between 8% and 15%.
BR-14	AI-generated content must be approved by the hotel partner before it is scheduled or published; content edited after approval needs a new approval.
BR-15	The platform manager can only request a partner suspension; the admin makes the account decision.
BR-16	A hotel partner cannot grant Admin or Platform Manager permissions and cannot access the staff of another partner.
BR-17	A password has at least 8 characters with at least one letter and one digit. A password reset link can be used once and expires after 15 minutes.
BR-18	Personalized recommendations require a signed-in customer and are based on preferences, search behaviour and booking history. Recommendations for guests who are not signed in are not provided in this version.
BR-19	A partner account can sign in once its email is verified. Each hotel has its own verification request and status: draft, pending review, changes requested, approved or rejected. The platform manager approves, rejects or requests changes per hotel, and the last two need a reason; the partner becomes approved when its first hotel is approved. Resubmitting creates a new request and keeps the history. Until a hotel is approved it is not listed, and for that hotel the partner can only edit the onboarding

	information and follow the verification status. The platform manager can ask an active hotel to verify again, with a reason; editing an active hotel does not need a new approval.
BR-20	A refund amount calculated by BR-08 is approved by default. Hotel staff check the request and forward it to the hotel partner, who can reject it only when the booking has already been refunded or the request is a duplicate, when the booking status does not allow a refund (checked in or no-show), or when fraud is suspected and evidence is attached. The reason is written to the audit log and sent to the customer. A request made before the check-in time and still undecided at that time is approved automatically.
BR-21	A partner can own several hotels. Each hotel is a separate tenant with its own rooms, prices, staff, bookings and wallet. A staff account belongs to exactly one hotel, and the partner works on one selected hotel at a time; only the dashboard can also show all hotels together.
BR-22	The hotel share of a paid booking goes to the pending balance of the hotel wallet and becomes available after the guest checks out and the hold period set by the platform has passed. A partner can withdraw at most the available balance, only to the beneficiary account declared by the partner, and each hotel has at most one open request at a time. The platform manager processes requests within 2 working days: a paid request records the transfer reference, and a rejected request returns the held amount to the available balance.
BR-23	A new account stays inactive until the 6-digit code sent to its email is entered. The code is valid for 10 minutes, requesting a new code cancels the previous one, and a code is cancelled after 5 wrong entries.
BR-24	After 5 wrong passwords in a row an account is locked for 15 minutes; the lock ends by itself and is not an account status. Login is limited to 5 requests per minute and email sending to 3 requests per 15 minutes. A user can see the signed-in devices and sign out of one session or of all devices.
BR-25	Staff accounts are created only by invitation from the partner, who chooses the hotel and the position; platform manager accounts are created only by invitation from the admin. The invitation link is valid for 72 hours and can be used once; the invited person sets the password, and until then the account is in the invited state.
BR-26	Each hotel must provide the business license, the fire safety (PCCC) certificate and the security and order (ANTT) certificate; the star rating certificate and other documents are optional. Files are PDF, JPG or PNG up to 10 MB, stored privately, and only the owner of the hotel and the platform manager can open them. The beneficiary account number is encrypted when stored and masked when shown.
BR-27	Each hotel declares its total number of rooms and a default base price for new room types, and cannot have more physical rooms than declared. A room type that still has physical rooms cannot be deleted. Only an AVAILABLE room can be booked; MAINTENANCE, CLEANING and OCCUPIED rooms are not ready. Locking a room moves it to MAINTENANCE and unlocking returns it to AVAILABLE; a room with a guest cannot be locked.
BR-28	Staff permissions come in five groups: Booking, Front desk, Customer care, Refund and Marketing. A staff member without any group can only view the data of the hotel.
BR-29	Amounts in VND are stored with two decimal places and rounded once, half up. A percentage discount is calculated on the base price. All amounts are calculated by the back end; clients only display them.

5.2 Common Requirements

- Dates are shown as dd/mm/yyyy and times in the Asia/Ho_Chi_Minh time zone.

- Money is shown in Vietnamese dong with dot as thousand separator, for example 2.250.000 ₫; a price always states whether it is per night or for the whole stay.
- Amounts in VND are stored with two decimal places and rounded once (half up); a percentage discount is calculated on the base price. Every amount is calculated by the back end and the client only displays it (BR-29).
- Every list has search, filters and pagination with 10 rows per page by default.
- Required fields are marked with *; inputs are validated on the client and again on the server.
- Every API response uses one envelope: { success, message, data, meta? } on success and { success: false, message, code, details? } on error. The client chooses the MSGxx message from the code, not from the message text.
- Data is not physically deleted: records are deactivated or soft-deleted, and every change is written to the audit log.
- Booking, payment and refund statuses are shown as separate labels with text and colour.
- The interface language can be switched between Vietnamese and English at any time.

5.3 Application Messages List

#	Message code	Message Type	Context	Content
1	MSG01	In line	No result matches the search criteria	No results found. Try changing the dates or filters.
2	MSG02	In red, under the text box	A required field is empty	The {field} field is required.
3	MSG03	In red, under the text box	Input is longer than the maximum length	Exceed max length of {max_length}.
4	MSG04	In red, under the text box	Email format is invalid	Please enter a valid email address.
5	MSG05	In line	Wrong email or password when logging in	Incorrect email or password. Please check again.
6	MSG06	In line	Login with a deactivated account	Your account has been deactivated. Please contact support.
7	MSG07	In red, under the text box	Email already registered	This email is already registered.
8	MSG08	In red, under the text box	Password does not meet the rules	Password must have at least 8 characters, including a letter and a digit.
9	MSG09	In line	Password reset requested	If the email is registered, a reset link has been sent to it.
10	MSG10	Toast message	Data saved successfully	Saved successfully.
11	MSG11	Pop-up	Before a delete, cancel or deactivate action	Are you sure you want to {action}? This cannot be undone.
12	MSG12	In line	Room no longer available at checkout	This room is no longer available for the selected dates.

13	MSG13	In line	Price changed since the room was selected	The price has changed. Please review the new total before continuing.
14	MSG14	In red, under the text box	Voucher or promotion code invalid	This code is invalid, expired or not eligible for this booking.
15	MSG15	In line	Payment result is being verified	We are verifying your payment. Please do not pay again.
16	MSG16	In line	Payment failed or cancelled	Payment was not completed. You can try again before {deadline}.
17	MSG17	Toast message	Booking confirmed	Your booking {code} is confirmed.
18	MSG18	Toast message	Booking cancelled with a refund due	Booking cancelled. Your refund of {amount} is being processed.
19	MSG19	In line	Review not allowed yet	You can review this hotel after your stay is completed.
20	MSG20	In line	QR voucher invalid at check-in	This voucher is invalid or the booking was cancelled.
21	MSG21	In line	Refund amount exceeds the refundable amount	The refund cannot exceed {max_amount}.
22	MSG22	In line	Scheduling unapproved content	Only approved content can be scheduled.
23	MSG23	In line	File upload failed or wrong type	Upload failed. Allowed types: {types}, maximum {size} MB.
24	MSG24	In line	AI service temporarily unavailable	The assistant is temporarily unavailable. You can still search normally or contact the hotel.
25	MSG25	Full page	User opens a screen without permission	You do not have permission to access this page.
26	MSG26	In line	Too many failed login attempts	Too many failed attempts. Please try again in {minutes} minutes.
27	MSG27	In line	Hotel not yet approved	This hotel is waiting for verification. You can update its information or follow the verification status.
28	MSG28	In line	Login before the email is verified	Please enter the code we sent to {email} to activate your account.
29	MSG29	In red, under the text box	Email code wrong or expired	The code is wrong or has expired. You can request a new code.
30	MSG30	In red, under the text box	Withdrawal amount above the available balance	The amount cannot be more than the available balance of {amount}.

5.4 Other Requirements

- Datasets used to build and evaluate the recommender (Kaggle TripAdvisor Hotel Reviews and, if added, HotelRec) are used only for research and testing, following their licences.
- Hotels, rooms and business documents used in demonstrations are sample data created by the team; payments in demonstrations are real bank transfers of small amounts through SePay.